

Improving Short-Horizon Market-Making with Order Book and Derivatives Signals

Can derivatives-implied signals improve short-horizon market-making decisions beyond limit order book information alone?

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Section	Content	Output
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Numbering: Labels printed in the outputs below are the notebook's own, and where they differ from the report the report numbering is authoritative. Table 8 here is report Table 14, and Appendix F here is report Appendix E.

Section 0: Setup and Imports

```
In [1]: import os
os.environ.setdefault('KMP_DUPLICATE_LIB_OK', 'TRUE')
import torch
import torch.nn as nn

import glob
import re
import warnings
from datetime import time, date

warnings.filterwarnings('ignore')

import numpy as np
import pandas as pd
import pandas_market_calendars as mcal
import databento as db
import xgboost

from sklearn.linear_model import LogisticRegression
from sklearn.neural_network import MLPClassifier
from sklearn.ensemble import HistGradientBoostingClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.isotonic import IsotonicRegression
from sklearn.metrics import (roc_auc_score, balanced_accuracy_score,
                             brier_score_loss, roc_curve)
from xgboost import XGBClassifier
import shap
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats as sps

# paths
ROOT = os.path.expanduser(r'~\Desktop\Imperial College London\MSc Risk Management &
Financial Engineering\Applied Project\Applied Project Analysis')
LOB_PROC = os.path.join(ROOT, 'Data', 'Processed', 'Databento')
OM_PROC = os.path.join(ROOT, 'Data', 'Processed', 'OptionMetrics')
OFI_DIR = os.path.join(ROOT, 'Data', 'Processed', 'Databento_OFI')
PROC_DIR = os.path.join(ROOT, 'Data', 'Processed')
FIG_DIR = os.path.join(ROOT, 'Figures')
DBN_DIR = os.path.join(ROOT, 'Data', 'Raw', 'Databento')
OM_DIR = os.path.join(ROOT, 'Data', 'Raw', 'OptionMetrics')
CBOE_CSV = os.path.join(ROOT, 'Data', 'Raw', 'Cboe', 'cboe_vix_daily_2024_2025.csv')
```

```

for d in (LOB_PROC, OM_PROC, OFI_DIR, PROC_DIR, FIG_DIR):
    os.makedirs(d, exist_ok=True)

# constants
SPX_SECID = 108105

EX_DIV_DATES = {
    '2024-03-15', '2024-06-21', '2024-09-20', '2024-12-20',
    '2025-03-21', '2025-06-20', '2025-09-19', '2025-12-19',
}

LOB_FEATURES = ['spread', 'imbalance', 'depth', 'queue_imb', 'signed_flow',
'realised_vol']
# production base drops queue_imb (order counts unpopulated in EQUUS.MINI),
# the locked six stay unchanged for the M1 to M8 grid
LOB_PROD_BASE = ['spread', 'imbalance', 'depth', 'signed_flow', 'realised_vol']
DERIV_FEATURES = ['iv_atm_30', 'delta_iv', 'skew', 'term_structure', 'gamma', 'vega']
ALL_FEATURES = LOB_FEATURES + DERIV_FEATURES

TRAIN_START, TRAIN_END = '2024-01-02', '2024-11-29'
VAL_START, VAL_END = '2024-12-02', '2025-04-30'
TEST_START, TEST_END = '2025-05-01', '2025-08-29'
ROB_START, ROB_END = '2025-09-01', '2025-12-31'
SEEDS = (42, 7, 123)
N_BOOT = 1000

# figure style constants, applied to every figure in the notebook
ORDER8 = ['M1', 'M2', 'M3', 'M4', 'M5', 'M6', 'M7', 'M8']
MAIN6 = ['M1', 'M2', 'M3', 'M4', 'M5', 'M6']
NAME = {'M1': 'M1 LR LOB', 'M2': 'M2 XGB LOB', 'M3': 'M3 MLP Ens LOB',
'M4': 'M4 LR LOB+D', 'M5': 'M5 XGB LOB+D', 'M6': 'M6 MLP Ens LOB+D',
'M7': 'M7 HistGB LOB', 'M8': 'M8 HistGB LOB+D'}
CMAP = plt.get_cmap('tab10')
COLOR = {m: CMAP(i) for i, m in enumerate(ORDER8)}
LS = {m: (':' if m in ('M1', 'M2', 'M3', 'M7') else '-') for m in ORDER8}
PRETTY = {'spread': 'Spread', 'imbalance': 'Imbalance', 'depth': 'Depth',
'queue_imb': 'Queue Imbalance', 'signed_flow': 'Signed Flow',
'realised_vol': 'Realised Vol', 'iv_atm_30': 'IV ATM 30D',
'delta_iv': 'Delta IV', 'skew': 'Skew', 'term_structure': 'Term Structure',
'gamma': 'Gamma', 'vega': 'Vega'}

nyse = mcal.get_calendar('NYSE')

assert xgboost.__version__.startswith('2.'), 'pinned xgboost<3 required (see Section 7
notes)'
print(f'databento {db.__version__} xgboost {xgboost.__version__} torch
{torch.__version__}')
print(f'LOB_PROC : {LOB_PROC}')
print(f'FIG_DIR : {FIG_DIR}')

```

```

databento 0.79.0 xgboost 2.1.4 torch 2.12.0+cpu
LOB_PROC : .\Data\Processed\Databento
FIG_DIR : .\Figures

```

Section 1: Data Verification

Quick sanity checks confirming file completeness and OptionMetrics cutoff before running the pipeline.

```
In [2]: dbn_files = sorted(glob.glob(os.path.join(DBN_DIR, '*.dbn.zst')))
print(f'Databento files : {len(dbn_files)}')
print(f'  first : {os.path.basename(dbn_files[0])}')
print(f'  last  : {os.path.basename(dbn_files[-1])}')

print()
for name in ['vsurfd2024', 'vsurfd2025', 'opprcd2024', 'opprcd2025', 'secprd2024',
'secprd2025']:
    df = pd.read_parquet(os.path.join(OM_DIR, f'{name}.parquet'))
    df = df[df['secid'] == SPX_SECID]
    print(f'{name:12s}  {len(df):>8,} rows  {df["date"].min()} -> {df["date"].max()}')

vs25 = pd.read_parquet(os.path.join(OM_DIR, 'vsurfd2025.parquet'))
assert str(vs25[vs25['secid'] == SPX_SECID]['date'].max()) == '2025-08-29', 'cutoff
mismatch'
print('\nvsurfd2025 MAX(date) == 2025-08-29  OK')
```

Databento files : 502

first : equs-mini-20240102.mbp-1.dbn.zst
last : equs-mini-20251231.mbp-1.dbn.zst

vsurfd2024	94,248 rows	2024-01-02 -> 2024-12-31
vsurfd2025	61,710 rows	2025-01-02 -> 2025-08-29
opprcd2024	5,889,228 rows	2024-01-02 -> 2024-12-31
opprcd2025	4,066,473 rows	2025-01-02 -> 2025-08-29
secprd2024	252 rows	2024-01-02 -> 2024-12-31
secprd2025	165 rows	2025-01-02 -> 2025-08-29

vsurfd2025 MAX(date) == 2025-08-29 OK

Section 2: LOB Feature Extraction

MBP-1 is top-of-book (L1) data, the consolidated best bid and offer plus trades. All microstructure features in this project are therefore top-of-book features. Queue-position and multi-level-depth features would require an MBP-10 or full-depth feed and are outside scope.

What this section does

Processes each daily `.dbn.zst` Databento file independently, builds the 10-second feature grid, and writes one parquet file per trading day to `Data/Processed/Databento/`. The loop is resume-safe. Any day already written is skipped instantly.

Why process day by day

The full raw dataset is approximately 130 GB. Loading it all at once would exhaust memory. The processed parquets are tiny by comparison (a few KB each), so the 494-file concatenation later is fast.

Input: `Data/Raw/Databento/equs-mini-YYYYMMDD.mbp-1.dbn.zst` (502 files)

Output: Data/Processed/Databento/lob_YYYY-MM-DD.parquet (494 files, 8 ex-div dates excluded)

Key implementation notes

- SDK v0.79: `to_df()` auto-converts fixed-point prices to float64 in dollars.
- `ts_event` column is already `datetime64[ns, UTC]`. Use `.dt.tz_convert('America/New_York')` only.
- `bid_sz_00` and `ask_sz_00` are `uint32`. Subtract after casting to `int64` to avoid wraparound.
- `bid_ct_00` and `ask_ct_00` are always zero in EQUUS.MINI (synthetic NBBO does not carry per-venue order counts). The `queue_imb` feature is hardcoded to 0.0.
- `side='N'` on trade records means unknown aggressor, approximately 29% of trades. Contribution to signed flow is zero.

Half-day sessions: On NYSE early-close days (13:00 ET) the synthetic EQUUS.MINI BBO continues to carry one-sided quote states after the close, with spreads up to \$433 and mids displaced by up to 30%, so binning uniformly to 16:00 would contaminate six sessions (2024-07-03, 2024-11-29, 2024-12-24, 2025-07-03, 2025-11-28, 2025-12-24, one of which sits in the primary test window). The loop therefore trims each session to its actual market close from the NYSE calendar.

```
In [3]: def compute_lob_features(quotes: pd.DataFrame, trades: pd.DataFrame, date_val) ->
pd.DataFrame:
    """
    Build the 10-second LOB feature grid for one regular-session trading day.

    quotes    : action != 'T' rows; must have 'ts_et' column (ET, tz-aware)
    trades    : action == 'T' rows; same
    date_val  : datetime.date for this day

    Returns DataFrame indexed by ts_et (10 s bin left edge, ET).
    """
    if quotes.empty:
        return pd.DataFrame()

    q = quotes.set_index('ts_et').sort_index()

    # Last BBO snapshot in each 10 s bin, forward-fill any gaps.
    # bid_ct_00/ask_ct_00 excluded: always 0 in EQUUS.MINI (synthetic NBBO
    # does not carry per-venue order counts), queue_imb set to 0.0 below.
    agg = (
        q.resample('10s', label='left', closed='left')[
            ['bid_px_00', 'ask_px_00', 'bid_sz_00', 'ask_sz_00']
        ]
        .last()
        .ffill()
        .dropna(subset=['bid_px_00', 'ask_px_00'])
    )

    # Restrict to regular session 09:30 to 16:00 ET, .copy() prevents SettingWithCopyWarning
    agg = agg[(agg.index.time >= time(9, 30)) & (agg.index.time < time(16, 0))].copy()
    if agg.empty:
        return pd.DataFrame()

    bid, ask = agg['bid_px_00'], agg['ask_px_00']
```

```

agg['mid']          = (bid + ask) / 2
agg['half_spread'] = (ask - bid) / 2
agg['spread']       = ask - bid

# Cast to int64 before subtraction: bid_sz_00/ask_sz_00 are uint32, without
# the cast, (bid - ask) wraps around when bid < ask, producing values ~4e9
# instead of a negative number, and imbalance ends up in the millions.
bid_sz  = agg['bid_sz_00'].astype(np.int64)
ask_sz  = agg['ask_sz_00'].astype(np.int64)
total_sz = bid_sz + ask_sz
agg['imbalance'] = np.where(total_sz > 0, (bid_sz - ask_sz) / total_sz, 0.0)
agg['depth']     = total_sz.astype(float)

# queue_imb: order-count fields are not populated in EQU5.MINI,
# set to 0.0. Models will learn zero importance for this feature.
agg['queue_imb'] = 0.0

# Realised vol: rolling 30-tick std on raw mid-return stream, sampled to 10 s grid.
# fill_method=None avoids pandas FutureWarning; no leading NAs to fill anyway.
q_mid  = ((q['bid_px_00'] + q['ask_px_00']) / 2).sort_index()
rv_tick = q_mid.pct_change(fill_method=None).rolling(30, min_periods=10).std()
rv_bins = rv_tick.resample('10s', label='left', closed='left').last().ffill()
agg['realised_vol'] = rv_bins.reindex(agg.index)

# Signed order flow: B=+1 (buyer aggressor), A=-1 (seller), N=0 (unknown)
if not trades.empty:
    t = trades.set_index('ts_et').sort_index()
    sign_map = {'B': 1.0, 'A': -1.0, 'N': 0.0}
    t = t.assign(signed_size=t['side'].map(sign_map).fillna(0.0) * t['size'])
    flow = t['signed_size'].resample('10s', label='left', closed='left').sum()
    agg['signed_flow'] = flow.reindex(agg.index).fillna(0.0)
else:
    agg['signed_flow'] = 0.0

agg['date'] = date_val
keep = ['date', 'mid', 'half_spread', 'spread',
        'imbalance', 'depth', 'queue_imb', 'signed_flow', 'realised_vol']
return agg[keep].copy()

```

```

In [4]: sched_full  = nyse.schedule(start_date='2024-01-02', end_date='2025-12-31')
trading_days = set(sched_full.index.date)
# Trim each session to its actual market close, early-close days end at 13:00 ET.
# The synthetic EQU5.MINI BBO can carry abnormal one-sided states after 13:00 early closes
# (spreads up to $433) binning to 16:00 contaminated six days.
close_map = {d.date(): c.tz_convert('America/New_York').time()
              for d, c in sched_full['market_close'].items()}

processed = skipped = already_done = 0

for f in dbn_files:
    # Parse date from filename: equ5-mini-YYYYMMDD.mbp-1.dbn.zst
    date_str = os.path.basename(f).split('.')[0].split('-')[-1] # 'YYYYMMDD'
    d = date(int(date_str[:4]), int(date_str[4:6]), int(date_str[6:]))

    out_path = os.path.join(LOB_PROC, f'lob_{d}.parquet')
    if os.path.exists(out_path):
        already_done += 1
        continue

```

```

if d not in trading_days or str(d) in EX_DIV_DATES:
    skipped += 1
    continue

df = db.DBNStore.from_file(f).to_df()

# ts_event is already datetime64[ns, UTC], just convert timezone
df['ts_et'] = df['ts_event'].dt.tz_convert('America/New_York')

t_col = df['ts_et'].dt.time
df = df[(t_col >= time(9, 30)) & (t_col < close_map.get(d, time(16, 0)))]

quotes = df[df['action'] != 'T'].copy()
trades = df[df['action'] == 'T'].copy()

feats = compute_lob_features(quotes, trades, d)
if feats.empty:
    skipped += 1
    continue

feats.to_parquet(out_path)
processed += 1
if (processed + already_done) % 50 == 0:
    print(f' {processed + already_done} days processed...')

print(f'LOB loop complete, processed: {processed}, already on disk: {already_done}, skipped: {skipped}')

# Spot-check one output
sample = pd.read_parquet(os.path.join(LOB_PROC, 'lob_2024-01-02.parquet'))
print(f'\nlob_2024-01-02.parquet shape={sample.shape}')
print(sample.head(3))

```

LOB loop complete, processed: 0, already on disk: 494, skipped: 8

lob_2024-01-02.parquet shape=(2340, 9)

	date	mid	half_spread	spread \
ts_et				
2024-01-02 09:30:00-05:00	2024-01-02	472.215	0.005	0.01
2024-01-02 09:30:10-05:00	2024-01-02	472.495	0.005	0.01
2024-01-02 09:30:20-05:00	2024-01-02	472.555	0.005	0.01

	imbalance	depth	queue_imb	signed_flow \
ts_et				
2024-01-02 09:30:00-05:00	-0.990050	201.0	0.0	-301.0
2024-01-02 09:30:10-05:00	-0.669421	605.0	0.0	1905.0
2024-01-02 09:30:20-05:00	-0.333333	600.0	0.0	-9.0

	realised_vol
ts_et	
2024-01-02 09:30:00-05:00	0.000004
2024-01-02 09:30:10-05:00	0.000002
2024-01-02 09:30:20-05:00	0.000002

Section 3: OptionMetrics Feature Extraction

What this section does

Loads the raw OptionMetrics IvyDB parquets, applies quality filters, constructs the four derivatives features from `vsurfd` (IV surface), computes daily mean near-ATM Greeks from `opprcd`, and saves a single combined daily feature table to `Data/Processed/OptionMetrics/deriv_features.parquet`.

Input: `Data/Raw/OptionMetrics/vsurfd2024.parquet`, `vsurfd2025.parquet`, `opprcd2024.parquet`, `opprcd2025.parquet`

Output: `Data/Processed/OptionMetrics/deriv_features.parquet` (416 rows, 6 features, one row per trading day from 2024-01-03 to 2025-08-29)

Key design decisions

- OptionMetrics IV is end-of-day. It is broadcast to all 10-second intervals of the following trading session. This deliberately tests whether prior-day options-market regime information improves intraday quoting. Intraday IV is not available.
- `vsurfd` stores delta as an integer in the range minus 90 to plus 90 (e.g. 50 means delta 0.50), not to be confused with `opprcd` which stores delta as a decimal in the range minus 1 to plus 1.
- Dispersion quality filter. Top 5% of the `dispersion` column excluded (rows with unreliable IV estimates), with the cutoff estimated on the training window alone and then applied to every period.
- `delta_iv` is the first difference of `iv_atm_30`. The first calendar date loses one observation, leaving 416 rows.

```
In [5]: # vsurfd, IV surface features
vsurfd = pd.concat([
    pd.read_parquet(os.path.join(OM_DIR, 'vsurfd2024.parquet')),
    pd.read_parquet(os.path.join(OM_DIR, 'vsurfd2025.parquet')),
], ignore_index=True)
vsurfd = vsurfd[vsurfd['secid'] == SPX_SECID].copy()
vsurfd['date'] = pd.to_datetime(vsurfd['date']).dt.date

# quality filter: the cutoff is estimated on the TRAINING window only and then
# applied to every period, so no validation or test observation influences it
disp_95 = vsurfd[vsurfd['date'] <= pd.Timestamp(TRAIN_END).date()]
['dispersion'].quantile(0.95)
vsurfd = vsurfd[vsurfd['dispersion'] <= disp_95].copy()
print(f'vsurfd after quality filter: {len(vsurfd):,} rows (train-window dispersion 95th pct = {disp_95:.4f})')

# iv_atm_30: delta=50 (INTEGER), days=30, calls
iv_atm30 = (
    vsurfd[(vsurfd['days'] == 30) & (vsurfd['delta'] == 50) & (vsurfd['cp_flag'] == 'C')]
    .set_index('date')[['impl_volatility']]
    .rename(columns={'impl_volatility': 'iv_atm_30'})
)

# skew = IV(put d=-25, 30d) - IV(call d=25, 30d)
iv_put25 = vsurfd[(vsurfd['days'] == 30) & (vsurfd['delta'] == -25)].set_index('date')
```



```

['impl_volatility'].rename('iv_put25')
iv_call25 = vsurfd[(vsurfd['days'] == 30) & (vsurfd['delta'] == 25)].set_index('date')
['impl_volatility'].rename('iv_call25')

# term_structure = IV(ATM 30d) - IV(ATM 91d)
iv_atm91 = (
    vsurfd[(vsurfd['days'] == 91) & (vsurfd['delta'] == 50) & (vsurfd['cp_flag'] == 'C')]
    .set_index('date')['impl_volatility'].rename('iv_atm_91')
)

deriv_vs = iv_atm30.join(iv_put25, how='left').join(iv_call25, how='left').join(iv_atm91,
how='left')
deriv_vs['skew'] = deriv_vs['iv_put25'] - deriv_vs['iv_call25']
deriv_vs['term_structure'] = deriv_vs['iv_atm_30'] - deriv_vs['iv_atm_91']
deriv_vs['delta_iv'] = deriv_vs['iv_atm_30'].diff() # day-over-day change

deriv_vs = deriv_vs[['iv_atm_30', 'delta_iv', 'skew', 'term_structure']].dropna()
print(f'vsurfd features: {deriv_vs.shape} ({deriv_vs.index.min()} ->
{deriv_vs.index.max()})')
print(deriv_vs.head(3))

```

vsurfd after quality filter: 149,354 rows (train-window dispersion 95th pct = 0.0524)

vsurfd features: (416, 4) (2024-01-03 -> 2025-08-29)

	iv_atm_30	delta_iv	skew	term_structure
date				
2024-01-03	0.122433	0.004383	0.024252	-0.010142
2024-01-04	0.123530	0.001097	0.025173	-0.009358
2024-01-05	0.117013	-0.006517	0.024110	-0.011428

```

In [6]: # opprcd, near-ATM Greeks (delta is DECIMAL [-1, +1], unlike vsurfd)
opprcd = pd.concat([
    pd.read_parquet(os.path.join(OM_DIR, 'opprcd2024.parquet')),
    pd.read_parquet(os.path.join(OM_DIR, 'opprcd2025.parquet')),
], ignore_index=True)
opprcd = opprcd[opprcd['secid'] == SPX_SECID].copy()
opprcd['date'] = pd.to_datetime(opprcd['date']).dt.date

near_atm = opprcd[opprcd['delta'].abs().between(0.40, 0.60)]
print(f'Near-ATM opprcd rows: {len(near_atm):,}')

greeks = near_atm.groupby('date')[['gamma', 'vega']].mean()
print(f'Greeks daily rows: {len(greeks):,}')
print(greeks.describe().round(4))

# combine and save to Processed/OptionMetrics/
deriv_features = deriv_vs.join(greeks, how='inner')
print(f'\nCombined deriv_features: {deriv_features.shape} ({deriv_features.index.min()} ->
{deriv_features.index.max()})')
print(deriv_features.head(3))

out_path = os.path.join(OM_PROC, 'deriv_features.parquet')
deriv_features.to_parquet(out_path)
print(f'\nSaved: {out_path}')

```

Near-ATM opprcd rows: 898,581

Greeks daily rows: 417

	gamma	vega
count	417.0000	417.0000
mean	0.0015	1201.1610

std	0.0003	119.7010
min	0.0006	915.3374
25%	0.0013	1107.2656
50%	0.0014	1206.1915
75%	0.0017	1284.9585
max	0.0021	1486.4449

Combined deriv_features: (416, 6) (2024-01-03 -> 2025-08-29)

	iv_atm_30	delta_iv	skew	term_structure	gamma	\
date						
2024-01-03	0.122433	0.004383	0.024252	-0.010142	0.001906	
2024-01-04	0.123530	0.001097	0.025173	-0.009358	0.001884	
2024-01-05	0.117013	-0.006517	0.024110	-0.011428	0.001955	

	vega
date	
2024-01-03	1010.894941
2024-01-04	1002.020083
2024-01-05	1009.973320

Saved: .\Data\Processed\OptionMetrics\deriv_features.parquet

Section 4: Combined Feature Grid

What this section does

Loads all 494 per-day LOB parquets, concatenates them, and merges the derivatives features using a NYSE calendar lookup. The result is a single in-memory DataFrame with all LOB and derivatives columns aligned on the 10-second grid.

Alignment rule

OptionMetrics end-of-day data for trading date D is joined to all LOB rows on trading date D+1 (the next NYSE business day). A naive `date + 1 calendar day` would mis-align 93 sessions, namely Mondays and post-holiday Tuesdays where the gap is 3 or 4 calendar days. The `om_lookup` dictionary maps each LOB date to its NYSE predecessor using the full exchange calendar.

Coverage

Derivatives coverage runs to 2025-08-29, and 952,740 of the 1,149,480 LOB rows (82.9%) carry all six derivatives features. LOB rows from 2025-09-01 onward have no derivatives counterpart and are kept with NaN derivatives columns for the robustness window.

Output: in-memory DataFrame `lob` with columns from both `LOB_FEATURES` and `DERIV_FEATURES`

```
In [7]: lob_parquets = sorted(glob.glob(os.path.join(LOB_PROC, 'lob_*.parquet')))
print(f'LOB parquet files: {len(lob_parquets)}')

lob = pd.concat([pd.read_parquet(f) for f in lob_parquets]).sort_index()
print(f'Total LOB rows: {len(lob):,} ({lob["date"].min()} -> {lob["date"].max()})')

# Load saved derivatives features
deriv_features = pd.read_parquet(os.path.join(OM_PROC, 'deriv_features.parquet'))
```

```

deriv_features.index = deriv_features.index.astype(str)
print(f'Derivatives features loaded: {deriv_features.shape}')

# Calendar-based OM -> next-trading-day mapping
# Lob_date T uses OM data from T-1 (previous NYSE trading day)
# 93 cases where gap > 1 calendar day (weekends, holiday weekends), naive date+1 fails these
sched_combined = nyse.schedule(start_date='2024-01-02', end_date='2025-08-29')
td_combined     = [d.date() for d in sched_combined.index]
om_lookup       = {str(td_combined[i]): str(td_combined[i - 1]) for i in range(1,
len(td_combined))}

lob['date_str'] = lob['date'].astype(str)
lob['om_date']  = lob['date_str'].map(om_lookup)
lob = lob.join(deriv_features.rename_axis('om_date'), on='om_date', how='left')
lob = lob.drop(columns=['date_str', 'om_date'])

n_full = lob[DERIV_FEATURES].notna().all(axis=1).sum()
print(f'Rows with full derivatives data: {n_full:,} / {len(lob):,}
({100*n_full/len(lob):.1f}%)')
print(lob[ALL_FEATURES].describe().round(4))

```

LOB parquet files: 494

Total LOB rows: 1,149,480 (2024-01-02 -> 2025-12-31)

Derivatives features loaded: (416, 6)

Rows with full derivatives data: 952,740 / 1,149,480 (82.9%)

	spread	imbalance	depth	queue_imb	signed_flow \
count	1.149480e+06	1.149480e+06	1.149480e+06	1149480.0	1.149480e+06
mean	2.740000e-02	-1.240000e-02	9.180513e+02	0.0	-1.034740e+01
std	2.640000e-02	4.943000e-01	8.005163e+02	0.0	1.316296e+03
min	-1.000000e-02	-1.000000e+00	2.000000e+00	0.0	-1.679430e+05
25%	2.000000e-02	-4.013000e-01	4.400000e+02	0.0	-1.100000e+02
50%	2.000000e-02	0.000000e+00	7.240000e+02	0.0	0.000000e+00
75%	3.000000e-02	3.714000e-01	1.150000e+03	0.0	1.090000e+02
max	7.390000e+00	9.996000e-01	7.843000e+04	0.0	1.403190e+05

	realised_vol	iv_atm_30	delta_iv	skew	term_structure \
count	1.149480e+06	952740.0000	952740.0000	952740.0000	952740.0000
mean	0.000000e+00	0.1403	0.0001	0.0426	-0.0054
std	0.000000e+00	0.0418	0.0170	0.0219	0.0147
min	0.000000e+00	0.0938	-0.1563	0.0098	-0.0272
25%	0.000000e+00	0.1138	-0.0062	0.0262	-0.0136
50%	0.000000e+00	0.1286	-0.0003	0.0411	-0.0100
75%	0.000000e+00	0.1525	0.0054	0.0536	-0.0003
max	3.700000e-03	0.4137	0.1047	0.1699	0.0991

	gamma	vega
count	952740.0000	952740.0000
mean	0.0015	1200.5216
std	0.0003	119.5165
min	0.0006	915.3374
25%	0.0013	1105.4424
50%	0.0014	1206.1915
75%	0.0017	1283.7745
max	0.0021	1486.4449

Section 5: Target Variable

What this section does

Constructs the binary adverse-selection indicator `y_t` for each 10-second observation and appends it to the feature grid.

Definition

$$y_t = \mathbf{1}[(mid_{t+10s} - mid_t) \times q_t \geq \frac{1}{2} half_spread_t]$$

where $q_t = \text{sign}(\text{signed_flow}_t)$.

- $q_t = +1$. Net buying pressure in the bin. The market maker sold at the ask. Adverse if price rises by at least half the spread.
- $q_t = -1$. Net selling pressure. The market maker bought at the bid. Adverse if price falls by at least half the spread.
- $q_t = 0$. No directional flow. No clear exposure, so $y_t = 0$ by convention.

Decision timing

Each row summarises information observed over the preceding 10-second bin. `mid` is the last BBO snapshot in the bin, `signed_flow` aggregates the bin's trades, and every other feature is computed from data inside the bin. The decision is made at the bin close, and the target measures the mid change from the close of the feature bin to the close of the next bin. Features therefore strictly precede the markout window and no feature overlaps the interval being predicted.

Threshold interpretation

With the multiplier at 0.5, the event is triggered by an adverse move of half the half-spread, so `y` is an *early adverse-selection indicator* (partial spread erosion), not a full break-even loss, which would require a move of at least one half-spread. Robustness of the information-set conclusion to the break-even (1.0) and severe (2.0) multipliers is reported in Section 15.5.

Feature availability at decision time

Feature Group	Frequency	Source	Known at Decision Time
Spread, Imbalance, Depth	10s Bin Close	Databento MBP-1	Yes, Last Snapshot of the Completed Bin
Signed Flow	Completed 10s Bin	Databento Trades	Yes, at Bin Close
OFI, Gross Volume, Quote Intensity	Completed 10s Bin	Databento MBP-1	Yes, at Bin Close
Running Realised Vol (rv_run)	Session So Far	Databento MBP-1	Yes, Causal Cumulative

Feature Group	Frequency	Source	Known at Decision Time
IV, Skew, Term Structure, Greeks	Daily, Lagged One Day	OptionMetrics	Yes, Previous EOD
VIX Open	Daily, Same Morning	Cboe	Yes, After 09:30 Print (Robustness Only)

Leakage prevention

`mid_next` (the mid at the close of the following 10-second bin, since each row's mid is the last snapshot inside its own bin) is computed with `groupby('date').shift(-1)`. This restricts the shift to within each trading day and prevents the last bin of day D from looking forward into day D+1.

Class balance

Approximately 37% of observations are $y_t = 1$. This is a mild imbalance. `Scale_pos_weight` for XGBoost is approximately 1.58.

Parameter

The threshold multiplier `tau = 0.5` is the primary setting. Sensitivity to tau is tested in Appendix A.

```
In [8]: def build_target(df: pd.DataFrame, tau: float = 0.5) -> pd.DataFrame:
    out = df.copy()
    out['mid_next'] = out.groupby('date')['mid'].shift(-1)
    out['mid_change'] = out['mid_next'] - out['mid']
    out['quote_dir'] = np.sign(out['signed_flow'])
    out['y'] = np.where(
        out['quote_dir'] != 0,
        (out['mid_change'] * out['quote_dir'] >= tau * out['half_spread']).astype(int),
        0,
    )
    return out.dropna(subset=['mid_next'])

lob_target = build_target(lob)
print(f'Feature grid: {len(lob_target):,} rows ({lob_target["date"].min()} ->
{lob_target["date"].max()})')
print(f'y=1 (adverse selection) rate: {lob_target["y"].mean():.3f}')
print(lob_target['y'].value_counts())

# Appendix B, descriptive statistics
print('\nAppendix B, Feature descriptive statistics')
print(lob_target[ALL_FEATURES].describe().round(4))
```

Feature grid: 1,148,986 rows (2024-01-02 -> 2025-12-31)

y=1 (adverse selection) rate: 0.369

y

0 724838

1 424148

Name: count, dtype: int64

Appendix B, Feature descriptive statistics

	spread	imbalance	depth	queue_imb	signed_flow \
count	1.148986e+06	1.148986e+06	1.148986e+06	1148986.0	1.148986e+06

mean	2.740000e-02	-1.240000e-02	9.179544e+02	0.0	-1.017870e+01
std	2.560000e-02	4.942000e-01	8.003929e+02	0.0	1.311311e+03
min	-1.000000e-02	-1.000000e+00	2.000000e+00	0.0	-1.679430e+05
25%	2.000000e-02	-4.013000e-01	4.400000e+02	0.0	-1.100000e+02
50%	2.000000e-02	0.000000e+00	7.230000e+02	0.0	0.000000e+00
75%	3.000000e-02	3.714000e-01	1.150000e+03	0.0	1.090000e+02
max	7.390000e+00	9.996000e-01	7.843000e+04	0.0	1.403190e+05

	realised_vol	iv_atm_30	delta_iv	skew	term_structure \
count	1.148986e+06	952331.0000	952331.0000	952331.0000	952331.0000
mean	0.000000e+00	0.1403	0.0001	0.0426	-0.0054
std	0.000000e+00	0.0418	0.0170	0.0219	0.0147
min	0.000000e+00	0.0938	-0.1563	0.0098	-0.0272
25%	0.000000e+00	0.1138	-0.0062	0.0262	-0.0136
50%	0.000000e+00	0.1286	-0.0003	0.0411	-0.0100
75%	0.000000e+00	0.1525	0.0054	0.0536	-0.0003
max	3.700000e-03	0.4137	0.1047	0.1699	0.0991

	gamma	vega
count	952331.0000	952331.0000
mean	0.0015	1200.5213
std	0.0003	119.5165
min	0.0006	915.3374
25%	0.0013	1105.4424
50%	0.0014	1206.1915
75%	0.0017	1283.7745
max	0.0021	1486.4449

Section 6: Train, Validation and Test Splits with Feature Scaling

What this section does

Applies the locked chronological train-validation-test split, fits `StandardScaler` on the training window only, and creates scaled arrays for all three splits. Two separate scalers are maintained. One for LOB-only features (M1, M2, M3) and one for the full feature set (M4, M5, M6).

Splits

Split	Dates	Rows	Purpose
Train	2024-01-02 to 2024-11-29	531,132	Model fitting
Validation	2024-12-02 to 2025-04-30	232,820	Hyperparameter tuning
Primary test	2025-05-01 to 2025-08-29	193,057	Main results, the volatile period after the April 2025 tariff shock
Robustness	2025-09-01 to 2025-12-31	191,977	OOS stability, LOB only (no derivatives available)

Note: Report Table 8 rescales `realised_vol` by 10^5 , so the raw-unit `describe` below rounds that column to 0.0000.

Why a single chronological split

The test set contains 193,057 observations over a genuine stress period (tariff shock, elevated volatility). That is sufficient for stable metric estimation. K-fold cross-validation would add complexity without meaningfully improving reliability within the 3,000-word constraint.

Scaling discipline

`StandardScaler` is fit exclusively on the training window and applied to validation and test without refitting. Fitting on the full dataset before splitting would constitute look-ahead bias. After hyperparameter selection, the final models are re-trained on train plus validation combined using a fresh scaler fit on that larger window.

LOB+Deriv row count

The LOB+Deriv training set holds 526,454 rows against 531,132 for LOB-only. A complete derivatives vector needs both the previous-trading-day mapping and the first-difference IV feature, so the first two LOB sessions of the window are not fully populated, which is the 4,678-row gap (two 2,339-row sessions).

Scaler consistency for final evaluation: The final models are re-trained on train plus validation with scalers fit on that combined window, so every matrix they are evaluated on must be transformed with those same scalers. Evaluating a final model on matrices scaled with the train-only scaler systematically distorts every test metric (the LOB-only XGBoost pooled AUC moves from 0.529 to 0.636 once the scalers match). The cell below therefore also defines `_f` evaluation matrices built with the train+val scalers, and all final-model evaluation downstream uses them.

```
In [9]: def date_slice(df, start, end):
        mask = (df['date'].astype(str) >= start) & (df['date'].astype(str) <= end)
        return df[mask].copy()

train = date_slice(lob_target, TRAIN_START, TRAIN_END)
val    = date_slice(lob_target, VAL_START, VAL_END)
test   = date_slice(lob_target, TEST_START, TEST_END)

for name, s in [('Train', train), ('Val', val), ('Test', test)]:
    print(f'{name:6s} {len(s):>8,} rows {s["date"].min()} -> {s["date"].max()} y=1={s["y"].mean():.3f}')

trainval = pd.concat([train, val], ignore_index=True)
print(f'Train+Val: {len(trainval):,} rows')

# LOB-only scaling
scaler_lob = StandardScaler()
X_tr_lob = scaler_lob.fit_transform(train[LOB_FEATURES])
X_va_lob = scaler_lob.transform(val[LOB_FEATURES])
X_te_lob = scaler_lob.transform(test[LOB_FEATURES])
y_train, y_val, y_test = train['y'].values, val['y'].values, test['y'].values

scaler_lob_tv = StandardScaler()
X_tv_lob = scaler_lob_tv.fit_transform(trainval[LOB_FEATURES])
y_tv     = trainval['y'].values

# LOB+Deriv scaling (rows with full derivatives only)
```

```

train_d    = train.dropna(subset=ALL_FEATURES)
val_d      = val.dropna(subset=ALL_FEATURES)
test_d     = test.dropna(subset=ALL_FEATURES)
trainval_d = pd.concat([train_d, val_d], ignore_index=True)

scaler_full = StandardScaler()
X_tr_full = scaler_full.fit_transform(train_d[ALL_FEATURES])
X_va_full = scaler_full.transform(val_d[ALL_FEATURES])
X_te_full = scaler_full.transform(test_d[ALL_FEATURES])
y_train_d, y_val_d, y_test_d = train_d['y'].values, val_d['y'].values, test_d['y'].values

scaler_full_tv = StandardScaler()
X_tv_full = scaler_full_tv.fit_transform(trainval_d[ALL_FEATURES])
y_tv_d     = trainval_d['y'].values

print(f'\nLOB-only   train:{len(X_tr_lob):>7,}   val:{len(X_va_lob):>7,}   test:
{len(X_te_lob):>7,}')
print(f'\nLOB+Deriv  train:{len(X_tr_full):>7,}   val:{len(X_va_full):>7,}   test:
{len(X_te_full):>7,}')

# final-model evaluation matrices, scaled with the same train+val scalers
# the final models were trained on
# Final models are trained on train+val-scaled features, so every matrix
# they are evaluated on must use the SAME train+val scalers.
# Evaluating them on train-only-scaled matrices was the bug that invalidated
# all results before the corrected run (LOB-only XGB: 0.529 -> 0.636).
X_va_lob_f = scaler_lob_tv.transform(val[LOB_FEATURES])
X_te_lob_f = scaler_lob_tv.transform(test[LOB_FEATURES])
X_va_full_f = scaler_full_tv.transform(val_d[ALL_FEATURES])
X_te_full_f = scaler_full_tv.transform(test_d[ALL_FEATURES])

# Appendix B of the report, descriptive statistics on the primary test window
print('\nAppendix B, feature descriptive statistics, primary test window')
print(test_d[ALL_FEATURES].describe().round(4).T)

```

```

Train    531,132 rows  2024-01-02 -> 2024-11-29  y=1=0.389
Val      232,820 rows  2024-12-02 -> 2025-04-30  y=1=0.352
Test     193,057 rows  2025-05-01 -> 2025-08-29  y=1=0.388
Train+Val: 763,952 rows

```

```

LOB-only   train:531,132   val:232,820   test:193,057
LOB+Deriv  train:526,454   val:232,820   test:193,057

```

Appendix B, feature descriptive statistics, primary test window

	count	mean	std	min	25% \
spread	193057.0	0.0235	0.0370	0.0000	0.0100
imbalance	193057.0	-0.0077	0.4626	-0.9993	-0.3665
depth	193057.0	930.8536	610.6123	5.0000	528.0000
queue_imb	193057.0	0.0000	0.0000	0.0000	0.0000
signed_flow	193057.0	-11.4458	1278.4715	-167943.0000	-196.0000
realised_vol	193057.0	0.0000	0.0000	0.0000	0.0000
iv_atm_30	193057.0	0.1418	0.0229	0.1060	0.1264
delta_iv	193057.0	-0.0010	0.0109	-0.0362	-0.0057
skew	193057.0	0.0521	0.0114	0.0355	0.0431
term_structure	193057.0	-0.0127	0.0052	-0.0221	-0.0167
gamma	193057.0	0.0012	0.0001	0.0010	0.0012
vega	193057.0	1338.4896	81.6953	1150.0924	1281.1348

	50%	75%	max
spread	0.0200	0.0300	7.3900
imbalance	0.0000	0.3426	0.9990
depth	790.0000	1216.0000	22536.0000
queue_imb	0.0000	0.0000	0.0000
signed_flow	0.0000	200.0000	35191.0000
realised_vol	0.0000	0.0000	0.0032
iv_atm_30	0.1386	0.1487	0.2083
delta_iv	-0.0009	0.0034	0.0267
skew	0.0486	0.0598	0.0821
term_structure	-0.0134	-0.0103	0.0019
gamma	0.0013	0.0013	0.0015
vega	1361.9779	1401.8023	1486.4449

Section 7: Model Training (M1 to M8)

What this section does

Tunes and fits all eight model configurations. The primary 2x3 grid plus the HistGradientBoosting robustness family.

	Logistic Regression	XGBoost	MLP Ensemble	HistGB
LOB Only	M1	M2	M3	M7
LOB + Derivatives	M4	M5	M6	M8

Model specifications

M1/M4 Logistic Regression. sklearn `LogisticRegression`, L2 penalty, `class_weight='balanced'`, `C` tuned over {0.01, 0.1, 1.0, 10.0}. The interpretable linear baseline.

M2/M5 XGBoost. `XGBClassifier` with `scale_pos_weight` set to the class ratio, grid over `max_depth` in {3, 5} and `learning_rate` in {0.05, 0.1}, early stopping on validation AUC with patience 30. All XGBoost results come from the pinned environment xgboost 2.1.4 (`pip install "xgboost<3"`). 3.x produces different ensembles from identical seeds and its `base_score` serialisation breaks SHAP 0.49.

M3/M6 MLP seed ensembles. sklearn `MLPClassifier`, architecture (128, 64), ReLU, `alpha` tuned over {1e-4, 1e-3, 1e-2}, early stopping. The final M3 and M6 are averages of three networks (seeds 42, 7, 123) at the tuned alpha. Seed ensembling is not optional decoration. Section 15 shows the single-seed information-set gain varying by roughly a factor of two across seeds (within-day -0.0031, -0.0061 and -0.0033), so any single-seed comparison is unreliable. A torch-based sequence model is evaluated separately in Section 21.

Class imbalance: Logistic regression and XGBoost use class-weighted loss, `class_weight='balanced'` and `scale_pos_weight` respectively. `scale_pos_weight` is computed on the training window and held fixed at the train+val refit so the tuned configuration is unchanged (the train+val ratio differs by under 2%, and the choice involves no leakage). The MLP has no class-weighted loss, while the TCN of Section 21 uses `BCEWithLogitsLoss` with `pos_weight` set to the training-window class ratio. The MLP is one of the three pre-committed families and is handled through the

imbalance-insensitive metrics (AUC, balanced accuracy) and the same validation-selected thresholding and calibration protocol as the tree models, while the TCN is evaluated separately as a sequence-model robustness check outside that protocol.

M7/M8 HistGradientBoosting. sklearn's LightGBM-style histogram booster, tuned over learning rate {0.05, 0.1} and leaf count {31, 63}. An additional boosting family confirming that family-specific quirks do not drive the information-set conclusion.

Two-stage protocol (the scaler discipline)

Stage one tunes on the training window with train-only scaling and selects hyperparameters on validation. Stage two refits the chosen configuration on train plus validation with the train+val scalers, and every test or robustness matrix those final models score is transformed with those same train+val scalers. Calibration and threshold selection instead use the stage-three training-only models described below. Evaluating final models on matrices scaled differently from their training matrices was the most consequential bug found in this project (Section 16). The `_f` matrices from Section 6 enforce consistency.

Two tiers of fits: Stage two produces the final predictive models, refit on train plus validation, which are the source of every headline AUC in Sections 8 to 12. Stage three produces an economic tier with the same tuned hyperparameters but fitted on the training window only. Its validation predictions are genuinely out of sample, so every quantity later selected on validation, the balanced-accuracy threshold in Section 8 and the isotonic calibration maps and decision thresholds in Section 13, is chosen without a model scoring data it trained on.

```
In [10]: # stage one, hyperparameter tuning on train-only scaling
def tune_lr(X_tr, ytr, X_v, yv):
    best = (0.0, 1.0)
    for C in (0.01, 0.1, 1.0, 10.0):
        m = LogisticRegression(C=C, class_weight='balanced', max_iter=1000, random_state=42)
        m.fit(X_tr, ytr)
        a = roc_auc_score(yv, m.predict_proba(X_v)[:, 1])
        if a > best[0]:
            best = (a, C)
    return best[1], best[0]

def tune_xgb(X_tr, ytr, X_v, yv, spw, label):
    best = (0.0, None, 400)
    for d in (3, 5):
        for lr in (0.05, 0.1):
            m = XGBClassifier(max_depth=d, learning_rate=lr, subsample=0.8,
                              n_estimators=400, scale_pos_weight=spw, eval_metric='auc',
                              early_stopping_rounds=30, random_state=42, verbosity=0)
            m.fit(X_tr, ytr, eval_set=[(X_v, yv)], verbose=False)
            a = roc_auc_score(yv, m.predict_proba(X_v)[:, 1])
            print(f' [{label}] depth={d} lr={lr}: val AUC={a:.4f} iter={m.best_iteration}',
                  flush=True)
            if a > best[0]:
                best = (a, {'max_depth': d, 'learning_rate': lr}, max(m.best_iteration + 1,
1))
    return best
```

```

def tune_mlp(X_tr, ytr, X_v, yv, label):
    best = (0.0, 1e-4)
    for alpha in (1e-4, 1e-3, 1e-2):
        m = MLPClassifier(hidden_layer_sizes=(128, 64), alpha=alpha, activation='relu',
                           max_iter=200, early_stopping=True, validation_fraction=0.1,
                           n_iter_no_change=7, random_state=42)
        m.fit(X_tr, ytr)
        a = roc_auc_score(yv, m.predict_proba(X_v)[: , 1])
        print(f' [{label}] alpha={alpha}: val AUC={a:.4f}', flush=True)
        if a > best[0]:
            best = (a, alpha)
    return best[1], best[0]

def tune_hgb(X_tr, ytr, X_v, yv, label):
    best = (0.0, None)
    for lr in (0.05, 0.1):
        for ln in (31, 63):
            m = HistGradientBoostingClassifier(learning_rate=lr, max_leaf_nodes=ln,
                                                max_iter=300, random_state=42,
                                                early_stopping=True, validation_fraction=0.1,
                                                n_iter_no_change=20)
            m.fit(X_tr, ytr)
            a = roc_auc_score(yv, m.predict_proba(X_v)[: , 1])
            print(f' [{label}] lr={lr} leaves={ln}: val AUC={a:.4f}', flush=True)
            if a > best[0]:
                best = (a, {'learning_rate': lr, 'max_leaf_nodes': ln})
    return best[1]

sp_lob = float((y_train == 0).sum()) / max((y_train == 1).sum(), 1)
sp_full = float((y_train_d == 0).sum()) / max((y_train_d == 1).sum(), 1)

best_C1, _ = tune_lr(X_tr_lob, y_train, X_va_lob, y_val)
best_C4, _ = tune_lr(X_tr_full, y_train_d, X_va_full, y_val_d)
print(f'LR selected: M1 C={best_C1} M4 C={best_C4}', flush=True)

_, params2, ep2 = tune_xgb(X_tr_lob, y_train, X_va_lob, y_val, sp_lob, 'M2')
_, params5, ep5 = tune_xgb(X_tr_full, y_train_d, X_va_full, y_val_d, sp_full, 'M5')
print(f'XGB selected: M2 {params2} n={ep2} M5 {params5} n={ep5}', flush=True)

alpha3, _ = tune_mlp(X_tr_lob, y_train, X_va_lob, y_val, 'M3')
alpha6, _ = tune_mlp(X_tr_full, y_train_d, X_va_full, y_val_d, 'M6')
print(f'MLP selected: alpha3={alpha3} alpha6={alpha6}', flush=True)

params7 = tune_hgb(X_tr_lob, y_train, X_va_lob, y_val, 'M7')
params8 = tune_hgb(X_tr_full, y_train_d, X_va_full, y_val_d, 'M8')
print(f'HistGB selected: M7 {params7} M8 {params8}', flush=True)

```

```

LR selected: M1 C=10.0 M4 C=0.01
[M2] depth=3 lr=0.05: val AUC=0.6723 iter=142
[M2] depth=3 lr=0.1: val AUC=0.6724 iter=73
[M2] depth=5 lr=0.05: val AUC=0.6713 iter=31
[M2] depth=5 lr=0.1: val AUC=0.6719 iter=34
[M5] depth=3 lr=0.05: val AUC=0.6745 iter=112
[M5] depth=3 lr=0.1: val AUC=0.6750 iter=89

```

```

[M5] depth=5 lr=0.05: val AUC=0.6745 iter=103
[M5] depth=5 lr=0.1: val AUC=0.6741 iter=22
XGB selected: M2 {'max_depth': 3, 'learning_rate': 0.1} n=74 M5 {'max_depth': 3, 'learning_rate': 0.1} n=90
[M3] alpha=0.0001: val AUC=0.6651
[M3] alpha=0.001: val AUC=0.6619
[M3] alpha=0.01: val AUC=0.6557
[M6] alpha=0.0001: val AUC=0.6668
[M6] alpha=0.001: val AUC=0.6544
[M6] alpha=0.01: val AUC=0.6634
MLP selected: alpha3=0.0001 alpha6=0.0001
[M7] lr=0.05 leaves=31: val AUC=0.6705
[M7] lr=0.05 leaves=63: val AUC=0.6700
[M7] lr=0.1 leaves=31: val AUC=0.6707
[M7] lr=0.1 leaves=63: val AUC=0.6696
[M8] lr=0.05 leaves=31: val AUC=0.6729
[M8] lr=0.05 leaves=63: val AUC=0.6723
[M8] lr=0.1 leaves=31: val AUC=0.6720
[M8] lr=0.1 leaves=63: val AUC=0.6712
HistGB selected: M7 {'learning_rate': 0.1, 'max_leaf_nodes': 31} M8 {'learning_rate': 0.05, 'max_leaf_nodes': 31}

```

```

In [11]: # stage two, final fits on train+val with train+val scaling
m1_final = LogisticRegression(C=best_C1, class_weight='balanced', max_iter=1000,
                              random_state=42).fit(X_tv_lob, y_tv)
m4_final = LogisticRegression(C=best_C4, class_weight='balanced', max_iter=1000,
                              random_state=42).fit(X_tv_full, y_tv_d)
m2_final = XGBClassifier(**params2, subsample=0.8, n_estimators=ep2,
                          scale_pos_weight=sp_lob, random_state=42,
                          verbosity=0).fit(X_tv_lob, y_tv)
m5_final = XGBClassifier(**params5, subsample=0.8, n_estimators=ep5,
                          scale_pos_weight=sp_full, random_state=42,
                          verbosity=0).fit(X_tv_full, y_tv_d)
print('LR and XGB final fits done', flush=True)

mlp3_models, mlp6_models = [], []
for seed in SEEDS:
    a = MLPClassifier(hidden_layer_sizes=(128, 64), alpha=alpha3, activation='relu',
                      max_iter=200, early_stopping=True, validation_fraction=0.05,
                      n_iter_no_change=7, random_state=seed).fit(X_tv_lob, y_tv)
    b = MLPClassifier(hidden_layer_sizes=(128, 64), alpha=alpha6, activation='relu',
                      max_iter=200, early_stopping=True, validation_fraction=0.05,
                      n_iter_no_change=7, random_state=seed).fit(X_tv_full, y_tv_d)
    mlp3_models.append(a)
    mlp6_models.append(b)
    print(f'MLP seed {seed} done', flush=True)

m7_final = HistGradientBoostingClassifier(**params7, max_iter=300, random_state=42,
                                           early_stopping=True, validation_fraction=0.1,
                                           n_iter_no_change=20).fit(X_tv_lob, y_tv)
m8_final = HistGradientBoostingClassifier(**params8, max_iter=300, random_state=42,
                                           early_stopping=True, validation_fraction=0.1,
                                           n_iter_no_change=20).fit(X_tv_full, y_tv_d)
print('HistGB final fits done', flush=True)

# paired evaluation matrices on test_d and val_d rows (train+val scalers),
# so LOB-only and LOB+Deriv models are compared one-to-one on identical rows
X_te_lob_p = scaler_lob_tv.transform(test_d[LOB_FEATURES])

```

```

X_va_lob_p = scaler_lob_tv.transform(val_d[LOB_FEATURES])

P_TE = {
    'M1': m1_final.predict_proba(X_te_lob_p)[: , 1],
    'M2': m2_final.predict_proba(X_te_lob_p)[: , 1],
    'M3': np.mean([m.predict_proba(X_te_lob_p)[: , 1] for m in mlp3_models], axis=0),
    'M4': m4_final.predict_proba(X_te_full_f)[: , 1],
    'M5': m5_final.predict_proba(X_te_full_f)[: , 1],
    'M6': np.mean([m.predict_proba(X_te_full_f)[: , 1] for m in mlp6_models], axis=0),
    'M7': m7_final.predict_proba(X_te_lob_p)[: , 1],
    'M8': m8_final.predict_proba(X_te_full_f)[: , 1],
}

P_VA = {
    'M1': m1_final.predict_proba(X_va_lob_p)[: , 1],
    'M2': m2_final.predict_proba(X_va_lob_p)[: , 1],
    'M3': np.mean([m.predict_proba(X_va_lob_p)[: , 1] for m in mlp3_models], axis=0),
    'M4': m4_final.predict_proba(X_va_full_f)[: , 1],
    'M5': m5_final.predict_proba(X_va_full_f)[: , 1],
    'M6': np.mean([m.predict_proba(X_va_full_f)[: , 1] for m in mlp6_models], axis=0),
    'M7': m7_final.predict_proba(X_va_lob_p)[: , 1],
    'M8': m8_final.predict_proba(X_va_full_f)[: , 1],
}

print('Single-seed MLP reference (pooled test AUC, motivates the ensembles):')
print(' M3 seeds: ' + ' '.join(
    f'{s}={roc_auc_score(y_test_d, m.predict_proba(X_te_lob_p)[: , 1]):.4f}'
    for s, m in zip(SEEDS, mlp3_models)))
print(' M6 seeds: ' + ' '.join(
    f'{s}={roc_auc_score(y_test_d, m.predict_proba(X_te_full_f)[: , 1]):.4f}'
    for s, m in zip(SEEDS, mlp6_models)))

```

LR and XGB final fits done

MLP seed 42 done

MLP seed 7 done

MLP seed 123 done

HistGB final fits done

Single-seed MLP reference (pooled test AUC, motivates the ensembles):

M3 seeds: 42=0.6288 7=0.6276 123=0.6286

M6 seeds: 42=0.6279 7=0.6258 123=0.6254

In [12]: *# stage three, economic-tier fits on the training window only, with the
same tuned hyperparameters as the final models and train-only scalers.
Their validation predictions are genuinely out of sample, so every
threshold and calibration map selected on validation later in the
notebook is chosen without a model scoring its own training data.
The train+val final models remain the source of every headline AUC.*

```

X_va_lob_e = scaler_lob.transform(val_d[LOB_FEATURES])
X_te_lob_e = scaler_lob.transform(test_d[LOB_FEATURES])

m1_econ = LogisticRegression(C=best_C1, class_weight='balanced', max_iter=1000,
                             random_state=42).fit(X_tr_lob, y_train)
m4_econ = LogisticRegression(C=best_C4, class_weight='balanced', max_iter=1000,
                             random_state=42).fit(X_tr_full, y_train_d)
m2_econ = XGBClassifier(**params2, subsample=0.8, n_estimators=ep2,
                        scale_pos_weight=sp_lob, random_state=42,
                        verbosity=0).fit(X_tr_lob, y_train)
m5_econ = XGBClassifier(**params5, subsample=0.8, n_estimators=ep5,
                        scale_pos_weight=sp_full, random_state=42,

```

```

        verbosity=0).fit(X_tr_full, y_train_d)
print('Economic-tier LR and XGB fits done', flush=True)

mlp3_econ, mlp6_econ = [], []
for seed in SEEDS:
    a = MLPClassifier(hidden_layer_sizes=(128, 64), alpha=alpha3, activation='relu',
                      max_iter=200, early_stopping=True, validation_fraction=0.05,
                      n_iter_no_change=7, random_state=seed).fit(X_tr_lob, y_train)
    b = MLPClassifier(hidden_layer_sizes=(128, 64), alpha=alpha6, activation='relu',
                      max_iter=200, early_stopping=True, validation_fraction=0.05,
                      n_iter_no_change=7, random_state=seed).fit(X_tr_full, y_train_d)
    mlp3_econ.append(a)
    mlp6_econ.append(b)
    print(f'Economic-tier MLP seed {seed} done', flush=True)

m7_econ = HistGradientBoostingClassifier(**params7, max_iter=300, random_state=42,
                                         early_stopping=True, validation_fraction=0.1,
                                         n_iter_no_change=20).fit(X_tr_lob, y_train)
m8_econ = HistGradientBoostingClassifier(**params8, max_iter=300, random_state=42,
                                         early_stopping=True, validation_fraction=0.1,
                                         n_iter_no_change=20).fit(X_tr_full, y_train_d)
print('Economic-tier HistGB fits done', flush=True)

P_VA_E = {
    'M1': m1_econ.predict_proba(X_va_lob_e)[: , 1],
    'M2': m2_econ.predict_proba(X_va_lob_e)[: , 1],
    'M3': np.mean([m.predict_proba(X_va_lob_e)[: , 1] for m in mlp3_econ], axis=0),
    'M4': m4_econ.predict_proba(X_va_full)[: , 1],
    'M5': m5_econ.predict_proba(X_va_full)[: , 1],
    'M6': np.mean([m.predict_proba(X_va_full)[: , 1] for m in mlp6_econ], axis=0),
    'M7': m7_econ.predict_proba(X_va_lob_e)[: , 1],
    'M8': m8_econ.predict_proba(X_va_full)[: , 1],
}
P_TE_E = {
    'M1': m1_econ.predict_proba(X_te_lob_e)[: , 1],
    'M2': m2_econ.predict_proba(X_te_lob_e)[: , 1],
    'M3': np.mean([m.predict_proba(X_te_lob_e)[: , 1] for m in mlp3_econ], axis=0),
    'M4': m4_econ.predict_proba(X_te_full)[: , 1],
    'M5': m5_econ.predict_proba(X_te_full)[: , 1],
    'M6': np.mean([m.predict_proba(X_te_full)[: , 1] for m in mlp6_econ], axis=0),
    'M7': m7_econ.predict_proba(X_te_lob_e)[: , 1],
    'M8': m8_econ.predict_proba(X_te_full)[: , 1],
}
print('Economic-tier validation and test predictions built', flush=True)

```

Economic-tier LR and XGB fits done
 Economic-tier MLP seed 42 done
 Economic-tier MLP seed 7 done
 Economic-tier MLP seed 123 done
 Economic-tier HistGB fits done
 Economic-tier validation and test predictions built

Section 8: Evaluation

Pooled versus within-day AUC, the central measurement decision

Pooled AUC ranks all test observations jointly, so a model is rewarded both for ranking intervals within a session and for ranking whole days by their adverse-selection base rate (which varies between roughly 0.25 and 0.45 across the test window). A day-constant predictor can improve pooled AUC by capturing between-day risk even though it cannot rank intervals within a session, and Section 15 demonstrates this with a placebo. Within-day AUC, the mean of per-day AUCs, is immune to between-day ranking by construction and isolates the actual market-making question. Can the model rank the 10-second intervals it will face within a session. Both metrics are reported throughout, and information-set conclusions rest on the within-day column.

Table 3: Pooled AUC, within-day AUC with day-bootstrap CIs, and balanced accuracy at a threshold selected on the economic tier's out-of-sample validation predictions. The pooled bootstrap resamples individual 10-second observations, which treats a day-dependent panel as independent, so the pooled interval printed below is a legacy diagnostic and pooled AUC is reported as a descriptive point estimate. All intraday AUC inference in the report uses the day-block bootstrap on within-day quantities, which resamples whole days. The day-level forecasting tests of Section 17 are separate procedures at daily frequency. The report calls this the day-bootstrap and reserves moving-block for the block-length-10 procedure applied to daily forecast errors in Section 17.

Table 4: Information-set gains per family, pooled and within-day, with the share of days the gain is positive and the Diebold-Mariano statistic on Brier loss differentials (Newey-West, 6 lags, positive means LOB+Derivatives has lower loss). The DM column is labelled DM (Brier) because it is a probability-loss diagnostic, not a ranking test. It is computed on the 193k-row 10-second loss series with six Newey-West lags, too few for intraday dependence with day-constant regressors, so it is retained here as a diagnostic only and is not reported. The daily-frequency forecast comparison of Section 17 is the relevant diagnostic, and because those models are nested its raw squared-error test is conservative, which is discussed there. DM and AUC answer different questions, DM tests probability placement (a day-level skill with day-constant features), AUC tests discrimination, and the two can disagree in sign without contradiction.

Reading the output: Every non-linear within-day gain is null or negative. The LR pooled gain is large (about +0.037) but its within-day gain is not statistically distinguishable from zero, which identifies it as pure between-day base-rate placement. This pattern, pooled yes but within-day no, is the headline finding of the project.

In [13]:

```
# Table 3, pooled + within-day AUC, joint bootstrap
days_te = test_d['date'].astype(str).values
uniq_days = np.unique(days_te)
day_idx = {d: np.where(days_te == d)[0] for d in uniq_days}

def daily_auc_series(y, p):
    out = {}
    for d in uniq_days:
```

```

        ix = day_idx[d]
        if y[ix].min() == y[ix].max():
            continue
        out[d] = roc_auc_score(y[ix], p[ix])
    return pd.Series(out)

def boot_mean_ci(series, n_boot=2000, seed=42):
    rng = np.random.RandomState(seed)
    vals = np.asarray(series)
    means = np.array([vals[rng.randint(0, len(vals), len(vals))].mean()
                      for _ in range(n_boot)])
    return np.percentile(means, [2.5, 97.5])

def best_threshold(y_true, p):
    grid = np.arange(0.05, 0.95, 0.01)
    scores = [balanced_accuracy_score(y_true, (p >= t).astype(int)) for t in grid]
    return float(grid[int(np.argmax(scores))])

rng = np.random.RandomState(42)
n_te = len(y_test_d)
auc_draws = np.empty((N_BOOT, len(ORDER8)))
for b in range(N_BOOT):
    idx = rng.randint(0, n_te, n_te)
    y_b = y_test_d[idx]
    for j, m in enumerate(ORDER8):
        auc_draws[b, j] = roc_auc_score(y_b, P_TE[m][idx])

DAILY = {m: daily_auc_series(y_test_d, P_TE[m]) for m in ORDER8}
# the balanced-accuracy threshold is selected on the economic-tier
# validation predictions, which are out of sample, and then applied
# unchanged to the final models' test predictions
TAU_BA = {m: best_threshold(y_val_d, P_VA_E[m]) for m in ORDER8}

table3_rows = []
for j, m in enumerate(ORDER8):
    lo, hi = np.percentile(auc_draws[:, j], [2.5, 97.5])
    wlo, whi = boot_mean_ci(DAILY[m].values)
    table3_rows.append({
        'Model': NAME[m],
        'Pooled AUC': roc_auc_score(y_test_d, P_TE[m]),
        'Pooled CI': f'[{lo:.4f}, {hi:.4f}]',
        'Within-Day AUC': DAILY[m].mean(),
        'Within-Day CI': f'[{wlo:.4f}, {whi:.4f}]',
        'Balanced Acc': balanced_accuracy_score(y_test_d, (P_TE[m] >=
TAU_BA[m]).astype(int)),
        'Tau': TAU_BA[m],
    })
table3 = pd.DataFrame(table3_rows).set_index('Model')
print(f'Table 3, Predictive Performance (test {TEST_START} to {TEST_END}, N={n_te:,})')
print(table3.round(4).to_string())

```

Table 3, Predictive Performance (test 2025-05-01 to 2025-08-29, N=193,057)

	Pooled AUC	Pooled CI	Within-Day AUC	Within-Day CI	Balanced Acc
Tau					
Model					

M1 LR LOB 0.53	0.5155	[0.5130, 0.5182]	0.5050	[0.5016, 0.5085]	0.4993
M2 XGB LOB 0.46	0.6327	[0.6302, 0.6351]	0.6103	[0.5853, 0.6390]	0.6157
M3 MLP Ens LOB 0.19	0.6298	[0.6274, 0.6323]	0.6081	[0.5832, 0.6368]	0.6144
M4 LR LOB+D 0.46	0.5524	[0.5499, 0.5552]	0.5059	[0.5024, 0.5094]	0.5131
M5 XGB LOB+D 0.48	0.6318	[0.6293, 0.6342]	0.6100	[0.5851, 0.6390]	0.6157
M6 MLP Ens LOB+D 0.22	0.6283	[0.6257, 0.6306]	0.6055	[0.5804, 0.6342]	0.6140
M7 HistGB LOB 0.19	0.6334	[0.6309, 0.6358]	0.6109	[0.5856, 0.6397]	0.6157
M8 HistGB LOB+D 0.30	0.6293	[0.6267, 0.6318]	0.6093	[0.5839, 0.6384]	0.6157

```
In [14]: # Table 4, information-set gains + Diebold-Mariano
def dm_test(y_true, p_a, p_b, h_lag=6):
    d = (p_a - y_true) ** 2 - (p_b - y_true) ** 2
    dbar, nn = d.mean(), len(d)
    dc = d - dbar
    s = (dc ** 2).mean()
    for l in range(1, h_lag + 1):
        s += 2 * (1 - l / (h_lag + 1)) * (dc[l:] * dc[:-l]).mean()
    return dbar / np.sqrt(s / nn)

PAIRS = [('M1', 'M4'), ('M2', 'M5'), ('M3', 'M6'), ('M7', 'M8')]
table4_rows = []
for a, b in PAIRS:
    i, j = ORDER8.index(a), ORDER8.index(b)
    dpool = auc_draws[:, j] - auc_draws[:, i]
    plo, phi = np.percentile(dpool, [2.5, 97.5])
    pp = roc_auc_score(y_test_d, P_TE[b]) - roc_auc_score(y_test_d, P_TE[a])
    dw = (DAILY[b] - DAILY[a]).dropna()
    wlo, whi = boot_mean_ci(dw.values)
    table4_rows.append({
        'Comparison': f'{a} to {b}',
        'Pooled Gain': f'{pp:+.4f}',
        'Pooled CI': f'[{plo:+.4f}, {phi:+.4f}]',
        'Within-Day Gain': f'{dw.mean():+.4f}',
        'Within-Day CI': f'[{wlo:+.4f}, {whi:+.4f}]',
        'Days Positive': f'{100 * (dw > 0).mean():.0f}%',
        'DM (Brier)': f'{dm_test(y_test_d.astype(float), P_TE[a], P_TE[b]):+.1f}',
    })
table4 = pd.DataFrame(table4_rows).set_index('Comparison')
print('Table 4, Information-Set Gains from Adding Derivatives')
print(table4.to_string())
```

Table 4, Information-Set Gains from Adding Derivatives

	Pooled Gain		Pooled CI		Within-Day Gain	Within-Day CI	Days Positive	D
M (Brier)								
Comparison								
M1 to M4 +56.6	+0.0369	[+0.0329, +0.0409]			+0.0008	[-0.0012, +0.0027]		55%
M2 to M5 +50.6	-0.0009	[-0.0017, +0.0001]			-0.0003	[-0.0011, +0.0004]		45%

M3 to M6 -14.4	-0.0015 [-0.0031, -0.0000]	-0.0025 [-0.0044, -0.0007]	35%
M7 to M8 -6.7	-0.0041 [-0.0055, -0.0025]	-0.0015 [-0.0029, +0.0000]	37%

Section 9: Feature Importance

Table 5 and the SHAP beeswarm quantify each feature's contribution to M5, the XGBoost model on the full feature set, via mean absolute SHAP over 5,000 test observations. Signed order flow dominates by more than an order of magnitude, the ATM IV level is the top derivatives feature, and the six derivatives features together receive about 5% of the attribution, consistent with their null within-day gain in Table 4.

The all-model importance grid extends the picture to every configuration with the appropriate method per family, using absolute standardised coefficients for the linear models, mean absolute SHAP for the boosted trees, and permutation AUC drop (50,000 test rows) for the MLP ensembles and any model SHAP cannot handle. The story is identical in every panel. Order-flow and spread features carry the model, derivatives features sit in the tail.

`queue_imb` is identically zero in EQUUS.MINI (the synthetic NBBO feed does not populate per-venue order counts) and correctly receives zero importance everywhere.

Why the linear LOB+D panel is the red one: In the all-model grid, M4 (logistic regression) is the only configuration that loads heavily on the derivatives features. A linear model cannot interact day-constant columns with intraday state, so the only use it can make of them is a day-level shift of the predicted probability, which is exactly the between-day base-rate placement behind its pooled-only gain. The flexible models (XGB, MLP, HistGB) extract the same day-level information from the order book itself and assign the derivatives block the roughly 5% residual attribution visible in their panels.

```
In [15]: # Table 5 and SHAP beeswarm for M5
n_shap = min(5000, len(X_te_full_f))
explainer = shap.TreeExplainer(m5_final)
shap_vals = explainer.shap_values(X_te_full_f[:n_shap])

plt.figure(figsize=(8, 5.5))
shap.summary_plot(shap_vals, X_te_full_f[:n_shap],
                  feature_names=[PRETTY[f] for f in ALL_FEATURES], show=False)
plt.title('M5 SHAP Feature Importance, Test Set')
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'shap_m5.png'), dpi=150, bbox_inches='tight')
plt.show()

table5 = (
    pd.DataFrame({'Feature': [PRETTY[f] for f in ALL_FEATURES],
                  'Mean Abs SHAP': np.abs(shap_vals).mean(axis=0),
                  'Group': ['Derivatives' if f in DERIV_FEATURES else 'LOB'
                           for f in ALL_FEATURES]})
    .sort_values('Mean Abs SHAP', ascending=False)
    .set_index('Feature')
)
```

```
print('Table 5, SHAP Feature Importance (M5)')
print(table5.round(4).to_string())
```

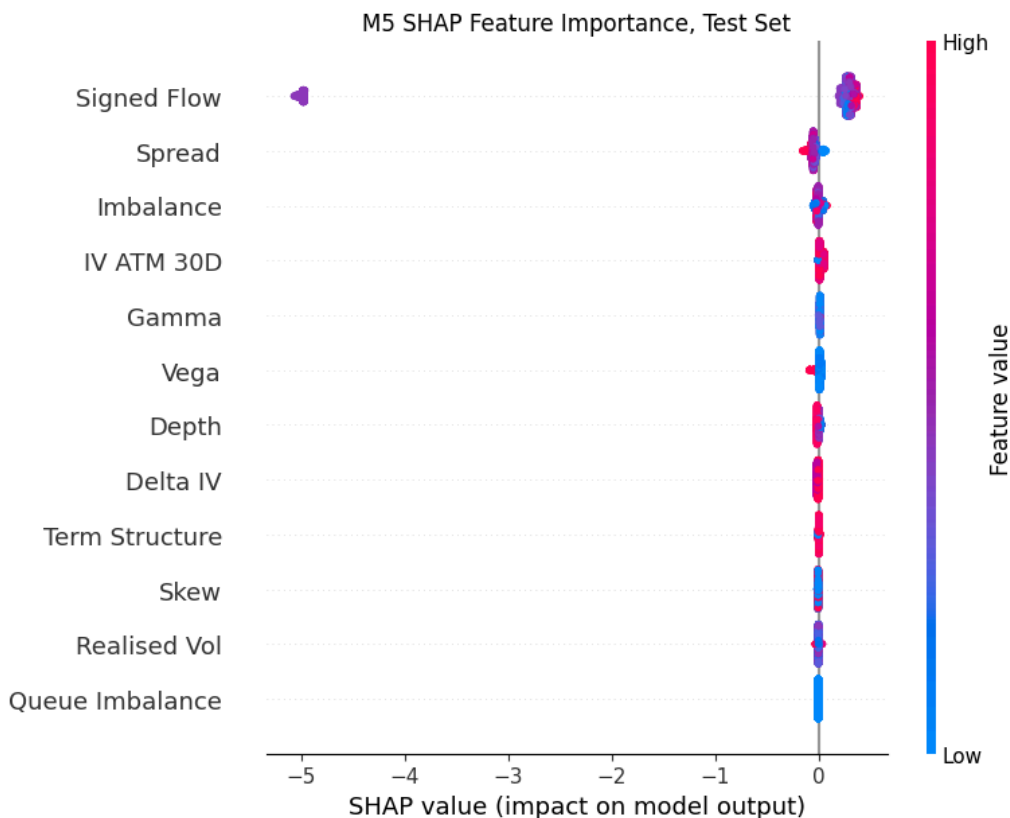


Table 5, SHAP Feature Importance (M5)

Feature	Mean Abs SHAP	Group
Signed Flow	1.0884	LOB
Spread	0.0492	LOB
Imbalance	0.0177	LOB
IV ATM 30D	0.0168	Derivatives
Gamma	0.0138	Derivatives
Vega	0.0130	Derivatives
Depth	0.0077	LOB
Delta IV	0.0066	Derivatives
Term Structure	0.0044	Derivatives
Skew	0.0026	Derivatives
Realised Vol	0.0019	LOB
Queue Imbalance	0.0000	LOB

```
In [16]: # feature importance for all eight models, method matched to family
IMP, METHOD = {}, {}

IMP['M1'] = pd.Series(np.abs(m1_final.coef_[0]), index=LOB_FEATURES)
IMP['M4'] = pd.Series(np.abs(m4_final.coef_[0]), index=ALL_FEATURES)
METHOD['M1'] = METHOD['M4'] = 'Absolute Standardised Coefficient'

for tag, mdl, X, feats in (('M2', m2_final, X_te_lob_p, LOB_FEATURES),
                           ('M5', m5_final, X_te_full_f, ALL_FEATURES)):
    sv = shap.TreeExplainer(mdl).shap_values(X[:5000])
    IMP[tag] = pd.Series(np.abs(sv).mean(axis=0), index=feats)
    METHOD[tag] = 'Mean Absolute SHAP'

for tag, mdl, X, feats in (('M7', m7_final, X_te_lob_p, LOB_FEATURES),
                           ('M8', m8_final, X_te_full_f, ALL_FEATURES)):
```

```

try:
    sv = shap.TreeExplainer mdl).shap_values(X[:5000])
    if isinstance(sv, list):
        sv = sv[1]
    IMP[tag] = pd.Series(np.abs(sv).mean(axis=0), index=feats)
    METHOD[tag] = 'Mean Absolute SHAP'
except Exception as e:
    print(f'{tag}: SHAP unsupported ({type(e).__name__}), using permutation',
flush=True)
    rng_i = np.random.RandomState(42)
    sub = rng_i.choice(len(X), 50000, replace=False)
    base = roc_auc_score(y_test_d[sub], mdl.predict_proba(X[sub])[:, 1])
    drops = []
    for j in range(X.shape[1]):
        Xp = X[sub].copy()
        Xp[:, j] = Xp[rng_i.permutation(len(sub)), j]
        drops.append(base - roc_auc_score(y_test_d[sub], mdl.predict_proba(Xp)[:, 1]))
    IMP[tag] = pd.Series(drops, index=feats)
    METHOD[tag] = 'Permutation AUC Drop'

rng_i = np.random.RandomState(42)
sub = rng_i.choice(len(X_te_lob_p), 50000, replace=False)
for tag, models, X, feats in (('M3', mlp3_models, X_te_lob_p, LOB_FEATURES),
                              ('M6', mlp6_models, X_te_full_f, ALL_FEATURES)):
    def ens_p(Xa, _models=models):
        return np.mean([m.predict_proba(Xa)[:, 1] for m in _models], axis=0)
    base = roc_auc_score(y_test_d[sub], ens_p(X[sub]))
    drops = []
    for j in range(X.shape[1]):
        Xp = X[sub].copy()
        Xp[:, j] = Xp[rng_i.permutation(len(sub)), j]
        drops.append(base - roc_auc_score(y_test_d[sub], ens_p(Xp)))
    IMP[tag] = pd.Series(drops, index=feats)
    METHOD[tag] = 'Permutation AUC Drop'
    print(f'{tag} permutation importance done', flush=True)

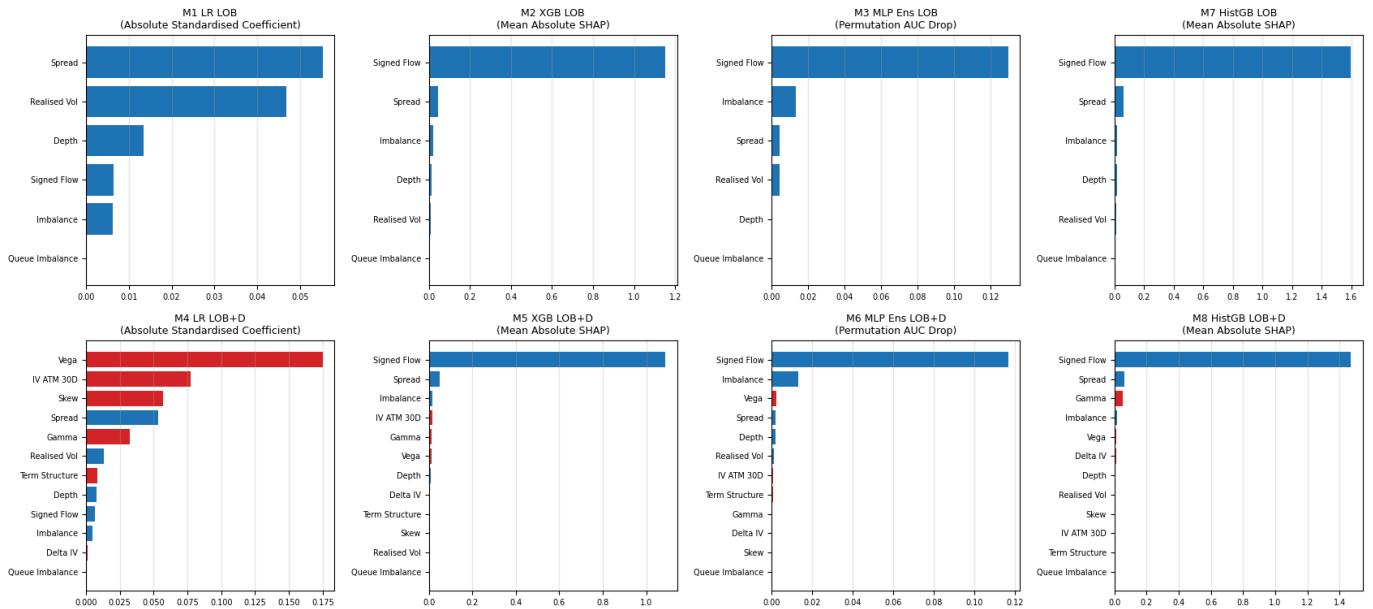
GRID_ORDER = ['M1', 'M2', 'M3', 'M7', 'M4', 'M5', 'M6', 'M8']
fig, axes = plt.subplots(2, 4, figsize=(17, 8))
for ax, tag in zip(axes.flat, GRID_ORDER):
    s = IMP[tag].sort_values()
    colors = ['tab:red' if f in DERIV_FEATURES else 'tab:blue' for f in s.index]
    ax.barh([PRETTY[f] for f in s.index], s.values, color=colors)
    ax.set_title(f'{NAME[tag]}\n({METHOD[tag]}', fontsize=9)
    ax.tick_params(labelsize=7)
    ax.grid(alpha=0.3, axis='x')
fig.suptitle('Feature Importance, All Models (Blue = LOB Features, Red = Derivatives
Features)')
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'feature_importance_all.png'), dpi=150,
bbox_inches='tight')
plt.show()

```

M3 permutation importance done

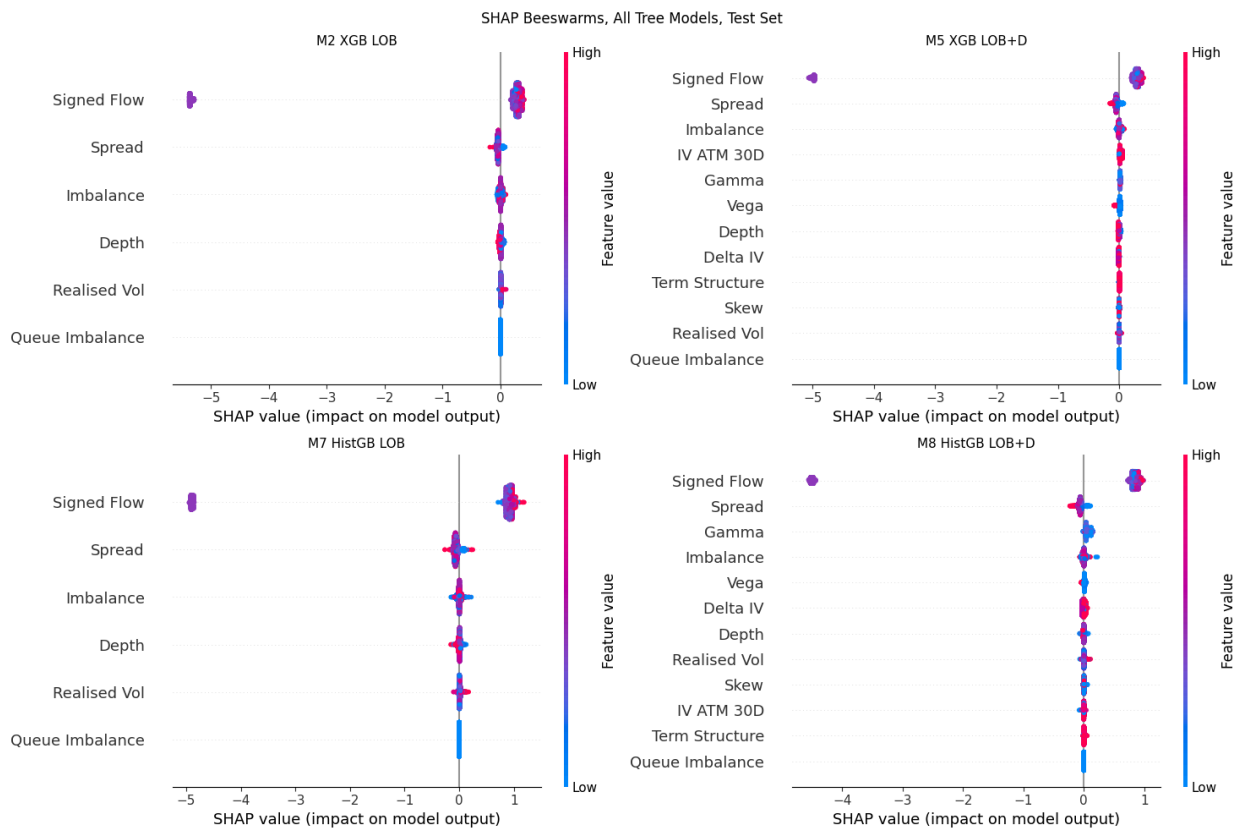
M6 permutation importance done

Feature Importance, All Models (Blue = LOB Features, Red = Derivatives Features)



```
In [17]: # SHAP beeswarms for every tree model (M2, M5, M7, M8)
TREE_MODELS = [('M2', m2_final, X_te_lob_p, LOB_FEATURES),
                ('M5', m5_final, X_te_full_f, ALL_FEATURES),
                ('M7', m7_final, X_te_lob_p, LOB_FEATURES),
                ('M8', m8_final, X_te_full_f, ALL_FEATURES)]

fig = plt.figure(figsize=(15, 10))
for k, (tag, mdl, X, feats) in enumerate(TREE_MODELS):
    ax = fig.add_subplot(2, 2, k + 1)
    plt.sca(ax)
    try:
        sv = shap.TreeExplainer(mdl).shap_values(X[:5000])
        if isinstance(sv, list):
            sv = sv[1]
        shap.summary_plot(sv, X[:5000], feature_names=[PRETTY[f] for f in feats],
                          show=False, plot_size=None, max_display=12)
        ax.set_title(NAME[tag], fontsize=11)
    except Exception as e:
        ax.text(0.5, 0.5, f'{NAME[tag]}\nSHAP unsupported ({type(e).__name__})',
               ha='center', va='center')
        ax.set_axis_off()
fig.suptitle('SHAP Beeswarms, All Tree Models, Test Set')
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'shap_all_trees.png'), dpi=150, bbox_inches='tight')
plt.show()
```



Section 10: Out-of-Sample Robustness (Table 8)

The OptionMetrics cutoff is 2025-08-29, so the September to December 2025 window has no derivatives data and tests only the LOB models. This is by design. It checks whether models fitted through April 2025 generalise to the September to December 2025 window with no further retraining and no rescaling beyond the locked train+val scaler.

The non-linear LOB models generalise strongly (pooled AUC above 0.73 against 0.63 on the primary test window). The diagnostic table below explains why the robustness window is easier rather than leaving the jump unexplained. Realised volatility is in fact higher in the robustness window (0.6788 against 0.5218), so the window is not calmer. What makes it easier is a lower adverse-event base rate and a lower share of active-flow bins, and Section 15.4 shows exposure prediction is a large part of headline discrimination, so a higher share of bins without directional signed flow can lift AUC by raising the share of mechanically safe observations. The number is therefore interpreted only as LOB-engine stability under a different regime, never as evidence about derivatives augmentation (which cannot be tested here for lack of OptionMetrics coverage). The linear model stays near chance, consistent with the useful signal being predominantly non-linear.

```
In [18]: # Table 8, OOS robustness on Sep to Dec 2025, consistent train+val scaling
def in_rob(f):
    m = re.search(r'lob_(\d{4}-\d{2}-\d{2})\.parquet', os.path.basename(f))
    return m and ROB_START <= m.group(1) <= ROB_END

rob_files = [f for f in sorted(glob.glob(os.path.join(LOB_PROC, 'lob_*.parquet')))]
rob_target = build_target(pd.concat([pd.read_parquet(f) for f in rob_files]))
print(f'Robustness rows: {len(rob_target):,} adverse rate={rob_target["y"].mean():.3f}')
```

```

X_rob_f = scaler_lob_tv.transform(rob_target[LOB_FEATURES])
y_rob = rob_target['y'].values
days_rob = rob_target['date'].astype(str).values
uniq_rob = np.unique(days_rob)

P_ROB = {
    'M1': m1_final.predict_proba(X_rob_f)[: , 1],
    'M2': m2_final.predict_proba(X_rob_f)[: , 1],
    'M3': np.mean([m.predict_proba(X_rob_f)[: , 1] for m in mlp3_models], axis=0),
    'M7': m7_final.predict_proba(X_rob_f)[: , 1],
}

table8_rows = []
for m, p in P_ROB.items():
    wd = np.mean([roc_auc_score(y_rob[days_rob == d], p[days_rob == d])
                  for d in uniq_rob
                  if y_rob[days_rob == d].min() != y_rob[days_rob == d].max()])
    table8_rows.append({'Model': NAME[m],
                       'Pooled AUC': roc_auc_score(y_rob, p),
                       'Within-Day AUC': wd})
table8 = pd.DataFrame(table8_rows).set_index('Model')
print('Table 8, Out-of-Sample Robustness (Sep to Dec 2025, LOB models)')
print(table8.round(4).to_string())

```

Robustness rows: 191,977 adverse rate=0.317

Table 8, Out-of-Sample Robustness (Sep to Dec 2025, LOB models)

	Pooled AUC	Within-Day AUC
Model		
M1 LR LOB	0.5099	0.5139
M2 XGB LOB	0.7318	0.7226
M3 MLP Ens LOB	0.7298	0.7209
M7 HistGB LOB	0.7324	0.7235

```

In [19]: # sample diagnostics, primary test window vs robustness window
def sample_diag(df, label):
    return pd.Series({
        'Rows': len(df),
        'Days': df['date'].astype(str).nunique(),
        'Adverse Rate': df['y'].mean(),
        'Active Flow Share': (df['quote_dir'] != 0).mean(),
        'Mean Spread USD': df['spread'].mean(),
        'Median Spread USD': df['spread'].median(),
        'Mean Abs Signed Flow': df['signed_flow'].abs().mean(),
        'Mean Realised Vol (1e-5)': df['realised_vol'].mean() * 1e5,
    }, name=label)

diag = pd.concat([sample_diag(test_d, 'Test May-Aug 2025'),
                  sample_diag(rob_target, 'Robustness Sep-Dec 2025')], axis=1)
print('Sample diagnostics, why the robustness window is easier')
print(diag.round(4).to_string())

```

Sample diagnostics, why the robustness window is easier

	Test May-Aug 2025	Robustness Sep-Dec 2025
Rows	193057.0000	191977.0000
Days	83.0000	83.0000
Adverse Rate	0.3878	0.3172
Active Flow Share	0.8583	0.6940

Mean Spread USD	0.0235	0.0266
Median Spread USD	0.0200	0.0200
Mean Abs Signed Flow	444.2684	151.3032
Mean Realised Vol (1e-5)	0.5218	0.6788

Section 11: Results Summary and Final Hyperparameters

Consolidated output of Sections 8 to 10 for the report, plus the Appendix C hyperparameter record. The calibration and decision-layer tables (6 and 7) follow in Section 13.

```
In [20]: print('Table 3, Predictive Performance')
print(table3.round(4).to_string())

print('\nTable 4, Information-Set Gains')
print(table4.to_string())

print('\nTable 5, SHAP Feature Importance (M5)')
print(table5.round(4).to_string())

print('\nTable 8, Out-of-Sample Robustness')
print(table8.round(4).to_string())

print('\nAppendix C, Final Hyperparameters')
for k, v in {
    'M1 LR C': best_C1,
    'M4 LR C': best_C4,
    'M2 XGB': f'{params2} n_estimators={ep2}',
    'M5 XGB': f'{params5} n_estimators={ep5}',
    'M3 MLP alpha': f'{alpha3} ensemble seeds {SEEDS}',
    'M6 MLP alpha': f'{alpha6} ensemble seeds {SEEDS}',
    'M7 HistGB': params7,
    'M8 HistGB': params8,
}.items():
    print(f' {k}: {v}')
```

Table 3, Predictive Performance

	Pooled AUC	Pooled CI	Within-Day AUC	Within-Day CI	Balanced Acc
Tau					
Model					
M1 LR LOB 0.53	0.5155	[0.5130, 0.5182]	0.5050	[0.5016, 0.5085]	0.4993
M2 XGB LOB 0.46	0.6327	[0.6302, 0.6351]	0.6103	[0.5853, 0.6390]	0.6157
M3 MLP Ens LOB 0.19	0.6298	[0.6274, 0.6323]	0.6081	[0.5832, 0.6368]	0.6144
M4 LR LOB+D 0.46	0.5524	[0.5499, 0.5552]	0.5059	[0.5024, 0.5094]	0.5131
M5 XGB LOB+D 0.48	0.6318	[0.6293, 0.6342]	0.6100	[0.5851, 0.6390]	0.6157
M6 MLP Ens LOB+D 0.22	0.6283	[0.6257, 0.6306]	0.6055	[0.5804, 0.6342]	0.6140
M7 HistGB LOB 0.19	0.6334	[0.6309, 0.6358]	0.6109	[0.5856, 0.6397]	0.6157
M8 HistGB LOB+D 0.30	0.6293	[0.6267, 0.6318]	0.6093	[0.5839, 0.6384]	0.6157

Table 4, Information-Set Gains

	Pooled Gain	Pooled CI	Within-Day Gain	Within-Day CI	Days Positive D
M (Brier)					
Comparison					
M1 to M4 +56.6	+0.0369	[+0.0329, +0.0409]	+0.0008	[-0.0012, +0.0027]	55%
M2 to M5 +50.6	-0.0009	[-0.0017, +0.0001]	-0.0003	[-0.0011, +0.0004]	45%
M3 to M6 -14.4	-0.0015	[-0.0031, -0.0000]	-0.0025	[-0.0044, -0.0007]	35%
M7 to M8 -6.7	-0.0041	[-0.0055, -0.0025]	-0.0015	[-0.0029, +0.0000]	37%

Table 5, SHAP Feature Importance (M5)

Feature	Mean Abs SHAP	Group
Signed Flow	1.0884	LOB
Spread	0.0492	LOB
Imbalance	0.0177	LOB
IV ATM 30D	0.0168	Derivatives
Gamma	0.0138	Derivatives
Vega	0.0130	Derivatives
Depth	0.0077	LOB
Delta IV	0.0066	Derivatives
Term Structure	0.0044	Derivatives
Skew	0.0026	Derivatives
Realised Vol	0.0019	LOB
Queue Imbalance	0.0000	LOB

Table 8, Out-of-Sample Robustness

	Pooled AUC	Within-Day AUC
Model		
M1 LR LOB	0.5099	0.5139
M2 XGB LOB	0.7318	0.7226
M3 MLP Ens LOB	0.7298	0.7209
M7 HistGB LOB	0.7324	0.7235

Appendix C, Final Hyperparameters

M1 LR C: 10.0
 M4 LR C: 0.01
 M2 XGB: {'max_depth': 3, 'learning_rate': 0.1} n_estimators=74
 M5 XGB: {'max_depth': 3, 'learning_rate': 0.1} n_estimators=90
 M3 MLP alpha: 0.0001 ensemble seeds (42, 7, 123)
 M6 MLP alpha: 0.0001 ensemble seeds (42, 7, 123)
 M7 HistGB: {'learning_rate': 0.1, 'max_leaf_nodes': 31}
 M8 HistGB: {'learning_rate': 0.05, 'max_leaf_nodes': 31}

Section 12: Diagnostics and Conditioning

Visual and conditional diagnostics on the final models. Every figure shows all relevant models and is saved to `Figures/`.

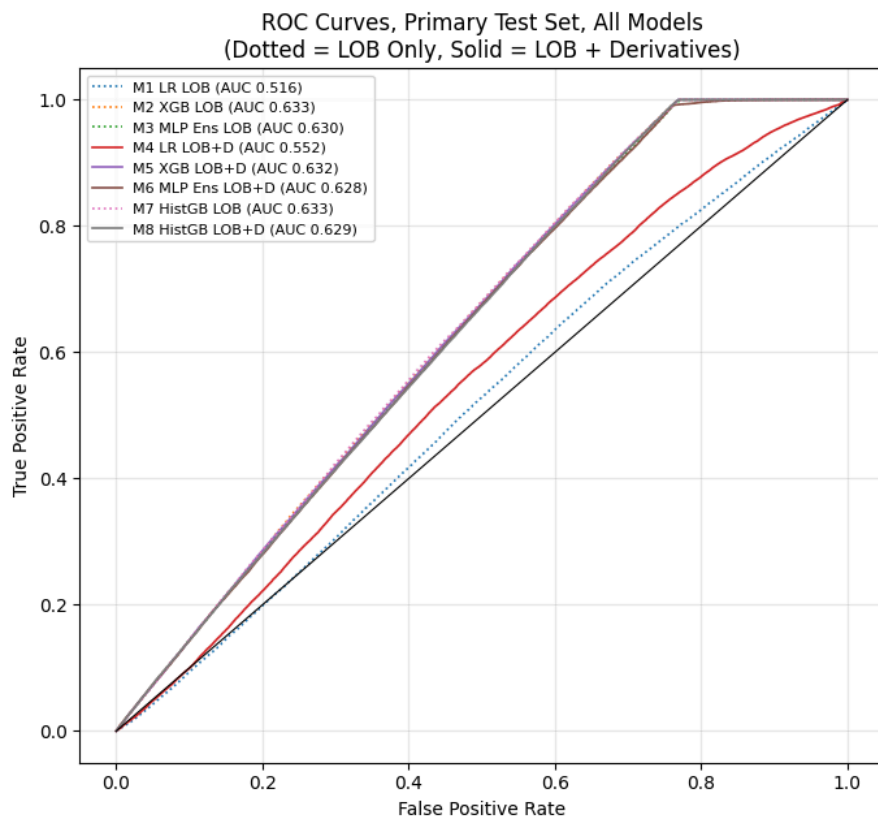
Subsection	Content	Figure
12.1	ROC curves, all eight models, curve AUCs verified against Table 3	<code>roc_curves.png</code>

Subsection	Content	Figure
12.2	Rolling weekly AUC through the test window	rolling_weekly_auc.png
12.3	Feature correlations and the pooled vs within-day comparison	feature_correlation.png , withinday_vs_pooled.png
12.4	Adverse rate by probability decile (Appendix E)	decile_lift.png
12.5	Regime and time-of-day conditioning (Appendix D)	printed tables

12.1 ROC curves

The curves look close to straight lines between the corners, which is the correct geometry for AUCs in the low 0.60s on a large sample. The legend AUCs are computed by trapezoidal integration of the plotted curves and asserted equal to the Table 3 values, so the curves and the table cannot drift apart. The visible gap separates the linear models from everything else. Within each family, the dotted (LOB only) and solid (LOB plus derivatives) curves overlap almost exactly, which is the graphical statement of the null result.

```
In [21]: fig, ax = plt.subplots(figsize=(7, 6.5))
for m in ORDER8:
    fpr, tpr, _ = roc_curve(y_test_d, P_TE[m])
    auc_plot = np.trapz(tpr, fpr)
    auc_tab = roc_auc_score(y_test_d, P_TE[m])
    assert abs(auc_plot - auc_tab) < 1e-6, f'{m} curve and table disagree'
    ax.plot(fpr, tpr, color=COLOR[m], ls=LS[m], lw=1.2,
            label=f'{NAME[m]} (AUC {auc_tab:.3f})')
ax.plot([0, 1], [0, 1], color='k', lw=0.8)
ax.set_xlabel('False Positive Rate')
ax.set_ylabel('True Positive Rate')
ax.set_title('ROC Curves, Primary Test Set, All Models\n(Dotted = LOB Only, Solid = LOB + Derivatives)')
ax.legend(fontsize=8)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'roc_curves.png'), dpi=150, bbox_inches='tight')
plt.show()
```



12.2 Rolling weekly AUC

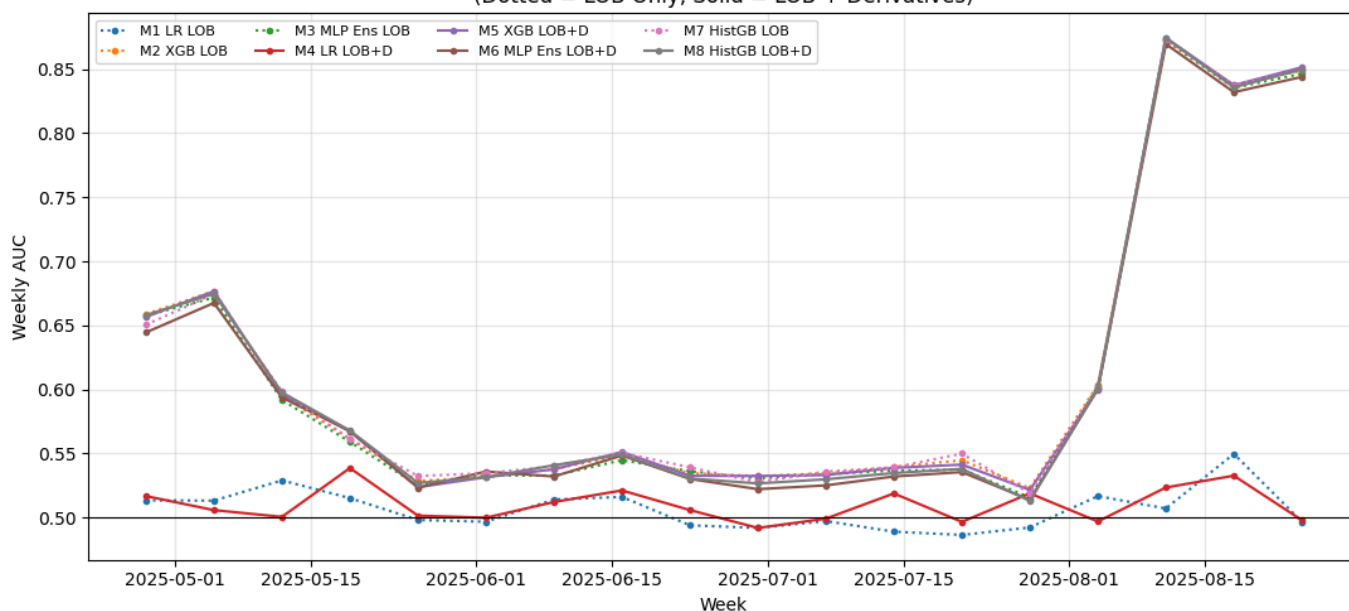
The test window opens immediately after the April and May 2025 tariff shock, so the early weeks are a stress regime and the later weeks a normalisation. Weekly AUC shows whether performance is stable through that transition. Weeks with fewer than 500 observations or one outcome class are skipped.

```
In [22]: week = pd.to_datetime(pd.Series(days_te)).dt.to_period('W').dt.start_time.values
weekly = {}
for wk in np.unique(week):
    ix = np.where(week == wk)[0]
    if len(ix) < 500 or y_test_d[ix].min() == y_test_d[ix].max():
        continue
    weekly[wk] = {m: roc_auc_score(y_test_d[ix], P_TE[m][ix]) for m in ORDER8}
wdf = pd.DataFrame(weekly).T.sort_index()
print(wdf.round(4).to_string())

fig, ax = plt.subplots(figsize=(11, 5.5))
for m in ORDER8:
    ax.plot(wdf.index, wdf[m], color=COLOR[m], ls=LS[m], marker='o', ms=3, label=NAME[m])
ax.axhline(0.5, color='k', lw=0.8)
ax.set_ylabel('Weekly AUC')
ax.set_xlabel('Week')
ax.set_title('Rolling Weekly AUC, Test Window, All Models\n(Dotted = LOB Only, Solid = LOB + Derivatives)')
ax.legend(ncol=4, fontsize=8)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'rolling_weekly_auc.png'), dpi=150, bbox_inches='tight')
plt.show()
```

	M1	M2	M3	M4	M5	M6	M7	M8
2025-04-28	0.5135	0.6588	0.6576	0.5168	0.6567	0.6446	0.6506	0.6581
2025-05-05	0.5133	0.6768	0.6717	0.5059	0.6767	0.6676	0.6745	0.6748
2025-05-12	0.5291	0.5956	0.5918	0.5006	0.5960	0.5938	0.5983	0.5977
2025-05-19	0.5152	0.5612	0.5591	0.5387	0.5672	0.5676	0.5614	0.5680
2025-05-26	0.4982	0.5287	0.5251	0.5016	0.5238	0.5234	0.5325	0.5271
2025-06-02	0.4968	0.5317	0.5330	0.5000	0.5316	0.5360	0.5347	0.5315
2025-06-09	0.5142	0.5379	0.5328	0.5122	0.5377	0.5321	0.5401	0.5409
2025-06-16	0.5162	0.5487	0.5449	0.5212	0.5513	0.5485	0.5506	0.5492
2025-06-23	0.4940	0.5347	0.5362	0.5059	0.5327	0.5302	0.5392	0.5305
2025-06-30	0.4920	0.5324	0.5306	0.4920	0.5325	0.5222	0.5277	0.5267
2025-07-07	0.4973	0.5346	0.5339	0.4992	0.5333	0.5252	0.5360	0.5299
2025-07-14	0.4891	0.5394	0.5367	0.5189	0.5388	0.5321	0.5392	0.5348
2025-07-21	0.4865	0.5449	0.5375	0.4967	0.5414	0.5355	0.5500	0.5381
2025-07-28	0.4923	0.5230	0.5158	0.5189	0.5216	0.5139	0.5190	0.5130
2025-08-04	0.5168	0.6036	0.6027	0.4970	0.6002	0.6004	0.6020	0.6011
2025-08-11	0.5071	0.8732	0.8725	0.5235	0.8744	0.8696	0.8732	0.8744
2025-08-18	0.5498	0.8375	0.8354	0.5327	0.8375	0.8320	0.8373	0.8358
2025-08-25	0.4968	0.8489	0.8465	0.4980	0.8515	0.8440	0.8497	0.8499

Rolling Weekly AUC, Test Window, All Models
(Dotted = LOB Only, Solid = LOB + Derivatives)



12.3 Feature correlations and pooled vs within-day AUC

The correlation matrix (training window) documents that the order-flow features are uncorrelated with the derivatives block, while the liquidity-state features are moderately correlated with it (`spread` against `gamma` -0.55 and `skew` +0.50), and that the derivatives block has strong internal structure. `queue_imb` is omitted because it is identically zero.

The paired bar chart restates Table 3 visually. The gap between each model's pooled and within-day bar illustrates the sensitivity of pooled AUC to between-day ranking, the part that day-constant features can inflate without intraday information.

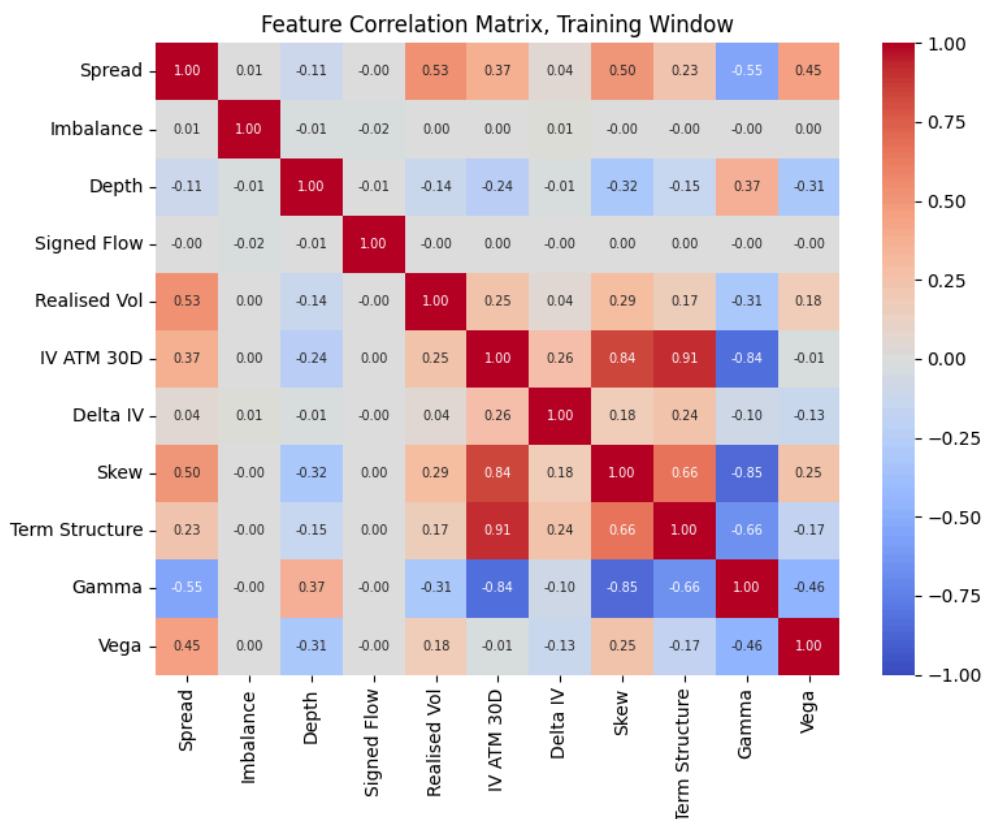
```
In [23]: corr_feats = [f for f in ALL_FEATURES if f != 'queue_imb']
corr = train_d[corr_feats].corr()
corr.index = [PRETTY[f] for f in corr_feats]
corr.columns = [PRETTY[f] for f in corr_feats]
```

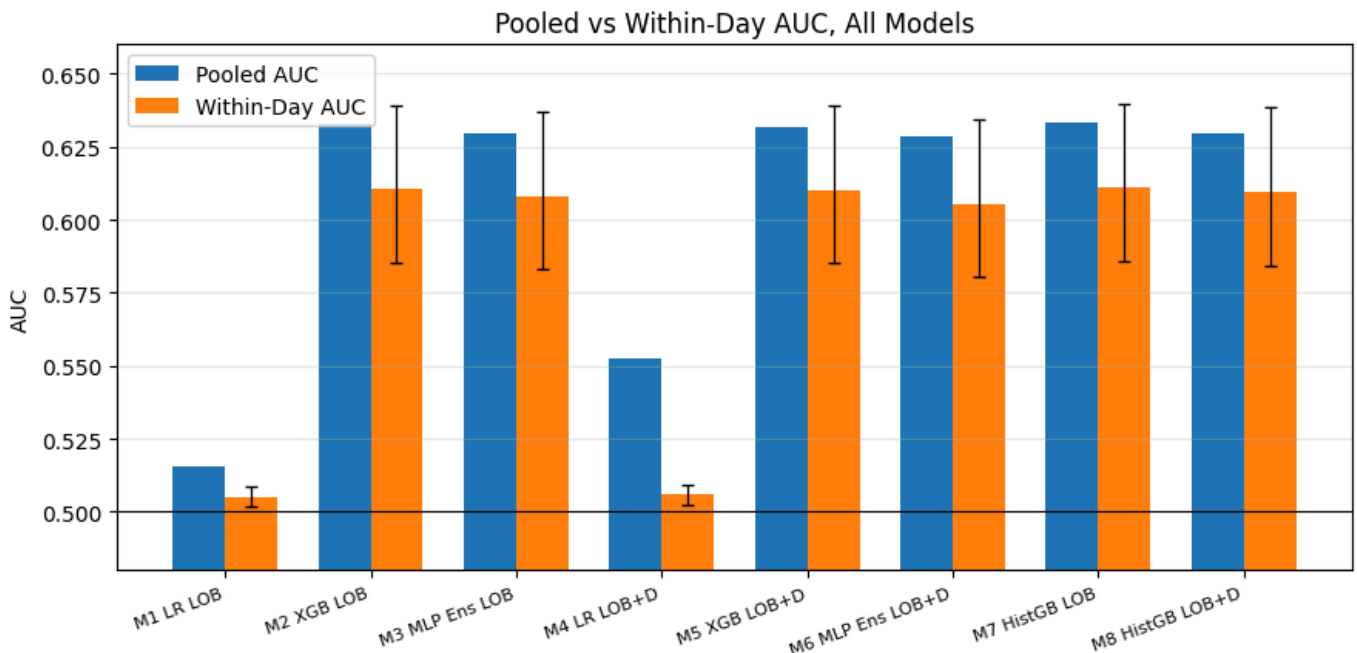
```

fig, ax = plt.subplots(figsize=(8, 6.5))
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm', center=0,
            annot_kws={'size': 7}, ax=ax, vmin=-1, vmax=1)
ax.set_title('Feature Correlation Matrix, Training Window')
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'feature_correlation.png'), dpi=150, bbox_inches='tight')
plt.show()

pooled_vals = [roc_auc_score(y_test_d, P_TE[m]) for m in ORDER8]
within_vals = [DAILY[m].mean() for m in ORDER8]
within_ci = np.array([boot_mean_ci(DAILY[m].values) for m in ORDER8])
yerr = np.vstack([np.array(within_vals) - within_ci[:, 0], within_ci[:, 1] -
np.array(within_vals)])
xb = np.arange(len(ORDER8))
fig, ax = plt.subplots(figsize=(9, 4.5))
ax.bar(xb - 0.18, pooled_vals, width=0.36, color='tab:blue', label='Pooled AUC')
ax.bar(xb + 0.18, within_vals, width=0.36, color='tab:orange', label='Within-Day AUC',
      yerr=yerr, capsize=3, error_kw={'lw': 0.9})
ax.axhline(0.5, color='k', lw=0.8)
ax.set_xticks(xb)
ax.set_xticklabels([NAME[m] for m in ORDER8], rotation=20, ha='right', fontsize=8)
ax.set_ylim(0.48, 0.66)
ax.set_ylabel('AUC')
ax.set_title('Pooled vs Within-Day AUC, All Models')
ax.legend()
ax.grid(alpha=0.3, axis='y')
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'withinday_vs_pooled.png'), dpi=150, bbox_inches='tight')
plt.show()

```



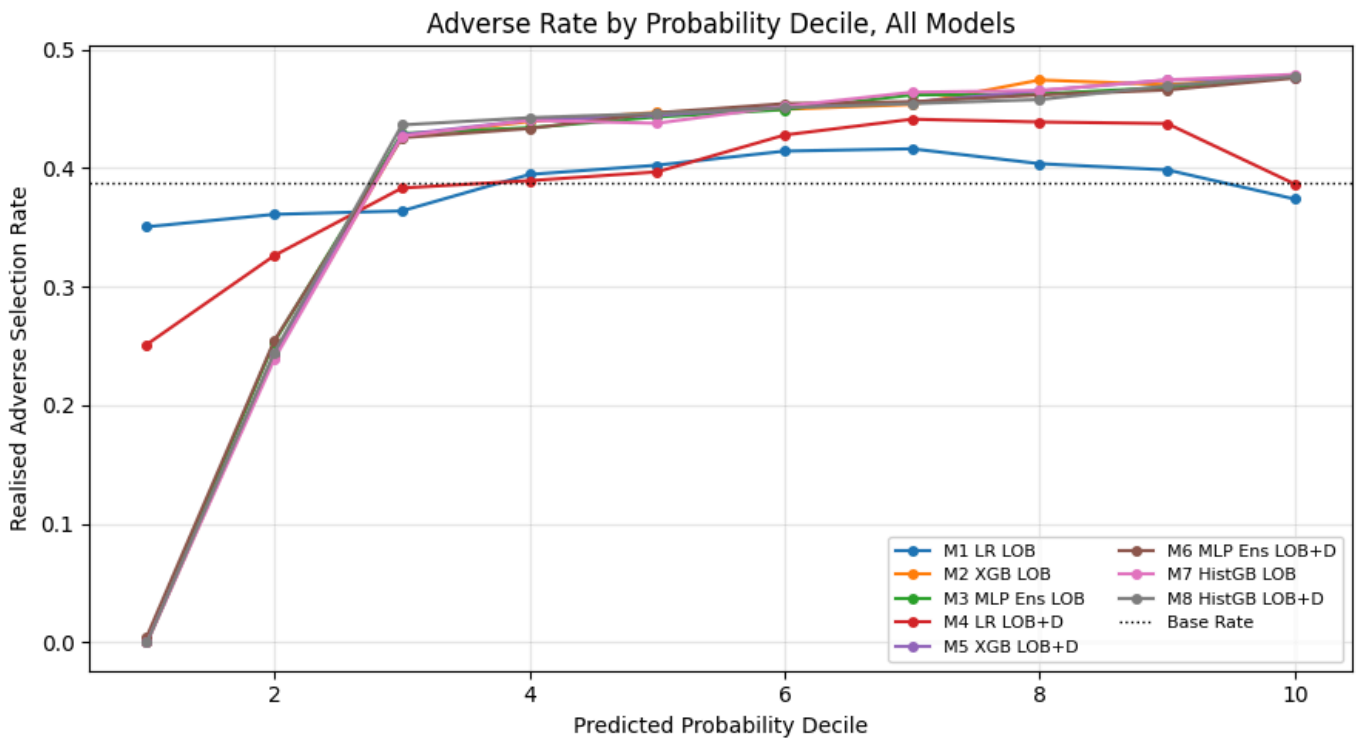


12.4 Adverse rate by probability decile (Appendix E)

A rank-quality check independent of calibration. Observations are sorted into deciles of each model's raw predicted probability, and the realised adverse rate is computed per decile. The non-linear models earn much of their discrimination by identifying clearly safe intervals. Their bottom decile carries an adverse rate near zero against a 0.37 base rate. Part of that bottom-decile purity is mechanical, intervals without net directional signed flow (no trades, side-unknown volume, or buys and sells netting out) have no defined exposure direction and are labelled zero by convention, which the models learn through the `signed_flow` feature. The top decile reaches roughly 0.48. The desk use case therefore includes tightening quotes in safe windows as well as widening in risky ones.

```
In [24]: fig, ax = plt.subplots(figsize=(9, 5))
decile_tab = {}
for m in ORDER8:
    qd = pd.qcut(P_TE[m], 10, labels=False, duplicates='drop')
    rates = [y_test_d[qd == k].mean() for k in range(int(qd.max()) + 1)]
    decile_tab[NAME[m]] = rates
    ax.plot(range(1, len(rates) + 1), rates, marker='o', ms=4,
            color=COLOR[m], label=NAME[m])
ax.axhline(y_test_d.mean(), color='k', ls=':', lw=1, label='Base Rate')
ax.set_xlabel('Predicted Probability Decile')
ax.set_ylabel('Realised Adverse Selection Rate')
ax.set_title('Adverse Rate by Probability Decile, All Models')
ax.legend(fontsize=8, ncol=2)
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'decile_lift.png'), dpi=150, bbox_inches='tight')
plt.show()

print('Appendix E, Adverse Rate by Decile')
print(pd.DataFrame(decile_tab, index=[f'D{i + 1}' for i in range(10)]).round(3).to_string())
```



Appendix E, Adverse Rate by Decile

	M1 LR LOB	M2 XGB LOB	M3 MLP Ens LOB	M4 LR LOB+D	M5 XGB LOB+D	M6 MLP Ens LOB+D	M7 HistGB LOB	M8 HistGB LOB+D
D1	0.350	0.000	0.000	0.251	0.000	0.005		
D2	0.361	0.241	0.253	0.326	0.243	0.255		
D3	0.364	0.426	0.429	0.383	0.428	0.425		
D4	0.395	0.439	0.434	0.389	0.440	0.433		
D5	0.402	0.447	0.443	0.397	0.444	0.446		
D6	0.414	0.449	0.449	0.428	0.452	0.454		
D7	0.416	0.454	0.462	0.441	0.455	0.456		
D8	0.404	0.474	0.462	0.439	0.465	0.462		
D9	0.398	0.470	0.468	0.437	0.474	0.466		
D10	0.374	0.477	0.478	0.386	0.476	0.476		

12.5 Regime and time-of-day conditioning (Appendix D)

If the derivatives features carried intraday signal anywhere, it should surface in some conditional pocket, such as high or low IV regimes (the Cremers and Weinbaum mechanism predicts more options information when the options market is active) or particular session buckets (an overnight-gap effect would concentrate at the open). The within-day paired gains are computed per day-level IV regime and the pooled gains per session bucket.

Neither conditioning reveals a pocket. The within-day gains hover at zero in both IV regimes for every non-linear family, and no session bucket shows a stable derivatives advantage. The day-constant nature of the features leaves them nothing to say about which intervals within the session are dangerous, in any regime, at any time of day.

```
In [25]: iv_day = test_d.groupby(test_d['date'].astype(str))['iv_atm_30'].first()
iv_med = test_d['iv_atm_30'].median()
hi_days = set(iv_day[iv_day > iv_med].index)

print('Appendix D Panel A, within-day paired gain by day-level IV regime')
for a, b in PAIRS:
    dw = (DAILY[b] - DAILY[a]).dropna()
    hi = dw[dw.index.isin(hi_days)]
    lo = dw[~dw.index.isin(hi_days)]
    print(f' {a} to {b}: high-IV days {hi.mean():+.4f} (n={len(hi)})      '
          f'low-IV days {lo.mean():+.4f} (n={len(lo)})')

hours = np.array([t.hour + t.minute / 60 for t in test_d.index.time])
print('\nAppendix D Panel B, pooled AUC gain by session bucket')
for bname, msk in (('Open 09:30-11:00', (hours >= 9.5) & (hours < 11)),
                  ('Midday 11:00-14:00', (hours >= 11) & (hours < 14)),
                  ('Close 14:00-16:00', (hours >= 14) & (hours < 16))):
    y_b = y_test_d[msk]
    row = f' {bname} (n={msk.sum():,}): '
    for a, b in PAIRS:
        d = roc_auc_score(y_b, P_TE[b][msk]) - roc_auc_score(y_b, P_TE[a][msk])
        row += f'{a}-{b} {d:+.4f} '
    print(row)
```

Appendix D Panel A, within-day paired gain by day-level IV regime

M1 to M4:	high-IV days +0.0015 (n=41)	low-IV days +0.0001 (n=42)
M2 to M5:	high-IV days +0.0003 (n=41)	low-IV days -0.0009 (n=42)
M3 to M6:	high-IV days -0.0014 (n=41)	low-IV days -0.0036 (n=42)
M7 to M8:	high-IV days -0.0002 (n=41)	low-IV days -0.0028 (n=42)

Appendix D Panel B, pooled AUC gain by session bucket

Open 09:30-11:00 (n=44,820):	M1-M4 +0.0413	M2-M5 -0.0001	M3-M6 -0.0022	M7-M8 -0.0032
Midday 11:00-14:00 (n=89,279):	M1-M4 +0.0277	M2-M5 -0.0012	M3-M6 -0.0022	M7-M8 -0.0039
Close 14:00-16:00 (n=58,958):	M1-M4 +0.0405	M2-M5 -0.0004	M3-M6 +0.0008	M7-M8 -0.0032

12.6 Latent regime conditioning (Markov switching)

The IV-median split in 12.5 conditions on an observed variable. A natural objection is that the relevant volatility regime is latent, and that derivatives information might earn its keep only inside a properly identified stress state. To close that door, a two-state Gaussian Markov-switching model (switching mean and variance) is fitted to daily log realised volatility, and the test days are classified by the filtered (causal, one-sided) regime probability. The within-day information-set gains are then recomputed inside each latent regime.

The switching parameters are estimated on the full daily sample for stability (494 daily observations split across two regimes leave little room for a pure out-of-sample fit), so the regime labels carry a mild look-ahead through the parameters, though the filtering itself is one-sided. For a conditioning diagnostic this is acceptable and it biases toward finding regime pockets, making the null below stronger.

Reading the output: The filtered high-volatility state lines up closely with the tariff-shock weeks and with the high-IV day split, and the within-day derivatives gains stay at or below zero inside both latent regimes for every non-linear family.

```
In [26]: # two-state Markov switching on daily log realised volatility
from statsmodels.tsa.regime_switching.markov_regression import MarkovRegression

day_rv_all = (lob_target.groupby(lob_target['date'].astype(str))['mid']
               .apply(lambda s: s.pct_change().std() * 1e4))
log_rv = np.log(day_rv_all.clip(lower=1e-6))
ms_mod = MarkovRegression(log_rv.values, k_regimes=2, trend='c', switching_variance=True)
ms_res = ms_mod.fit(search_reps=20)
filt = ms_res.filtered_marginal_probabilities
hi_state = int(np.argmax([ms_res.params[ms_mod.param_names.index(f'const[{k}]]')
                          for k in range(2)]))
p_hi = pd.Series(filt[:, hi_state], index=log_rv.index)
print(f'High-volatility latent state: regime {hi_state}')
print(f'Days classified high-vol (filtered p > 0.5): {(p_hi > 0.5).sum()} of {len(p_hi)}')

test_days_set = set(uniq_days)
hi_latent = {d for d in test_days_set if p_hi.get(d, 0) > 0.5}
print(f'Test window: {len(hi_latent)} high-vol days, {len(test_days_set) - len(hi_latent)}
low-vol days')
agree = np.mean([(d in hi_latent) == (d in hi_days) for d in test_days_set])
print(f'Agreement with the IV-median day split: {100 * agree:.0f}%')

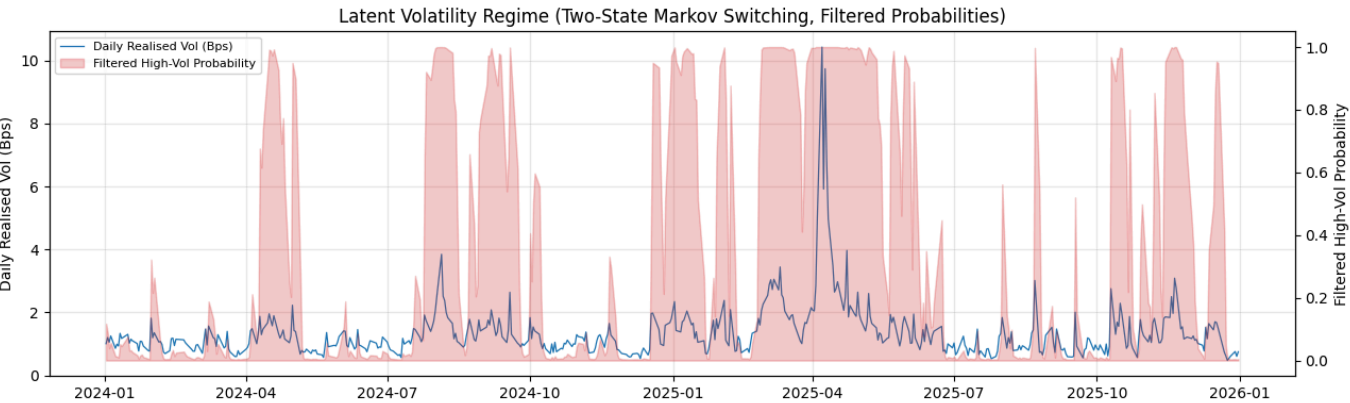
print('\nWithin-day information-set gains by latent regime')
for a, b in PAIRS:
    dw = (DAILY[b] - DAILY[a]).dropna()
    hi = dw[dw.index.isin(hi_latent)]
    lo_ = dw[~dw.index.isin(hi_latent)]
    print(f' {a} to {b}: high-vol {hi.mean():+.4f} (n={len(hi)})      '
          f'low-vol {lo_.mean():+.4f} (n={len(lo_)})')

fig, ax = plt.subplots(figsize=(13, 4))
x_days = pd.to_datetime(p_hi.index)
ax.plot(x_days, day_rv_all.values, color='tab:blue', lw=0.9, label='Daily Realised Vol
(Bps)')
ax2 = ax.twinx()
ax2.fill_between(x_days, 0, p_hi.values, color='tab:red', alpha=0.25,
                 label='Filtered High-Vol Probability')
ax.set_ylabel('Daily Realised Vol (Bps)')
ax2.set_ylabel('Filtered High-Vol Probability')
ax.set_title('Latent Volatility Regime (Two-State Markov Switching, Filtered
Probabilities)')
h1, l1 = ax.get_legend_handles_labels()
h2, l2 = ax2.get_legend_handles_labels()
ax.legend(h1 + h2, l1 + l2, fontsize=8, loc='upper left')
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'latent_regime.png'), dpi=150, bbox_inches='tight')
plt.show()
```

```
High-volatility latent state: regime 0
Days classified high-vol (filtered p > 0.5): 167 of 494
Test window: 24 high-vol days, 59 low-vol days
Agreement with the IV-median day split: 72%
```

Within-day information-set gains by latent regime

M1 to M4: high-vol +0.0015 (n=24) low-vol +0.0005 (n=59)
M2 to M5: high-vol +0.0004 (n=24) low-vol -0.0006 (n=59)
M3 to M6: high-vol -0.0017 (n=24) low-vol -0.0029 (n=59)
M7 to M8: high-vol -0.0006 (n=24) low-vol -0.0019 (n=59)



Section 13: Calibration and the Deployable Decision Rule

The raw probability outputs of the boosted models are poorly calibrated (compressed around the threshold), which makes fixed-threshold rules hard to interpret. This section fixes the decision layer and validates its economics with inference that respects the data structure.

Subsection	Question answered	Output
13.1	Are the probabilities trustworthy after calibration?	Table 6, reliability curves, probability histograms
13.2	What is the desk-ready operating point and is its gain real?	Table 7, day-block bootstrap CIs, threshold sweep

13.1 Probability calibration (Table 6)

The decision layer must not learn from predictions a model made on its own training data. The final models are refit on train plus validation, so their validation-window predictions are partially in sample and are unsuitable for fitting a calibration map or selecting a decision threshold. The economic tier fitted in Section 7 therefore uses a parallel set of models trained on the training window only, with the same tuned hyperparameters and train-only scalars. Their validation predictions are genuinely out of sample. The isotonic map for each model is fitted on those predictions and applied to the same model's test predictions. The calibration map, the decision threshold and the test evaluation therefore never see the validation window through a model that trained on it, and the calibrated and deployed predictions share one probability scale.

Quality is measured by the Brier score and the expected calibration error (ECE, 10 equal-width bins). Isotonic maps are monotone, so AUC is preserved up to ties. The headline predictive results of Sections 8 to 12 use the train-plus-validation final models on the untouched test window and are unaffected by this construction.

The calibrated probabilities concentrate near the test base rate. That is the honest statement of signal strength at 10 seconds. Discrimination is real but the implied probability shifts are modest, which is why the decision threshold must be chosen on economic grounds rather than at the classical 0.5.

```
In [27]: def ece(y_true, p, n_bins=10):
    bins = np.clip((p * n_bins).astype(int), 0, n_bins - 1)
    total = 0.0
    for b in range(n_bins):
        msk = bins == b
        if msk.sum():
            total += msk.mean() * abs(y_true[msk].mean() - p[msk].mean())
    return total

# isotonic maps fitted on the out-of-sample validation predictions of the
# economic-tier models and applied to the same models' test predictions
P_ISO, ISO, table6_rows = {}, {}, []
for m in ORDER8:
    iso = IsotonicRegression(out_of_bounds='clip').fit(P_VA_E[m], y_val_d)
    ISO[m] = iso
    P_ISO[m] = iso.predict(P_TE_E[m])
    table6_rows.append({
        'Model': NAME[m],
        'Brier Raw': brier_score_loss(y_test_d, P_TE_E[m]),
        'Brier Iso': brier_score_loss(y_test_d, P_ISO[m]),
        'ECE Raw': ece(y_test_d, P_TE_E[m]),
        'ECE Iso': ece(y_test_d, P_ISO[m]),
    })
table6 = pd.DataFrame(table6_rows).set_index('Model')
print('Table 6, Probability Calibration (Isotonic Fitted on Out-of-Sample Validation)')
print(table6.round(4).to_string())

fig, axes = plt.subplots(2, 4, figsize=(16, 7.5), sharex=True, sharey=True)
for ax, m in zip(axes.flat, ORDER8):
    for p, lab, c in ((P_TE_E[m], 'Raw', 'tab:blue'), (P_ISO[m], 'Isotonic', 'tab:red')):
        bins = np.clip((p * 10).astype(int), 0, 9)
        xs, ys = [], []
        for b in range(10):
            msk = bins == b
            if msk.sum() < 50:
                continue
            xs.append(p[msk].mean())
            ys.append(y_test_d[msk].mean())
        ax.plot(xs, ys, marker='o', ms=3, color=c, label=lab)
    ax.plot([0, 1], [0, 1], color='k', lw=0.8)
    ax.set_title(NAME[m], fontsize=10)
    ax.grid(alpha=0.3)
    ax.set_xlim(0, 1)
    ax.set_ylim(0, 1)
axes[0, 0].legend(fontsize=8)
fig.supxlabel('Predicted Probability')
fig.supylabel('Observed Frequency')
fig.suptitle('Reliability Curves, Test Set, All Models (Raw vs Isotonic Calibration)')
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'reliability_curves.png'), dpi=150, bbox_inches='tight')
plt.show()
```

```

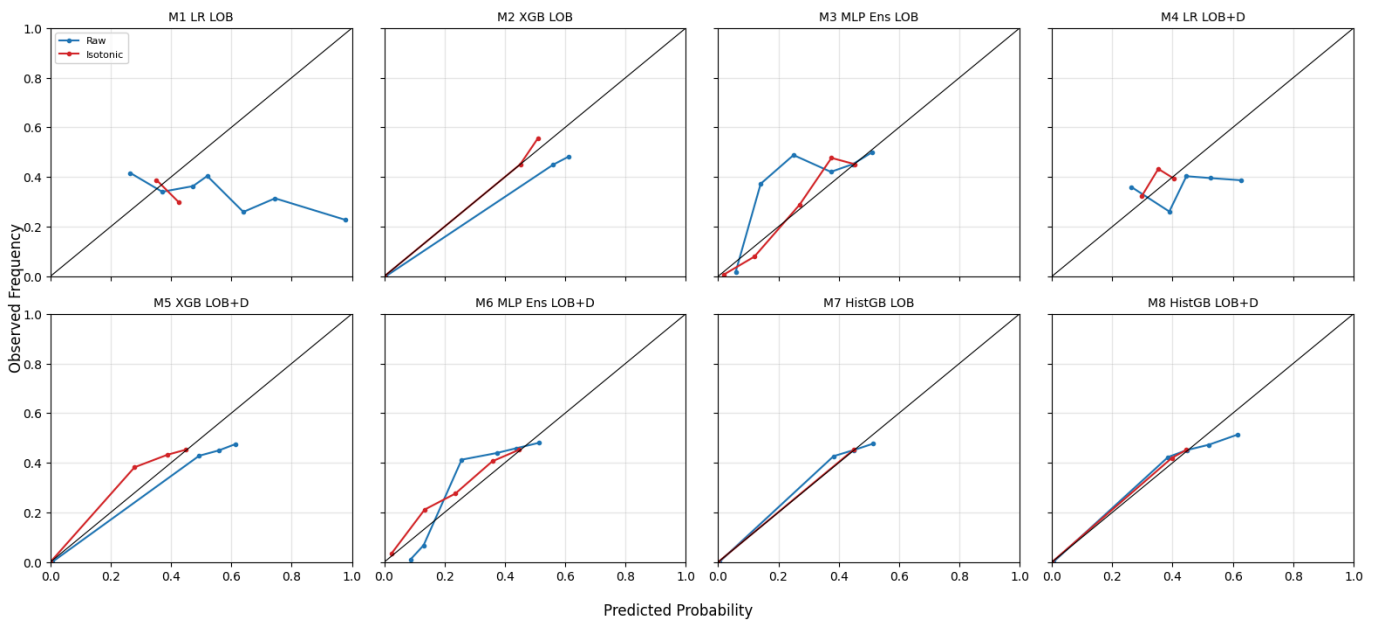
fig, axes = plt.subplots(2, 4, figsize=(16, 6), sharex=True)
for ax, m in zip(axes.flat, ORDER8):
    ax.hist(P_TE_E[m], bins=60, color='tab:blue', alpha=0.8)
    ax.set_title(NAME[m], fontsize=10)
    ax.grid(alpha=0.3)
fig.supxlabel('Predicted Probability')
fig.suptitle('Raw Probability Distributions, Test Set, All Models')
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'probability_histograms.png'), dpi=150,
bbox_inches='tight')
plt.show()

```

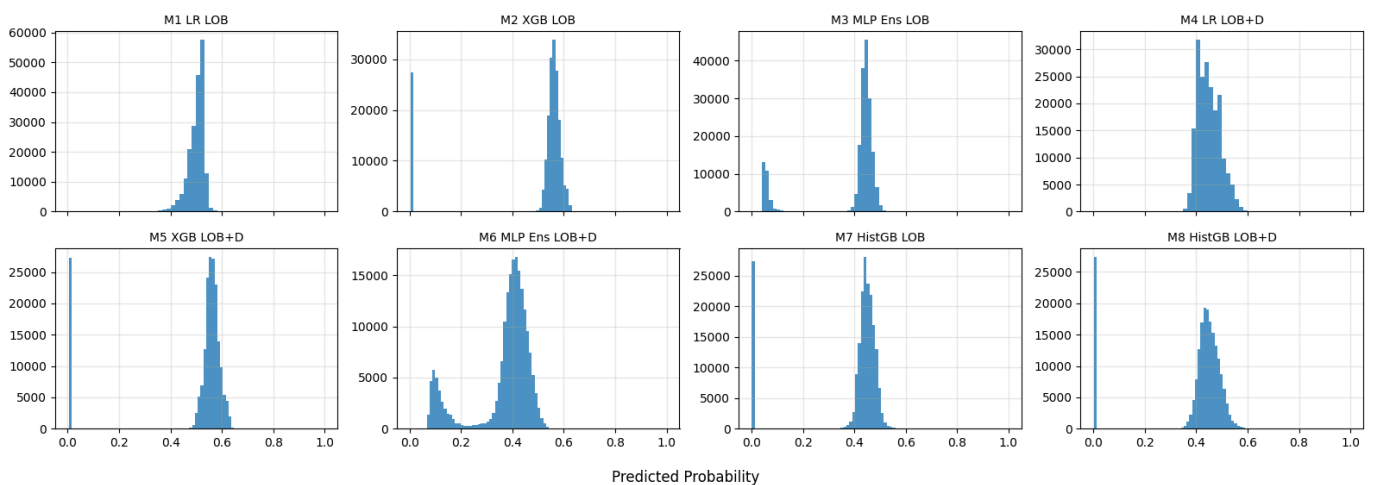
Table 6, Probability Calibration (Isotonic Fitted on Out-of-Sample Validation)

	Brier Raw	Brier Iso	ECE Raw	ECE Iso
Model				
M1 LR LOB	0.2505	0.2388	0.1134	0.0368
M2 XGB LOB	0.2230	0.2123	0.0964	0.0004
M3 MLP Ens LOB	0.2144	0.2135	0.0133	0.0033
M4 LR LOB+D	0.2402	0.2386	0.0631	0.0516
M5 XGB LOB+D	0.2232	0.2126	0.0959	0.0047
M6 MLP Ens LOB+D	0.2182	0.2153	0.0448	0.0074
M7 HistGB LOB	0.2124	0.2123	0.0035	0.0027
M8 HistGB LOB+D	0.2130	0.2125	0.0098	0.0055

Reliability Curves, Test Set, All Models (Raw vs Isotonic Calibration)



Raw Probability Distributions, Test Set, All Models



13.2 Deployable decision rule and day-block inference (Table 7)

The rule: Flag the next 10 seconds as high adverse-selection risk when the isotonic-calibrated probability is at or above τ^* . The threshold is selected on the calibrated out-of-sample validation predictions of the economic-tier models by maximising outlier-robust spread-capture gain subject to a flag rate of at most 60% on validation (a desk that quotes defensively most of the day forgoes the volume that pays for the system). The flag-rate cap binds at selection on validation and realised test coverage is allowed to differ, so the Flag Rate column reports the realised test value. Evaluation is on the outlier-robust test subset, excluding the first session minute and rows with spread above one dollar (Section 12.5 of the development history documented how a small set of wide synthetic-NBBO spreads otherwise dominates dollar aggregates).

Why day-block bootstrap: Observations within a day share the day's volatility and the same broadcast derivatives values, so an iid bootstrap understates variance. Days are resampled with replacement, 1,000 draws, identical draws across models so pairwise differences are valid.

Units disclaimer: All dollar figures here and downstream are improvements in the mean markout-based spread-capture proxy among the intervals a rule retains for quoting, relative to quoting every candidate interval. The proxy deducts the positive part of the adverse markout from the half-spread, so favourable subsequent price moves are not credited beyond full retention of the half-spread. These figures are not realised PnL. Fill probability, queue position, inventory, fees, rebates, hedging costs and latency are outside the locked scope and would all shrink the realised number.

Reading the output: M1 is degenerate. Its calibrated probabilities are nearly constant, so no admissible threshold exists on validation and the validation-quantile fallback flags essentially everything, leaving an uninterpretable sliver of unflagged intervals (its row is reported for completeness and excluded from comparisons). No threshold is ever read off the test distribution, so a rule that also degenerates on test is reported as unavailable rather than rescued. The Flag Rate column is the realised test-set coverage of each rule, and reading it alongside the gain column is essential. The gain statistic is confounded with selectivity, since retaining fewer intervals under an informative ranking tends to raise their conditional mean, and across the seven non-degenerate rules the gain is almost perfectly ordered by flag rate. The pairwise rows are therefore comparisons between rules sitting at different validation-selected operating points, and they confound information content with coverage rather than identifying an information-set effect. The day-block confidence intervals are valid for the difference at those fixed operating points and do not cover the additional uncertainty in where the threshold search lands.

```
In [28]: # robust subsets and spread-capture columns
def robust_frame(df):
    tod = df.index.time
    first_min = np.array([t.hour == 9 and t.minute == 30 for t in tod])
    rbm = (~first_min) & (df['spread'] <= 1.0).values
    out = df[rbm].copy()
    out['markout'] = (out['mid_change'] * out['quote_dir']).values
    # the proxy deducts the positive part of the adverse markout from the
    # half-spread, so favourable subsequent moves are not credited beyond
    # full retention of the half-spread
    out['spr_cap'] = out['half_spread'].values - np.clip(out['markout'].values, 0, None)
    return rbm, out
```

```

rb, tr_r = robust_frame(test_d)
rb_v, val_r = robust_frame(val_d)
print(f'Robust subsets: test {rb.sum():,}/{len(test_d):,} val
{rb_v.sum():,}/{len(val_d):,}')

def mm_stats(df, flagged):
    spr_cap = df['spr_cap'].values
    unfl = ~flagged
    if unfl.sum() == 0 or flagged.sum() == 0:
        return None
    return {'flag_rate': flagged.mean(),
            'd_spr_cap': spr_cap[unfl].mean() - spr_cap.mean()}

# threshold selection on calibrated OUT-OF-SAMPLE validation predictions
# (economic-tier models), applied once to the calibrated test predictions
P_VA_ISO = {m: ISO[m].predict(P_VA_E[m]) for m in ORDER8}
tau_star, rule = {}, {}
for m in ORDER8:
    pv = P_VA_ISO[m][rb_v]
    best_t, best_g = None, -np.inf
    for t in np.arange(0.05, 0.95, 0.01):
        flg = pv >= t
        if not (0.02 <= flg.mean() <= 0.60):
            continue
        s = mm_stats(val_r, flg)
        if s and s['d_spr_cap'] > best_g:
            best_g, best_t = s['d_spr_cap'], t
    if best_t is None:
        best_t = float(np.quantile(pv, 0.60))
        print(f'NOTE: {m} has no admissible threshold on validation, quantile fallback')
    tau_star[m] = best_t
    rule[m] = mm_stats(tr_r, P_ISO[m][rb] >= best_t)
    if rule[m] is None:
        # a rule that degenerates on test is reported as unavailable rather
        # than rescued with a threshold read off the test distribution
        tau_star[m] = np.nan
        print(f'NOTE: {m} degenerate on test at its validation threshold, no rule reported')

# day-block bootstrap, shared draws across models
days_r = tr_r['date'].astype(str).values
uniq_r = np.unique(days_r)
ridx = {d: np.where(days_r == d)[0] for d in uniq_r}
spr_cap_r = tr_r['spr_cap'].values
flags_r = {m: (P_ISO[m][rb] >= tau_star[m]) for m in ORDER8}
per_day = {}
for d in uniq_r:
    ix = ridx[d]
    st = {'n': len(ix), 'sum': spr_cap_r[ix].sum()}
    for m in ORDER8:
        u = ~flags_r[m][ix]
        st[f'nu_{m}'], st[f'su_{m}'] = u.sum(), spr_cap_r[ix][u].sum()
    per_day[d] = st
rngd = np.random.RandomState(42)
rule_draws = {m: np.empty(N_BOOT) for m in ORDER8}

```

```

for b in range(N_BOOT):
    smp = rngd.choice(uniq_r, len(uniq_r), replace=True)
    ntot = sum(per_day[d]['n'] for d in smp)
    ctot = sum(per_day[d]['sum'] for d in smp)
    for m in ORDER8:
        nu = sum(per_day[d][f'nu_{m}'] for d in smp)
        su = sum(per_day[d][f'su_{m}'] for d in smp)
        rule_draws[m][b] = (su / nu - ctot / ntot) if nu > 0 else np.nan

table7_rows = []
for m in ORDER8:
    lo, hi = np.nanpercentile(rule_draws[m], [2.5, 97.5])
    table7_rows.append({
        'Model': NAME[m],
        'Tau Star': round(tau_star[m], 2),
        'Flag Rate': round(rule[m]['flag_rate'], 3),
        'Gain USD': f"{rule[m]['d_spr_cap']:.5f}",
        'Day-Block CI': f'[{lo:.5f}, {hi:.5f}]',
    })
table7 = pd.DataFrame(table7_rows).set_index('Model')
print('Table 7, Deployable Decision Rule (Calibrated, Outlier-Robust Subset)')
print(table7.to_string())

print('\nPairwise derivatives gain in the decision layer (shared day draws)')
for a, b in (('M1', 'M4'), ('M2', 'M5'), ('M3', 'M6'), ('M7', 'M8')):
    diff = rule_draws[b] - rule_draws[a]
    lo, hi = np.nanpercentile(diff, [2.5, 97.5])
    pt = rule[b]['d_spr_cap'] - rule[a]['d_spr_cap']
    print(f' {a} to {b}: {pt:+.5f} CI [{lo:+.5f}, {hi:+.5f}] '
          f'Pr(gain>0)={(diff > 0).mean():.2f}')

# save the full prediction set for downstream sections and external checks
preds_out = test_d[['date', 'y']].copy()
for m in ORDER8:
    preds_out[f'p_{m}_raw'] = P_TE[m]
    preds_out[f'p_{m}_raw_econ'] = P_TE_E[m]
    preds_out[f'p_{m}_iso'] = P_ISO[m]
preds_out.to_parquet(os.path.join(PROC_DIR, 'preds_test_final.parquet'))
print('\nSaved preds_test_final.parquet (final-model raw, economic-tier raw and isotonic)')

```

Robust subsets: test 192,549/193,057 val 232,218/232,820

NOTE: M1 has no admissible threshold on validation, quantile fallback

Table 7, Deployable Decision Rule (Calibrated, Outlier-Robust Subset)

	Tau Star	Flag Rate	Gain USD	Day-Block CI
Model				
M1 LR LOB	0.35	1.000	-0.46470	[-1.66274, 0.18959]
M2 XGB LOB	0.44	0.494	0.00493	[0.00412, 0.00561]
M3 MLP Ens LOB	0.44	0.662	0.00899	[0.00783, 0.01000]
M4 LR LOB+D	0.33	0.475	0.00300	[0.00136, 0.00464]
M5 XGB LOB+D	0.43	0.679	0.00866	[0.00684, 0.01012]
M6 MLP Ens LOB+D	0.43	0.660	0.00928	[0.00816, 0.01020]
M7 HistGB LOB	0.44	0.529	0.00547	[0.00456, 0.00621]
M8 HistGB LOB+D	0.44	0.583	0.00600	[0.00468, 0.00705]

Pairwise derivatives gain in the decision layer (shared day draws)

M1 to M4: +0.46770 CI [-0.18621, +1.66444] Pr(gain>0)=0.81

M2 to M5: +0.00373 CI [+0.00240, +0.00494] Pr(gain>0)=1.00

M3 to M6: +0.00028 CI [-0.00057, +0.00123] Pr(gain>0)=0.75

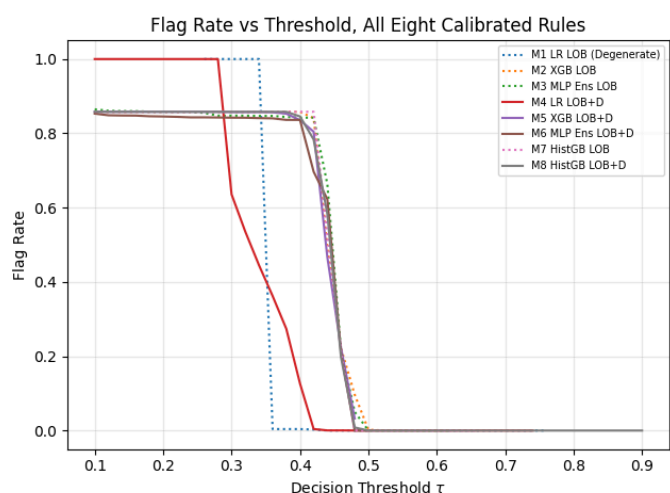
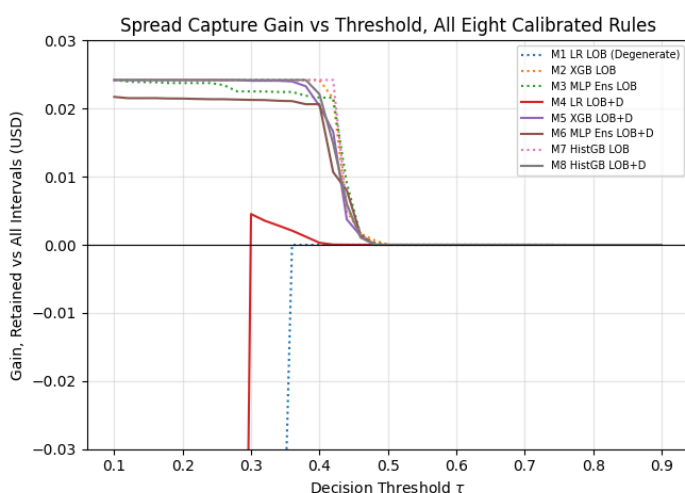
M7 to M8: +0.00053 CI [-0.00023, +0.00128] Pr(gain>0)=0.90

Saved preds_test_final.parquet (final-model raw, economic-tier raw and isotonic)

```
In [29]: # threshold sweep for all six calibrated rules
taus = np.round(np.arange(0.10, 0.91, 0.02), 2)

def mm_curve(p_iso):
    fl, dc = [], []
    for t in taus:
        s = mm_stats(tr_r, p_iso >= t)
        fl.append(s['flag_rate'] if s else np.nan)
        dc.append(s['d_spr_cap'] if s else np.nan)
    return np.array(fl), np.array(dc)

fig, axes = plt.subplots(1, 2, figsize=(13, 4.8))
for m in ORDER8:
    fl, dc = mm_curve(P_ISO[m][rb])
    lab = NAME[m] + (' (Degenerate)' if m == 'M1' else '')
    axes[0].plot(taus, dc, color=COLOR[m], ls=LS[m], label=lab)
    axes[1].plot(taus, fl, color=COLOR[m], ls=LS[m], label=lab)
axes[0].axhline(0, color='k', lw=0.8)
axes[0].set_ylim(-0.03, 0.03)
axes[0].set_xlabel(r'Decision Threshold  $\tau$ ')
axes[0].set_ylabel('Gain, Retained vs All Intervals (USD)')
axes[0].set_title('Spread Capture Gain vs Threshold, All Eight Calibrated Rules')
axes[1].set_xlabel(r'Decision Threshold  $\tau$ ')
axes[1].set_ylabel('Flag Rate')
axes[1].set_title('Flag Rate vs Threshold, All Eight Calibrated Rules')
for ax in axes:
    ax.legend(fontsize=7)
    ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'tau_sensitivity.png'), dpi=150, bbox_inches='tight')
plt.show()
```



Section 14: Economic Translation

This section adapts the useful subset of the stochastic-control market-making literature (Avellaneda and Stoikov, 2008, and the Gueant, Lehalle and Fernandez-Tapia line) to this project's empirical framework. The full control machinery (HJB value functions, optimal quoting surfaces, inventory tilting) does not transfer because this project has no inventory state and no fill-intensity model. Three analyses do transfer.

Subsection	Borrowed idea	Translation here
14.1	Backtest of the quoting rule	Day-by-day spread-capture improvement of each calibrated rule
14.2	Risk-sensitive quote widening	Break-even spread premium as a function of calibrated probability
14.3	Spread response to volatility	Spread, adverse rate and markout by IV quintile (Appendix F)

14.1 Daily backtest of the calibrated rules

For each test day, the spread-capture improvement of each rule over the flag-nothing baseline. This view equal-weights days, while Table 7 pools observations, and the two are different estimands. A rule can earn a larger pooled gain by being more active on benign days. Every non-degenerate rule accumulates steadily, and the ordering of the curves follows rule coverage rather than information set, so the day-weighted view inherits the same caveat as Table 7. The right panel differences rules that operate at different coverage levels, so its title names the model families rather than an identified derivatives effect and it is descriptive only. The M1 rule is degenerate and shown clipped for completeness.

```
In [30]: daily_bt = {}
for m in ORDER8:
    flagged = flags_r[m]
    g_all = tr_r.groupby(tr_r['date'].astype(str))['spr_cap'].mean()
    if (~flagged).sum() == 0:
        daily_bt[m] = pd.Series(dtype=float)
        continue
    g_un = tr_r[~flagged].groupby(tr_r['date'].astype(str)[~flagged])['spr_cap'].mean()
    daily_bt[m] = (g_un - g_all).dropna()
    print(f'{NAME[m]}: mean daily gain={daily_bt[m].mean():+.5f} USD   '
          f'hit rate={100 * (daily_bt[m] > 0).mean():.0f}%   '
          f'worst day={daily_bt[m].min():+.5f} USD')

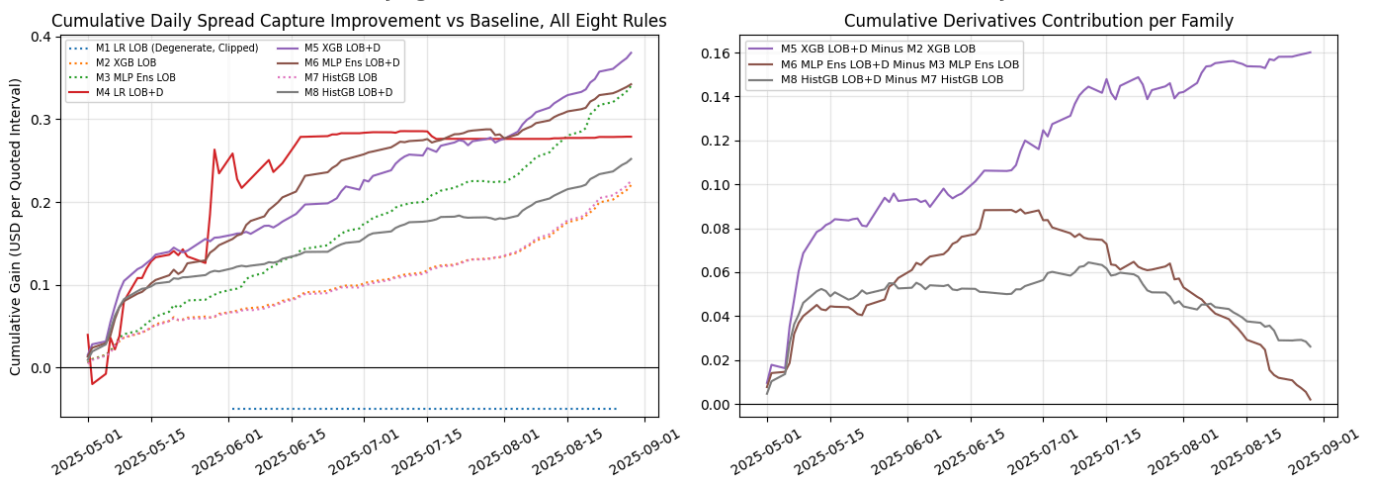
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
for m in ORDER8:
    if len(daily_bt[m]) == 0:
        continue
    lab = NAME[m]
    series = daily_bt[m].cumsum()
    if m == 'M1':
        lab += ' (Degenerate, Clipped)'
        series = series.clip(lower=-0.05)
    axes[0].plot(pd.to_datetime(daily_bt[m].index), series.values,
                  color=COLOR[m], ls=LS[m], label=lab)
```

```

axes[0].axhline(0, color='k', lw=0.8)
axes[0].set_ylim(-0.06, None)
axes[0].set_title('Cumulative Daily Spread Capture Improvement vs Baseline, All Eight Rules')
axes[0].set_ylabel('Cumulative Gain (USD per Quoted Interval)')
axes[0].legend(ncol=2, fontsize=7)
axes[0].grid(alpha=0.3)
for a, b in (('M2', 'M5'), ('M3', 'M6'), ('M7', 'M8')):
    idxu = daily_bt[a].index.union(daily_bt[b].index)
    diff = (daily_bt[b].reindex(idxu).fillna(0) - daily_bt[a].reindex(idxu).fillna(0))
    axes[1].plot(pd.to_datetime(idxu), diff.cumsum().values,
                 color=COLOR[b], label=f'{NAME[b]} Minus {NAME[a]}')
axes[1].axhline(0, color='k', lw=0.8)
axes[1].set_title('Cumulative Derivatives Contribution per Family')
axes[1].legend(fontsize=8)
axes[1].grid(alpha=0.3)
for ax in axes:
    ax.tick_params(axis='x', rotation=30)
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'backtest_cumulative.png'), dpi=150, bbox_inches='tight')
plt.show()

```

M1 LR LOB: mean daily gain=-0.55250 USD hit rate=44% worst day=-4.51917 USD
M2 XGB LOB: mean daily gain=+0.00265 USD hit rate=96% worst day=-0.00357 USD
M3 MLP Ens LOB: mean daily gain=+0.00410 USD hit rate=95% worst day=-0.00336 USD
M4 LR LOB+D: mean daily gain=+0.00344 USD hit rate=51% worst day=-0.05958 USD
M5 XGB LOB+D: mean daily gain=+0.00458 USD hit rate=84% worst day=-0.00597 USD
M6 MLP Ens LOB+D: mean daily gain=+0.00413 USD hit rate=93% worst day=-0.00692 USD
M7 HistGB LOB: mean daily gain=+0.00272 USD hit rate=94% worst day=-0.00204 USD
M8 HistGB LOB+D: mean daily gain=+0.00304 USD hit rate=89% worst day=-0.00201 USD

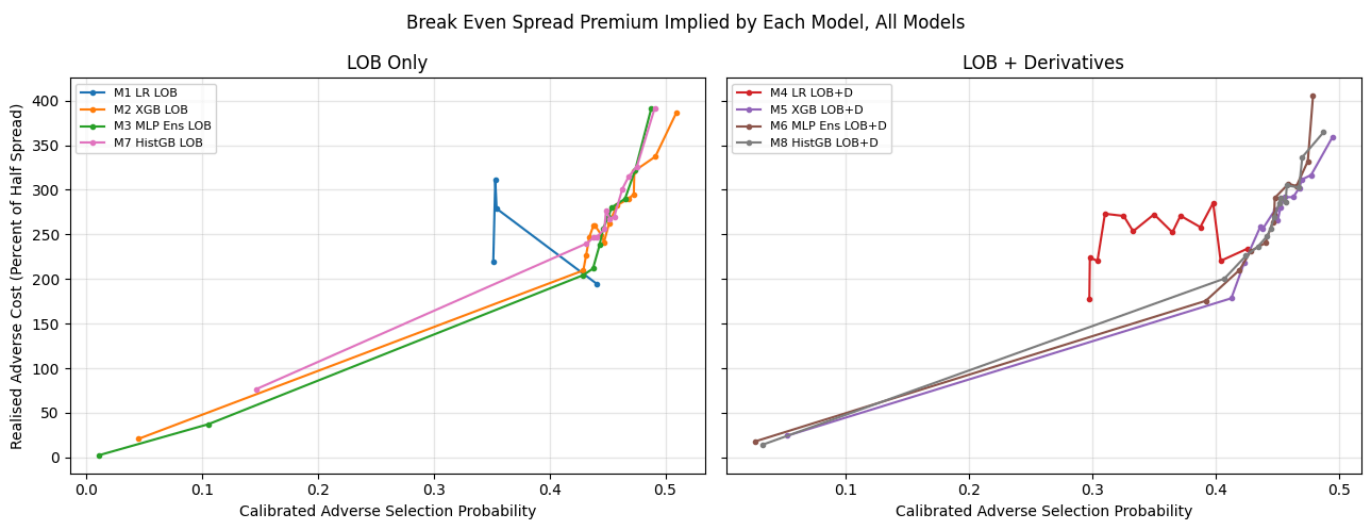


14.2 Break-even spread premium (the Avellaneda and Stoikov bridge)

In the Avellaneda and Stoikov objective, the quote distance from mid balances execution probability against expected adverse cost. The model supplies the missing empirical ingredient, the expected adverse cost conditional on the signal. Observations are binned by calibrated probability and the realised adverse cost per bin is expressed as a fraction of the prevailing half-spread, which is the break-even widening factor a quoting rule should apply. The premium shows a broadly increasing relationship with calibrated probability for every model with usable calibration. Quoting at the touch is roughly break-even when the model reads the environment as safe and costs a multiple of the half-spread when it

reads it as risky. This mapping from model output to a multiplicative spread adjustment is the most direct quoting prescription the analysis produces.

```
In [31]: pos_hs = tr_r['half_spread'].values > 0
cost_ratio = (np.clip(tr_r['markout'].values[pos_hs], 0, None)
              / tr_r['half_spread'].values[pos_hs])
fig, axes = plt.subplots(1, 2, figsize=(13, 5), sharey=True)
panels = [('LOB Only', ('M1', 'M2', 'M3', 'M7')),
          ('LOB + Derivatives', ('M4', 'M5', 'M6', 'M8'))]
for ax, (title, group) in zip(axes, panels):
    for m in group:
        p_iso = P_ISO[m][rb][pos_hs]
        bins = pd.qcut(p_iso, 20, labels=False, duplicates='drop')
        xs, ys = [], []
        for b in range(int(bins.max()) + 1):
            msk = bins == b
            xs.append(p_iso[msk].mean())
            ys.append(cost_ratio[msk].mean())
        ax.plot(xs, np.array(ys) * 100, marker='o', ms=3, color=COLOR[m], label=NAME[m])
    ax.set_title(title)
    ax.legend(fontsize=8)
    ax.grid(alpha=0.3)
    ax.set_xlabel('Calibrated Adverse Selection Probability')
axes[0].set_ylabel('Realised Adverse Cost (Percent of Half Spread)')
fig.suptitle('Break Even Spread Premium Implied by Each Model, All Models')
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'spread_premium_curve.png'), dpi=150, bbox_inches='tight')
plt.show()
```



14.3 Spread and adverse selection by IV quintile (Appendix F)

Adverse selection theory (Glosten and Milgrom, 1985) predicts that quoted spreads widen when informed-trading risk rises. The gradient across IV quintiles grounds the role of the IV level in standard theory. Mean absolute markout rises monotonically with IV and quoted spreads respond in the top quintiles. The adverse-event rate peaks in the middle quintiles because the event is defined relative to the half-spread, and in the top quintile spreads widen faster than prices move, which is the market already pricing the risk.

```
In [32]: y_r = y_test_d[rb]
q_iv = pd.qcut(tr_r['iv_atm_30'], 5, labels=False)
rows_iv = []
for qb in range(5):
    msk = (q_iv == qb).values
    rows_iv.append({
        'IV Quintile': qb + 1,
        'Mean IV ATM 30D': tr_r['iv_atm_30'].values[msk].mean(),
        'Mean Spread USD': tr_r['spread'].values[msk].mean(),
        'Adverse Rate': y_r[msk].mean(),
        'Mean Abs Markout Bps': (np.abs(tr_r['markout'].values[msk])
                                / tr_r['mid'].values[msk]).mean() * 1e4,
    })
print('Appendix F, Spread and Adverse Selection by IV Quintile')
print(pd.DataFrame(rows_iv).set_index('IV Quintile').round(4).to_string())
```

IV Quintile	Mean IV ATM 30D	Mean Spread USD	Adverse Rate	Mean Abs Markout Bps
1	0.1154	0.0223	0.2555	0.3751
2	0.1297	0.0176	0.4066	0.6447
3	0.1388	0.0179	0.4374	0.6715
4	0.1474	0.0235	0.4370	0.8813
5	0.1795	0.0350	0.4062	1.0008

14.4 Section 14 conclusions

1. Every non-degenerate calibrated rule improves the retained-interval spread-capture proxy, so the decision layer is worth running, but it cannot settle the information-set question. The measured gain is almost perfectly ordered by flag rate across rules, so pairwise differences between models at different validation-selected thresholds reflect coverage rather than information.
2. The model output maps directly into the Avellaneda and Stoikov quoting objective as a multiplicative spread-widening factor that rises broadly with calibrated probability.
3. The IV channel is confirmed in the cross-section. Absolute markouts rise monotonically with the IV level and quoted spreads respond, grounding the derivatives features in adverse-selection theory even though they carry no incremental intraday signal beyond the order book.
4. Stochastic-control extensions (inventory management, HJB-optimal quoting with the model-implied premium as input) are the natural next research step.

Section 15: The Validation Layer

The headline numbers in Sections 8 to 14 are the end point of a falsification chain that changed the project's answer twice. This section documents that chain, because the methodological lessons are part of the contribution, and re-runs the two falsification tests that remain informative on the corrected pipeline.

The chain of evidence:

1. **The naive result:** The first complete pipeline showed pooled AUC gains from derivatives of +0.008 (LR), +0.021 (XGB) and +0.038 (MLP), all formally significant under paired bootstrap and Diebold-

Mariano tests. An apparently clean yes.

2. **The placebo test:** Permuting the day-level derivatives mapping (each day receives another random day's true values, destroying all alignment) and retraining did not remove the gains. Information-free day-constant columns matched or beat the real features on pooled AUC. Mechanism. The test window has large day-to-day base-rate variation, pooled AUC rewards ranking days by base rate, and a day-constant column can let a flexible model do that. Day-constant columns also absorb day effects during training, freeing the LOB features for within-day structure.
3. **Seed fragility:** The apparent MLP derivatives gain reversed sign across initialisations. Under the corrected pipeline the per-seed within-day gains are -0.0031, -0.0061 and -0.0033, a spread of roughly a factor of two, so seed ensembling became mandatory (Section 7).
4. **The day-ID control and the scaler bug:** A control experiment (replacing derivatives with an arbitrary day identifier) produced inconsistent numbers and led to the discovery that every evaluation had trained final models on train+val-scaled features but scored them on train-only-scaled matrices. For tree models whose split thresholds live in scaled units this distorts everything. The LOB-only XGBoost moves from pooled AUC 0.529 to 0.636 once evaluation matrices match training matrices. The corrected protocol is the one used throughout this notebook.
5. **The clean answer:** On the corrected pipeline and half-day-cleaned data (Section 19), every non-linear within-day derivatives gain is null or negative, and the LR pooled gain is pure between-day placement.

Lessons: Consistent scaling between training and evaluation of final models. Pooled versus within-day metrics whenever day-constant features meet day-heterogeneous base rates. Seed ensembling for neural networks. Placebo permutation tests as routine practice.

15.1 The placebo test on the corrected pipeline

The placebo is re-run here under the corrected protocol. The derivatives day-mapping is permuted, the XGBoost LOB+Derivatives configuration is retrained from scratch (tuning frozen at the M5 selection for speed), and pooled and within-day AUC are evaluated consistently. Three permutations suffice to show the effect because the permutation variance is tiny.

Reading the output: Permuted, information-free derivatives reproduce the real M5's pooled AUC almost exactly, and its within-day AUC too. The placebo matches the real M5's performance closely, showing that this performance does not require correctly aligned derivatives information, and that pooled AUC could never have separated the two.

```
In [33]: # placebo, permute the day-level derivatives mapping and retrain M5's config
rng_pl = np.random.RandomState(7)
deriv_days = test_d['date'].astype(str).unique()
real_m5_pooled = roc_auc_score(y_test_d, P_TE['M5'])
real_m5_within = DAILY['M5'].mean()
print(f'Real M5: pooled={real_m5_pooled:.4f}  within-day={real_m5_within:.4f}\n')

# day-level derivatives table over the combined window
full_d = pd.concat([train_d, val_d, test_d])
day_key = full_d['date'].astype(str)
deriv_by_day = full_d.groupby(day_key, values)[DERIV_FEATURES].first()
```

```

for perm in range(3):
    shuffled = deriv_by_day.copy()
    shuffled[:] = deriv_by_day.sample(frac=1.0, random_state=perm).values
    plc = full_d.copy()
    plc[DERIV_FEATURES] = shuffled.reindex(day_key.values).values

    p_tr = plc[plc['date'].astype(str) <= TRAIN_END]
    p_va = plc[(plc['date'].astype(str) >= VAL_START) & (plc['date'].astype(str) <=
VAL_END)]
    p_te = plc[plc['date'].astype(str) >= TEST_START]
    p_tv = pd.concat([p_tr, p_va], ignore_index=True)

    sc_tv_p = StandardScaler().fit(p_tv[ALL_FEATURES])
    mp = XGBClassifier(**params5, subsample=0.8, n_estimators=ep5,
                        scale_pos_weight=sp_full, random_state=42, verbosity=0)
    mp.fit(sc_tv_p.transform(p_tv[ALL_FEATURES]), p_tv['y'].values)
    pp = mp.predict_proba(sc_tv_p.transform(p_te[ALL_FEATURES]))[:, 1]
    wd = daily_auc_series(p_te['y'].values, pp).mean()
    print(f'Placebo permutation {perm}: pooled={roc_auc_score(p_te["y"].values, pp):.4f} '
          f'within-day={wd:.4f}', flush=True)

```

Real M5: pooled=0.6318 within-day=0.6100

Placebo permutation 0: pooled=0.6307 within-day=0.6095

Placebo permutation 1: pooled=0.6327 within-day=0.6102

Placebo permutation 2: pooled=0.6318 within-day=0.6099

15.2 Seed dispersion of the MLPs

The per-seed test AUCs of the ensemble members, and the per-seed derivatives gain. The per-seed information-set gains vary by roughly a factor of two, which is why no single-seed MLP comparison is quotable and the shipped M3/M6 are three-seed averages.

```

In [34]: print('Per-seed pooled test AUC and per-seed M3 to M6 gain')
for s, a, b in zip(SEEDS, mlp3_models, mlp6_models):
    p3 = a.predict_proba(X_te_lob_p)[: , 1]
    p6 = b.predict_proba(X_te_full_f)[: , 1]
    auc3, auc6 = roc_auc_score(y_test_d, p3), roc_auc_score(y_test_d, p6)
    w3 = daily_auc_series(y_test_d, p3).mean()
    w6 = daily_auc_series(y_test_d, p6).mean()
    print(f' seed {s:3d}: M3 {auc3:.4f}/{w3:.4f} M6 {auc6:.4f}/{w6:.4f} '
          f'gain pooled {auc6 - auc3:+.4f} within {w6 - w3:+.4f}')

p3e, p6e = P_TE['M3'], P_TE['M6']
print(f' ensemble: M3 {roc_auc_score(y_test_d, p3e):.4f}/{DAILY["M3"].mean():.4f} '
      f'M6 {roc_auc_score(y_test_d, p6e):.4f}/{DAILY["M6"].mean():.4f} '
      f'gain within {(DAILY["M6"] - DAILY["M3"]).dropna().mean():+.4f}')

```

Per-seed pooled test AUC and per-seed M3 to M6 gain

seed 42:	M3 0.6288/0.6071	M6 0.6279/0.6040	gain pooled -0.0009 within -0.0031
seed 7:	M3 0.6276/0.6060	M6 0.6258/0.6000	gain pooled -0.0018 within -0.0061
seed 123:	M3 0.6286/0.6056	M6 0.6254/0.6023	gain pooled -0.0032 within -0.0033
ensemble:	M3 0.6298/0.6081	M6 0.6283/0.6055	gain within -0.0025

15.3 Joint pairwise comparison of all eight models

The joint bootstrap from Section 8 scores every model on the same resample draws, so all 28 pairwise pooled AUC differences have coherent confidence intervals. The matrix below reports the point difference (row minus column) with an asterisk where the 95% CI excludes zero. Those intervals resample individual observations rather than days, so the asterisks are retained as a legacy diagnostic and no information-set claim in the report rests on them. The boosted and ensembled models form a descriptive cluster in pooled point estimates that the linear models never approach, and the valid within-day information-set comparisons within that cluster are null or favour the LOB-only model.

```
In [35]: matrix = pd.DataFrame(index=[NAME[m] for m in ORDER8],
                                columns=[NAME[m] for m in ORDER8], dtype=object)
for i, a in enumerate(ORDER8):
    for j, b in enumerate(ORDER8):
        if i == j:
            matrix.iloc[i, j] = '.'
            continue
        d = auc_draws[:, i] - auc_draws[:, j]
        lo, hi = np.percentile(d, [2.5, 97.5])
        pt = roc_auc_score(y_test_d, P_TE[a]) - roc_auc_score(y_test_d, P_TE[b])
        star = '*' if lo > 0 or hi < 0 else ' '
        matrix.iloc[i, j] = f'{pt:+.4f}{star}'
print('Pairwise pooled AUC differences (row minus column, * = 95% CI excludes zero)')
print(matrix.to_string())
```

```
Pairwise pooled AUC differences (row minus column, * = 95% CI excludes zero)
      M1 LR LOB M2 XGB LOB M3 MLP Ens LOB M4 LR LOB+D M5 XGB LOB+D M6 MLP Ens LOB+D
M7 HistGB LOB M8 HistGB LOB+D
M1 LR LOB      .    -0.1172*    -0.1143*    -0.0369*    -0.1163*    -0.1127*
-0.1179*    -0.1138*
M2 XGB LOB      +0.1172*      .    +0.0029*    +0.0803*    +0.0009    +0.0044*
-0.0007    +0.0034*
M3 MLP Ens LOB  +0.1143*    -0.0029*      .    +0.0774*    -0.0020*    +0.0015*
-0.0036*    +0.0005
M4 LR LOB+D     +0.0369*    -0.0803*    -0.0774*      .    -0.0794*    -0.0758*
-0.0810*    -0.0769*
M5 XGB LOB+D     +0.1163*    -0.0009    +0.0020*    +0.0794*      .    +0.0036*
-0.0016*    +0.0025*
M6 MLP Ens LOB+D +0.1127*    -0.0044*    -0.0015*    +0.0758*    -0.0036*      .
-0.0051*    -0.0011
M7 HistGB LOB    +0.1179*    +0.0007    +0.0036*    +0.0810*    +0.0016*    +0.0051*
.    +0.0041*
M8 HistGB LOB+D  +0.1138*    -0.0034*    -0.0005    +0.0769*    -0.0025*    +0.0011
-0.0041*      .
```

15.4 Sample definition and activity-conditional evaluation

Two external-review objections are tested here. First, that crossed or abnormally wide quote states could distort labels and tree splits. The headline metrics are recomputed on test subsets excluding non-positive spreads and spreads above 25 cents and above one dollar. Second, and more substantively, that the target convention `y = 0` whenever a bin has no net directional signed flow hands every model a partly mechanical win, predicting *whether a bin will carry net directional signed flow at all* rather than whether

an active interval adversely selects. The headline metrics are recomputed on the active-flow subsample (`quote_dir != 0` , about 86% of test rows), and the XGBoost pair is additionally retrained from scratch on active rows only.

Reading the output: The spread filters change nothing (99.9% of rows survive and every AUC is identical to three decimals). The activity conditioning is the informative one. Within-day AUC drops from about 0.61 to roughly 0.50 to 0.52 across model classes, so a large share of the headline discrimination is exposure prediction, identifying the intervals without net directional signed flow that are mechanically safe, rather than direction-conditional adverse-selection ranking. Retraining on active rows does not recover the difference (the conditional problem is genuinely hard at 10 seconds with top-of-book features), and the derivatives gain conditional on activity stays negative after retraining without excluding zero. Three consequences. The headline AUC must be described as covering both exposure and conditional adverse selection. The safe-window identification skill is still economically real (the calibrated rules monetise it in Table 7). And the research-question answer is unchanged, since the derivatives null holds unconditionally and conditionally.

```
In [36]: # sample-definition robustness from the final predictions
def masked_eval(msk, label):
    print(f'{label}: n={msk.sum():,} ({100 * msk.mean():.1f}%), '
          f'base rate={y_test_d[msk].mean():.3f}')
    rows = {}
    for m in ORDER8:
        dsub = days_te[msk]
        wd = pd.Series({d: roc_auc_score(y_test_d[msk][dsub == d], P_TE[m][msk][dsub == d])
                        for d in np.unique(dsub)
                        if y_test_d[msk][dsub == d].min() != y_test_d[msk][dsub ==
d].max()})
        rows[m] = (roc_auc_score(y_test_d[msk], P_TE[m][msk]), wd)
        print(f' {NAME[m]:18s} pooled={rows[m][0]:.4f} within-day={wd.mean():.4f}')
        diff = (rows['M5'][1] - rows['M2'][1]).dropna()
        lo, hi = boot_mean_ci(diff.values)
        print(f' M2 to M5 within-day gain: {diff.mean():+.4f} [{lo:+.4f}, {hi:+.4f}] '
              f'days+={100 * (diff > 0).mean():.0f}%\n')

masked_eval(np.ones(len(test_d), bool), 'Full test sample')
masked_eval(((test_d['spread'] > 0) & (test_d['spread'] <= 0.25)).values,
            'Spread > 0 and <= $0.25')
masked_eval(((test_d['spread'] > 0) & (test_d['spread'] <= 1.00)).values,
            'Spread > 0 and <= $1.00')
masked_eval((test_d['quote_dir'] != 0).values,
            'Active flow only (quote_dir != 0)')
```

```
Full test sample: n=193,057 (100.0%), base rate=0.388
M1 LR LOB          pooled=0.5155 within-day=0.5050
M2 XGB LOB          pooled=0.6327 within-day=0.6103
M3 MLP Ens LOB      pooled=0.6298 within-day=0.6081
M4 LR LOB+D         pooled=0.5524 within-day=0.5059
M5 XGB LOB+D         pooled=0.6318 within-day=0.6100
M6 MLP Ens LOB+D     pooled=0.6283 within-day=0.6055
M7 HistGB LOB        pooled=0.6334 within-day=0.6109
M8 HistGB LOB+D      pooled=0.6293 within-day=0.6093
M2 to M5 within-day gain: -0.0003 [-0.0011, +0.0004] days+=45%
```



```

        if te_a['y'].values[days_a == d].min() !=
te_a['y'].values[days_a == d].max()))
    print(f'active-retrained {sname:6s}: pooled={roc_auc_score(te_a["y"].values, p):.4f} '
          f'within-day={DA_a[sname].mean():.4f} (trees={it}, n_test={len(te_a):,})',
flush=True)
diff_a = (DA_a['LOB+D'] - DA_a['LOB']).dropna()
lo, hi = boot_mean_ci(diff_a.values)
print(f'active-retrained derivatives gain: {diff_a.mean():+.4f} [{lo:+.4f}, {hi:+.4f}] '
      f'days+={100 * (diff_a > 0).mean():.0f}%')

```

active-retrained LOB : pooled=0.5222 within-day=0.5213 (trees=136, n_test=165,699)
active-retrained LOB+D : pooled=0.5187 within-day=0.5201 (trees=80, n_test=165,699)
active-retrained derivatives gain: -0.0011 [-0.0024, +0.0001] days+=41%

15.5 Label-threshold robustness

The committed target multiplier (0.5 times the half-spread) defines an early adverse-selection indicator. Here the target is rebuilt at the break-even multiplier (1.0) and the severe multiplier (2.0), and the XGBoost pair is retrained per threshold under the standard protocol.

Reading the output: At the break-even threshold the derivatives gain is not statistically distinguishable from zero. At the severe threshold it shows a small exploratory positive gain of +0.0021 with CI [+0.0010, +0.0034], consistent with severe adverse moves being more volatility-regime-driven and hence more visible to a day-level IV anchor. The gain arises in a post-hoc robustness test and is one fifth of a percent of AUC, so the headline null is robust to the label definition.

```

In [38]: # Label-threshold robustness, XGB pair retrained at tau_label = 1.0 and 2.0
for tau_label in (1.0, 2.0):
    fr_t = build_target(lob, tau=tau_label).dropna(subset=ALL_FEATURES)
    tr_t = date_slice(fr_t, TRAIN_START, TRAIN_END)
    va_t = date_slice(fr_t, VAL_START, VAL_END)
    te_t = date_slice(fr_t, TEST_START, TEST_END)
    tv_t = pd.concat([tr_t, va_t], ignore_index=True)
    sp_t = float((tr_t['y'] == 0).sum()) / max((tr_t['y'] == 1).sum(), 1)
    days_t = te_t['date'].astype(str).values
    DA_t = {}
    for sname, scols in (('LOB', LOB_FEATURES), ('LOB+D', ALL_FEATURES)):
        sc_tr = StandardScaler().fit(tr_t[scols])
        m = XGBClassifier(max_depth=3, learning_rate=0.05, subsample=0.8,
                          n_estimators=400, scale_pos_weight=sp_t, eval_metric='auc',
                          early_stopping_rounds=30, random_state=42, verbosity=0)
        m.fit(sc_tr.transform(tr_t[scols]), tr_t['y'].values,
              eval_set=[(sc_tr.transform(va_t[scols]), va_t['y'].values)], verbose=False)
        it = max(m.best_iteration + 1, 1)
        sc_tv = StandardScaler().fit(tv_t[scols])
        mf = XGBClassifier(max_depth=3, learning_rate=0.05, subsample=0.8,
                          n_estimators=it, scale_pos_weight=sp_t, random_state=42,
verbosity=0)
        mf.fit(sc_tv.transform(tv_t[scols]), tv_t['y'].values)
        p = mf.predict_proba(sc_tv.transform(te_t[scols]))[:, 1]
        DA_t[sname] = pd.Series({d: roc_auc_score(te_t['y'].values[days_t == d], p[days_t ==
d])

                                for d in np.unique(days_t)
                                if te_t['y'].values[days_t == d].min() !=
te_t['y'].values[days_t == d].max())})

```

```

print(f'tau_label={tau_label}: {sname:6s} pooled={roc_auc_score(te_t["y"].values,
p):.4f} '
      f'within-day={DA_t[sname].mean():.4f} (base rate {te_t["y"].mean():.3f})',
flush=True)
diff_t = (DA_t['LOB+D'] - DA_t['LOB']).dropna()
lo, hi = boot_mean_ci(diff_t.values)
print(f'tau_label={tau_label}: within-day derivatives gain {diff_t.mean():+.4f} '
      f'[{lo:+.4f}, {hi:+.4f}] days+={100 * (diff_t > 0).mean():.0f}%\n', flush=True)

```

```

tau_label=1.0: LOB      pooled=0.6337  within-day=0.6118  (base rate 0.365)
tau_label=1.0: LOB+D    pooled=0.6316  within-day=0.6125  (base rate 0.365)
tau_label=1.0: within-day derivatives gain +0.0007 [-0.0002, +0.0017]  days+=53%

tau_label=2.0: LOB      pooled=0.6542  within-day=0.6259  (base rate 0.313)
tau_label=2.0: LOB+D    pooled=0.6512  within-day=0.6280  (base rate 0.313)
tau_label=2.0: within-day derivatives gain +0.0021 [+0.0010, +0.0034]  days+=64%

```

Section 16: Final Corrected Results

Headline numbers (primary test window 2025-05-01 to 2025-08-29, half-day-cleaned data, consistent scaling): The cell below reproduces them from the saved prediction file to verify the tables without retraining.

Model	Pooled AUC	Within-Day AUC
M1 LR LOB	0.5155	0.5050
M2 XGB LOB	0.6327	0.6103
M3 MLP Ens LOB	0.6298	0.6081
M4 LR LOB+D	0.5524	0.5059
M5 XGB LOB+D	0.6318	0.6100
M6 MLP Ens LOB+D	0.6283	0.6055
M7 HistGB LOB	0.6334	0.6109
M8 HistGB LOB+D	0.6293	0.6093

The answer at the locked 10-second horizon:

Intraday signal. No. Every non-linear information-set gain is null or negative, pooled and within-day (M2 to M5 within-day -0.0003 with CI [-0.0011, +0.0004] and 45% of days positive, a textbook null, and M3 to M6 is significantly negative at -0.0025). The LR pooled gain (+0.0369) is entirely between-day placement, its within-day gain is +0.0008 and not significant. SHAP attributes about 5% of M5's output to all six derivatives features combined.

Day-level placement. Supported by the nested test, without economic confirmation. The decision layer of Section 13 cannot arbitrate the question. Its gain statistic is almost perfectly ordered by rule coverage, so the pairwise differences between models at different validation-selected thresholds measure selectivity rather than information. The positive evidence that remains is the forecasting channel of Section 17. EOD derivatives improve next-day adverse-selection forecasts out of sample, and Section 17.2 shows the

nested-model test rejects equal predictive accuracy on 209 evaluation days while the unadjusted test does not.

Out of sample (September to December 2025, LOB only). The non-linear LOB models generalise strongly (pooled AUC about 0.73), confirming the engine itself is sound and the test-window numbers are not overfit to the stressed test regime.

```
In [39]: # verification cell, reproduce the headline numbers from saved predictions
final_preds = pd.read_parquet(os.path.join(PROC_DIR, 'preds_test_final.parquet'))
y16 = final_preds['y'].values
d16 = final_preds['date'].astype(str).values
ud16 = np.unique(d16)
idx16 = {d: np.where(d16 == d)[0] for d in ud16}

def within_day_auc16(p):
    vals = [roc_auc_score(y16[idx16[d]], p[idx16[d]]) for d in ud16
            if y16[idx16[d]].min() != y16[idx16[d]].max()]
    return np.mean(vals)

print('Verification from preds_test_final.parquet')
print(f'{"Model":18s} {"Pooled":>8s} {"Within-Day":>11s}')
for m in ORDER8:
    p = final_preds[f'p_{m}_raw'].values
    print(f'{"NAME[m]:18s} {"roc_auc_score(y16, p):8.4f} {"within_day_auc16(p):11.4f}')
print('\nInformation-set gains (pooled / within-day)')
for a, b in PAIRS:
    pa, pb = final_preds[f'p_{a}_raw'].values, final_preds[f'p_{b}_raw'].values
    print(f'  {a} to {b}: {roc_auc_score(y16, pb) - roc_auc_score(y16, pa):+.4f} / '
          f'{within_day_auc16(pb) - within_day_auc16(pa):+.4f}')
```

Verification from preds_test_final.parquet

Model	Pooled	Within-Day
M1 LR LOB	0.5155	0.5050
M2 XGB LOB	0.6327	0.6103
M3 MLP Ens LOB	0.6298	0.6081
M4 LR LOB+D	0.5524	0.5059
M5 XGB LOB+D	0.6318	0.6100
M6 MLP Ens LOB+D	0.6283	0.6055
M7 HistGB LOB	0.6334	0.6109
M8 HistGB LOB+D	0.6293	0.6093

Information-set gains (pooled / within-day)

M1 to M4: +0.0369 / +0.0008
M2 to M5: -0.0009 / -0.0003
M3 to M6: -0.0015 / -0.0025
M7 to M8: -0.0041 / -0.0015

Section 17: The Fair Shot

Before accepting the 10-second null, derivatives information gets its best literature-motivated shot along the dimensions the locked design had not tested.

Dimension 1, option flow and positioning: The options-information literature locates the signal in trading activity rather than surface shape, citing option volume (Pan and Poteshman, 2006, Easley, O'Hara and Srinivas, 1998), put-call parity deviations (Cremers and Weinbaum, 2010) and relative option activity (Johnson and So, 2012). Five daily features are built from `opprcd` :

- the open-interest-weighted Cremers and Weinbaum call-minus-put IV spread over matched strike and expiry pairs
- the put-call volume ratio
- log total SPX option volume
- the ODTE volume share, the defining feature of the modern regime, averaging about half of SPX volume in this sample
- the day-over-day change in log open interest

Dimension 2, horizon: End-of-day information should matter more as the prediction horizon approaches the daily scale. The adverse-selection target is rebuilt at 10 seconds, 1 minute, 5 minutes and 30 minutes, and XGBoost is retrained per horizon under three feature sets, LOB only, LOB plus the six surface features, and LOB plus all eleven derivatives features.

Dimension 3, the day level itself: Next-day adverse-selection rate and next-day realised volatility are forecast from EOD predictors with expanding-window out-of-sample evaluation, the natural home of options information per the volatility-forecasting literature. The desk-relevant control is whether lagged day aggregates of the desk's own LOB data forecast just as well, in which case the feed would be redundant even at the day level.

```
In [40]: # extended option-flow features from opprcd (resume-safe, loads if already built)
EXT_PATH = os.path.join(OM_PROC, 'deriv_features_extended.parquet')
DERIV_EXT = ['cw_spread', 'pc_vol_ratio', 'log_opt_vol', 'dte0_share', 'd_log_oi']
DERIV11 = DERIV_FEATURES + DERIV_EXT

if os.path.exists(EXT_PATH):
    deriv_all_ext = pd.read_parquet(EXT_PATH)
    deriv_all_ext.index = deriv_all_ext.index.astype(str)
    print(f'Loaded extended features: {deriv_all_ext.shape}')
else:
    cols17 = ['secid', 'date', 'exdate', 'cp_flag', 'strike_price', 'volume',
              'open_interest', 'impl_volatility', 'delta', 'best_bid']
    op17 = pd.concat([
        pd.read_parquet(os.path.join(OM_DIR, 'opprcd2024.parquet'), columns=cols17),
        pd.read_parquet(os.path.join(OM_DIR, 'opprcd2025.parquet'), columns=cols17),
    ], ignore_index=True)
    op17 = op17[op17['secid'] == SPX_SECID].copy()
    op17['date'] = pd.to_datetime(op17['date']).dt.date.astype(str)
    op17['exdate'] = pd.to_datetime(op17['exdate']).dt.date.astype(str)

    g17 = op17.groupby('date')
    vol_tot = g17['volume'].sum()
    vol_put = op17[op17['cp_flag'] == 'P'].groupby('date')['volume'].sum()
    vol_call = op17[op17['cp_flag'] == 'C'].groupby('date')['volume'].sum()
    oi_tot = g17['open_interest'].sum()
    dte0 = op17[op17['exdate'] == op17['date']].groupby('date')['volume'].sum()
    ext = pd.DataFrame({
        'log_opt_vol': np.log(vol_tot.clip(lower=1)),
```

```

        'pc_vol_ratio': (vol_put / vol_call.clip(lower=1)).clip(0, 5),
        'dte0_share': (dte0 / vol_tot.clip(lower=1)).fillna(0.0),
        'd_log_oi': np.log(oi_tot.clip(lower=1)).diff(),
    })
    liq = op17[(op17['impl_volatility'].notna()) & (op17['best_bid'] > 0)
               & (op17['delta'].abs().between(0.20, 0.80))]
    calls = liq[liq['cp_flag'] == 'C'][['date', 'exdate', 'strike_price',
                                         'impl_volatility', 'open_interest']]
    puts = liq[liq['cp_flag'] == 'P'][['date', 'exdate', 'strike_price',
                                         'impl_volatility', 'open_interest']]
    pairs17 = calls.merge(puts, on=['date', 'exdate', 'strike_price'], suffixes=('_c',
                                         '_p'))
    pairs17['w'] = (pairs17['open_interest_c'] + pairs17['open_interest_p']).clip(lower=1)
    pairs17['ivdiff_w'] = (pairs17['impl_volatility_c'] - pairs17['impl_volatility_p']) *
pairs17['w']
    ext['cw_spread'] = pairs17.groupby('date').apply(lambda d: d['ivdiff_w'].sum() /
d['w'].sum())
    ext = ext.dropna()
    deriv6_17 = pd.read_parquet(os.path.join(OM_PROC, 'deriv_features.parquet'))
    deriv6_17.index = deriv6_17.index.astype(str)
    deriv_all_ext = deriv6_17.join(ext, how='inner')
    deriv_all_ext.to_parquet(EXT_PATH)
    print(f'Built and saved extended features: {deriv_all_ext.shape}')

print(f'0DTE share of SPX volume, sample mean: {deriv_all_ext["dte0_share"].mean():.2f}')
print(deriv_all_ext[DERIV_EXT].describe().round(4).to_string())

```

Built and saved extended features: (416, 11)

0DTE share of SPX volume, sample mean: 0.53

	cw_spread	pc_vol_ratio	log_opt_vol	dte0_share	d_log_oi
count	416.0000	416.0000	416.0000	416.0000	416.0000
mean	0.0022	1.3195	14.9975	0.5266	0.0005
std	0.0059	0.1422	0.1689	0.0706	0.0377
min	-0.0151	1.0381	14.5316	0.2647	-0.3006
25%	-0.0017	1.2337	14.8793	0.4782	0.0016
50%	0.0019	1.3127	14.9899	0.5202	0.0096
75%	0.0063	1.3864	15.1174	0.5737	0.0140
max	0.0490	2.7114	15.6108	0.6902	0.0346

```

In [41]: # horizon sweep, XGBoost per horizon and feature set, consistent scaling
sweep = lob.copy()
sweep['date_str'] = sweep['date'].astype(str)
sweep['om_date'] = sweep['date_str'].map(om_lookup)
sweep = sweep.join(deriv_all_ext[DERIV_EXT].rename_axis('om_date'), on='om_date',
how='left')
sweep = sweep.drop(columns=['om_date'])
sweep['quote_dir'] = np.sign(sweep['signed_flow'])

HORIZONS17 = [(1, '10s'), (6, '1m'), (30, '5m'), (180, '30m')]
FEATSETS17 = [('LOB', LOB_FEATURES),
              ('LOB+D6', LOB_FEATURES + DERIV_FEATURES),
              ('LOB+D11', LOB_FEATURES + DERIV11)]

def slice_ds(df, s, e):
    return df[(df['date_str'] >= s) & (df['date_str'] <= e)]

```

```

def fit_xgb_consistent(train_f, val_f, tv_f, test_f, cols, spw):
    sc_tr = StandardScaler().fit(train_f[cols])
    m = XGBClassifier(max_depth=3, learning_rate=0.05, subsample=0.8,
                      n_estimators=400, scale_pos_weight=spw, eval_metric='auc',
                      early_stopping_rounds=30, random_state=42, verbosity=0)
    m.fit(sc_tr.transform(train_f[cols]), train_f['y'].values,
          eval_set=[(sc_tr.transform(val_f[cols]), val_f['y'].values)], verbose=False)
    best_it = max(m.best_iteration + 1, 1)
    sc_tv = StandardScaler().fit(tv_f[cols])
    mf = XGBClassifier(max_depth=3, learning_rate=0.05, subsample=0.8,
                      n_estimators=best_it, scale_pos_weight=spw,
                      random_state=42, verbosity=0)
    mf.fit(sc_tv.transform(tv_f[cols]), tv_f['y'].values)
    return mf.predict_proba(sc_tv.transform(test_f[cols]))[:, 1], best_it

def daily_auc_generic(y, p, days):
    out = {}
    for d in np.unique(days):
        msk = days == d
        if y[msk].min() == y[msk].max():
            continue
        out[d] = roc_auc_score(y[msk], p[msk])
    return pd.Series(out)

def gain_ci(diff_series, seed=42):
    diff = diff_series.dropna().values
    rng_g = np.random.RandomState(seed)
    means = np.array([diff[rng_g.randint(0, len(diff), len(diff))].mean()
                      for _ in range(2000)])
    lo, hi = np.percentile(means, [2.5, 97.5])
    return diff.mean(), lo, hi, (diff > 0).mean()

HG = {}
for h, hname in HORIZONS17:
    print(f'\nHorizon {hname} (h={h} steps)', flush=True)
    fr = sweep.copy()
    fr['mid_h'] = fr.groupby('date_str')['mid'].shift(-h)
    fr['y'] = np.where(
        fr['quote_dir'] != 0,
        ((fr['mid_h'] - fr['mid']) * fr['quote_dir'] >= 0.5 *
fr['half_spread']).astype(int), 0)
    fr = fr.dropna(subset=['mid_h'] + DERIV11)
    tr_f, va_f = slice_ds(fr, TRAIN_START, TRAIN_END), slice_ds(fr, VAL_START, VAL_END)
    te_f = slice_ds(fr, TEST_START, TEST_END)
    tv_f = pd.concat([tr_f, va_f], ignore_index=True)
    spw = float((tr_f['y'] == 0).sum()) / max((tr_f['y'] == 1).sum(), 1)
    days17 = te_f['date_str'].values
    DA17 = {}
    for sname, scols in FEATSETS17:
        p, it = fit_xgb_consistent(tr_f, va_f, tv_f, te_f, scols, spw)
        DA17[sname] = daily_auc_generic(te_f['y'].values, p, days17)
        print(f' {sname:8s}: pooled={roc_auc_score(te_f["y"].values, p):.4f} '
              f'within-day={DA17[sname].mean():.4f} (trees={it})', flush=True)
    HG[hname] = {}
    for sname in ('LOB+D6', 'LOB+D11'):

```



```

m_, lo, hi, frac = gain_ci(DA17[sname] - DA17['LOB'])
HG[hname][sname] = (m_, lo, hi)
print(f'  within-day gain {sname} vs LOB: {m_:+.4f} [{lo:+.4f}, {hi:+.4f}] '
      f'days+={100 * frac:.0f}%', flush=True)

```

Horizon 10s (h=1 steps)

```

LOB      : pooled=0.6330  within-day=0.6105  (trees=164)
LOB+D6   : pooled=0.6308  within-day=0.6095  (trees=113)
LOB+D11  : pooled=0.6303  within-day=0.6088  (trees=107)
within-day gain LOB+D6 vs LOB: -0.0010 [-0.0019, -0.0002] days+=40%
within-day gain LOB+D11 vs LOB: -0.0017 [-0.0027, -0.0008] days+=36%

```

Horizon 1m (h=6 steps)

```

LOB      : pooled=0.6336  within-day=0.6096  (trees=22)
LOB+D6   : pooled=0.6338  within-day=0.6101  (trees=29)
LOB+D11  : pooled=0.6334  within-day=0.6106  (trees=26)
within-day gain LOB+D6 vs LOB: +0.0005 [-0.0001, +0.0011] days+=53%
within-day gain LOB+D11 vs LOB: +0.0010 [+0.0001, +0.0020] days+=59%

```

Horizon 5m (h=30 steps)

```

LOB      : pooled=0.6434  within-day=0.6176  (trees=55)
LOB+D6   : pooled=0.6459  within-day=0.6204  (trees=13)
LOB+D11  : pooled=0.6416  within-day=0.6149  (trees=12)
within-day gain LOB+D6 vs LOB: +0.0028 [+0.0000, +0.0056] days+=54%
within-day gain LOB+D11 vs LOB: -0.0028 [-0.0119, +0.0052] days+=53%

```

Horizon 30m (h=180 steps)

```

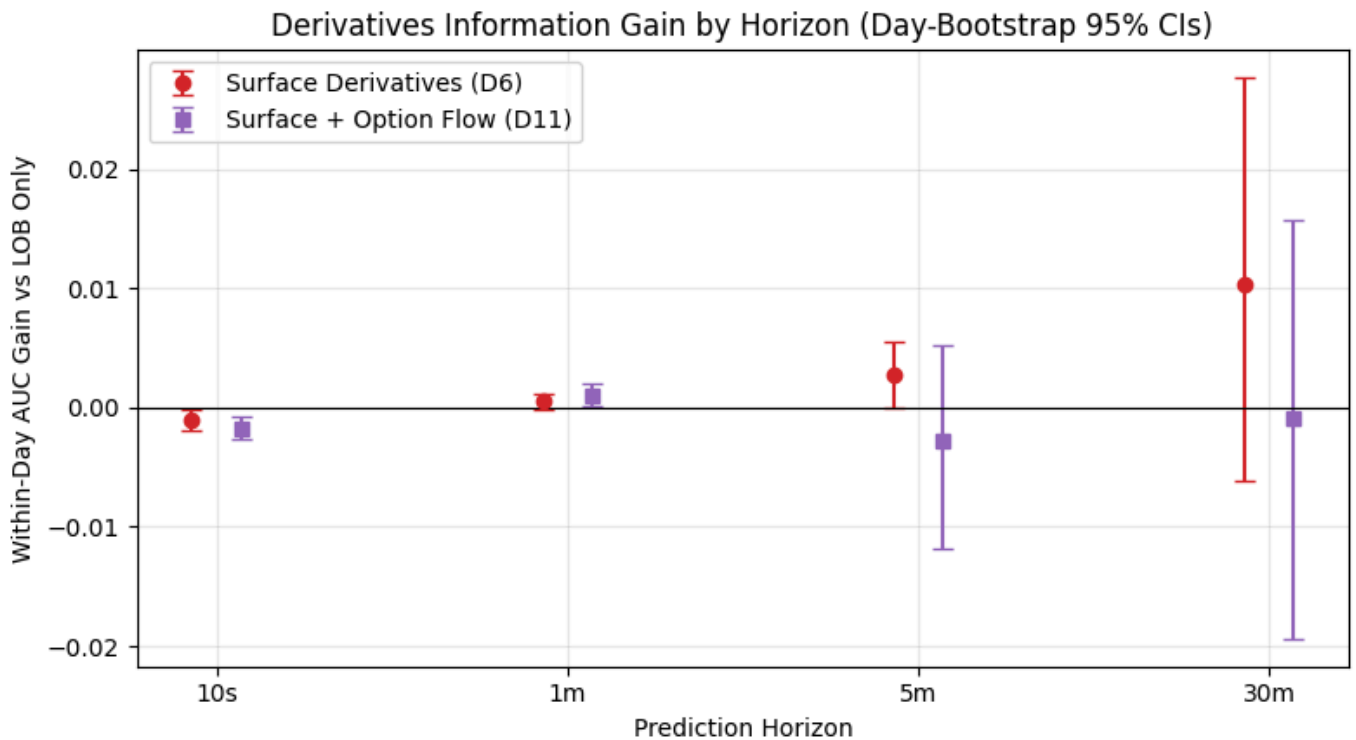
LOB      : pooled=0.6574  within-day=0.6294  (trees=9)
LOB+D6   : pooled=0.6622  within-day=0.6398  (trees=26)
LOB+D11  : pooled=0.6532  within-day=0.6285  (trees=31)
within-day gain LOB+D6 vs LOB: +0.0104 [-0.0062, +0.0276] days+=64%
within-day gain LOB+D11 vs LOB: -0.0009 [-0.0193, +0.0157] days+=63%

```

```

In [42]: # the horizon story as a picture
hor_names = [hn for _, hn in HORIZONS17]
x17 = np.arange(len(hor_names))
fig, ax = plt.subplots(figsize=(8, 4.5))
for sname, off, c, mk, lab in (('LOB+D6', -0.07, 'tab:red', 'o', 'Surface Derivatives
(D6)'),
                               ('LOB+D11', 0.07, 'tab:purple', 's', 'Surface + Option Flow
(D11)')):
    vals = [HG[hn][sname][0] for hn in hor_names]
    los = [HG[hn][sname][1] for hn in hor_names]
    his = [HG[hn][sname][2] for hn in hor_names]
    ax.errorbar(x17 + off, vals,
                yerr=[np.array(vals) - np.array(los), np.array(his) - np.array(vals)],
                fmt=mk, color=c, capsize=4, label=lab)
ax.axhline(0, color='k', lw=0.8)
ax.set_xticks(x17)
ax.set_xticklabels(hor_names)
ax.set_xlabel('Prediction Horizon')
ax.set_ylabel('Within-Day AUC Gain vs LOB Only')
ax.set_title('Derivatives Information Gain by Horizon (Day-Bootstrap 95% CIs)')
ax.legend()
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'horizon_gains.png'), dpi=150, bbox_inches='tight')
plt.show()

```

```
In [43]: # day-level forecasting, expanding-window out-of-sample
from sklearn.linear_model import LinearRegression

fr1 = sweep.copy()
fr1['mid_h'] = fr1.groupby('date_str')['mid'].shift(-1)
fr1['y'] = np.where(
    fr1['quote_dir'] != 0,
    ((fr1['mid_h'] - fr1['mid']) * fr1['quote_dir']) >= 0.5 *
fr1['half_spread']).astype(int), 0)
fr1 = fr1.dropna(subset=['mid_h'])
day_stats = fr1.groupby('date_str').agg(
    adv_rate=('y', 'mean'),
    d_spread=('spread', 'mean'),
    d_depth=('depth', 'mean'),
    d_absflow=('signed_flow', lambda s: s.abs().mean()),
)
day_stats['rv_bps'] = fr1.groupby('date_str')['mid'].apply(lambda s: s.pct_change().std() *
1e4)

deriv_for_day = fr1.groupby('date_str')[DERIV11].first()
lob_lag = day_stats[['d_spread', 'd_depth', 'd_absflow', 'rv_bps']].shift(1)
lob_lag.columns = ['lag_' + c for c in lob_lag.columns]
X_day_all = deriv_for_day.join(lob_lag).dropna()
Y_day_all = day_stats.loc[X_day_all.index, ['adv_rate', 'rv_bps']]
print(f'days: {len(X_day_all)} (expanding OOS from day 200)')

def oos_r2(X, y, start=200):
    # benchmark is the expanding historical mean, recomputed each step,
    # which is the standard out-of-sample R2 convention
    preds, actual, bench = [], [], []
    for i in range(start, len(X)):
        reg = LinearRegression().fit(X.iloc[:i], y.iloc[:i])
        preds.append(reg.predict(X.iloc[i:i + 1])[0])
```

```

        actual.append(y.iloc[i])
        bench.append(y.iloc[:i].mean())
    preds, actual, bench = np.array(preds), np.array(actual), np.array(bench)
    return 1 - ((actual - preds) ** 2).sum() / ((actual - bench) ** 2).sum()

lag_cols = [c for c in X_day_all.columns if c.startswith('lag_')]
SETS_DAY = {
    'LOB day lags only': lag_cols,
    'Surface derivatives (6)': DERIV_FEATURES,
    'Extended derivatives (5)': DERIV_EXT,
    'All derivatives (11)': DERIV11,
    'LOB lags + all derivatives': DERIV11 + lag_cols,
}
for target in ('adv_rate', 'rv_bps'):
    print(f'\nTarget: next-day {target}')
    for sname, scols in SETS_DAY.items():
        print(f' {sname:28s}: OOS R2 = {oos_r2(X_day_all[scols], Y_day_all[target]):+.3f}')

```

days: 409 (expanding OOS from day 200)

Target: next-day adv_rate

LOB day lags only	: OOS R2 = +0.296
Surface derivatives (6)	: OOS R2 = +0.129
Extended derivatives (5)	: OOS R2 = +0.018
All derivatives (11)	: OOS R2 = +0.168
LOB lags + all derivatives	: OOS R2 = +0.444

Target: next-day rv_bps

LOB day lags only	: OOS R2 = +0.571
Surface derivatives (6)	: OOS R2 = +0.594
Extended derivatives (5)	: OOS R2 = +0.256
All derivatives (11)	: OOS R2 = +0.605
LOB lags + all derivatives	: OOS R2 = +0.587

17.1 Fair-shot results

Option flow at 10 seconds: nothing. The eleven-feature set behaves like the six-feature set, and both are significantly negative at this horizon. The Pan and Poteshman flow channel does not rescue the intraday case.

Horizon sweep, no stable intraday increment at any tested horizon: both sets are significantly negative at 10 seconds, and at the longer horizons the within-day gains shift around zero. The only positive estimate whose CI excludes zero is tiny (D11 at one minute) and it does not replicate across the adjacent feature set, which is not stable enough across adjacent horizons and feature sets to treat as confirmatory.

Day level, the meaningful case: derivatives improve both day-level targets. For next-day adverse-selection conditions, the thing the desk actually needs for threshold placement, they add a material point-estimate improvement in out-of-sample power beyond the desk's own history (OOS R-squared roughly 0.30 to 0.44 when added to the LOB lags), and Section 17.2 tests it. For next-day realised volatility they also beat the desk's own lagged LOB day aggregates, 0.605 for the derivatives block alone against 0.571, so the day-level value is not confined to adverse-selection conditions. The decision layer

of Table 7 cannot corroborate this forecasting edge, since its gain statistic tracks rule coverage, so the day-level case for the feed rests on the forecasting evidence alone.

17.2 Inference on the day-level gain

The point improvement in next-day adverse-rate forecasting (OOS R-squared roughly 0.30 with the desk's own day lags alone, roughly 0.44 adding the derivatives) is the basis of the day-level claim, so it deserves the same inferential discipline as the intraday null. The gain is checked three ways, with a Diebold-Mariano test on the daily squared forecast errors (Newey-West, 5 lags), a moving-block bootstrap of the mean squared-error improvement (block length 10 days), and a leave-group-out attribution identifying which feature group carries the gain. The cumulative squared-error improvement path shows where in the evaluation window the gain accrues. A fourth check addresses the first of these. The comparison is nested, since the augmented model adds the derivatives features to the same day lags, and a test on the raw squared-error difference is undersized in that setting because the larger model carries estimation noise even when the added coefficients are zero in population. The MSPE-adjusted statistic of Clark and West (2007) removes that penalty and is the appropriate test here.

A ridge-regularised version of the same expanding forecast (RidgeCV over a wide penalty grid) is run alongside as an overfitting control, since 15 predictors on an expanding window starting at 200 days leaves room for plain least squares to chase noise.

Reading the output: The attribution is clean. Dropping the six surface features removes essentially the whole gain, while dropping the five option-flow features actually improves the forecast, so the day-level value lives in the surface, consistent with every other section. On 209 evaluation days the unadjusted DM statistic is +1.57 ($p = 0.115$) and the block-bootstrap CI narrowly includes zero, because the improvement concentrates in a handful of stress transitions. The Clark-West statistic, which is the one appropriate for a nested comparison, is +2.04 with a one-sided p of 0.021 and rejects equal predictive accuracy. The difference is consistent with the estimation-noise correction the adjustment makes. The Clark-West result supports incremental predictive content in the augmented forecast, which the leave-group-out attribution locates in the surface variables, while the raw comparison at this sample size cannot show that the augmented model lowers realised forecast error. The decision-layer translation of Table 7 still cannot corroborate the channel, since its gain statistic is ordered by rule coverage rather than by information set, so the economic conversion remains unvalidated.

```
In [44]: # inference on the day-level forecasting gain
Y_adv = Y_day_all['adv_rate']

def expanding_errors(cols, start=200):
    errs, dates_ = [], []
    for i in range(start, len(X_day_all)):
        reg = LinearRegression().fit(X_day_all[cols].iloc[:i], Y_adv.iloc[:i])
        errs.append(Y_adv.iloc[i] - reg.predict(X_day_all[cols].iloc[i:i + 1])[0])
        dates_.append(X_day_all.index[i])
    return pd.Series(errs, index=dates_)

e_lob = expanding_errors(lag_cols)
e_all = expanding_errors(lag_cols + DERIV11)
# expanding historical-mean benchmark, matching oos_r2 above
```

```

bench_adv = np.array([Y_adv.iloc[:i].mean() for i in range(200, len(X_day_all))])
denom = ((Y_adv.iloc[200:].values - bench_adv) ** 2).sum()
print(f'OOS R2, next-day adverse rate: LOB lags {1 - (e_lob ** 2).sum() / denom:+.3f} '
      f'LOB lags + derivatives {1 - (e_all ** 2).sum() / denom:+.3f}')

d_se = e_lob.values ** 2 - e_all.values ** 2
dbar, n_d = d_se.mean(), len(d_se)
dc = d_se - dbar
s_var = (dc ** 2).mean()
for l in range(1, 6):
    s_var += 2 * (1 - l / 6) * (dc[l:] * dc[:-l]).mean()
dm_day = dbar / np.sqrt(s_var / n_d)
print(f'DM on squared day-level forecast errors (NW lag 5): {dm_day:+.2f} '
      f'p={2 * (1 - sps.norm.cdf(abs(dm_day))):.3f}')

rng_d = np.random.RandomState(42)
L = 10
boots = []
for _ in range(2000):
    starts = rng_d.randint(0, n_d - L + 1, int(np.ceil(n_d / L)))
    idx = np.concatenate([np.arange(s_, s_ + L) for s_ in starts])[n_d:]
    boots.append(d_se[idx].mean())
lo, hi = np.percentile(boots, [2.5, 97.5])
print(f'Block-bootstrap (L=10) mean squared-error improvement: {dbar:.6f} [{lo:.6f}, {hi:.6f}']

# Clark-West MSPE adjustment. The comparison is nested, so the raw squared-error
# difference is undersized: the larger model carries estimation noise even when the
# added coefficients are zero in population. Clark and West (2007) add that noise back.
def expanding_preds(cols, start=200):
    ps = []
    for i in range(start, len(X_day_all)):
        reg = LinearRegression().fit(X_day_all[cols].iloc[:i], Y_adv.iloc[:i])
        ps.append(reg.predict(X_day_all[cols].iloc[i:i + 1])[0])
    return np.array(ps)

f_lob = expanding_preds(lag_cols)
f_all = expanding_preds(lag_cols + DERIV11)
cw = e_lob.values ** 2 - (e_all.values ** 2 - (f_lob - f_all) ** 2)
cwbar = cw.mean()
cwc = cw - cwbar
s_cw = (cwc ** 2).mean()
for l in range(1, 6):
    s_cw += 2 * (1 - l / 6) * (cwc[l:] * cwc[:-l]).mean()
t_cw = cwbar / np.sqrt(s_cw / len(cw))
print(f'Clark-West MSPE-adjusted statistic (NW lag 5): {t_cw:+.2f} '
      f'one-sided p={1 - sps.norm.cdf(t_cw):.3f}')

print('\nLeave-group-out attribution (OOS R2 of the full set minus a group)')
full_cols = lag_cols + DERIV11
for gname, drop in (('none (full set)', []), ('surface derivatives (6)', DERIV_FEATURES),
                   ('option flow (5)', DERIV_EXT), ('all derivatives (11)', DERIV11),
                   ('LOB day lags', lag_cols)):
    cols_g = [c for c in full_cols if c not in drop]
    e_g = expanding_errors(cols_g)
    print(f' drop {gname:26s}: OOS R2 = {1 - (e_g ** 2).sum() / denom:+.3f}')

fig, ax = plt.subplots(figsize=(9, 4.2))

```

```

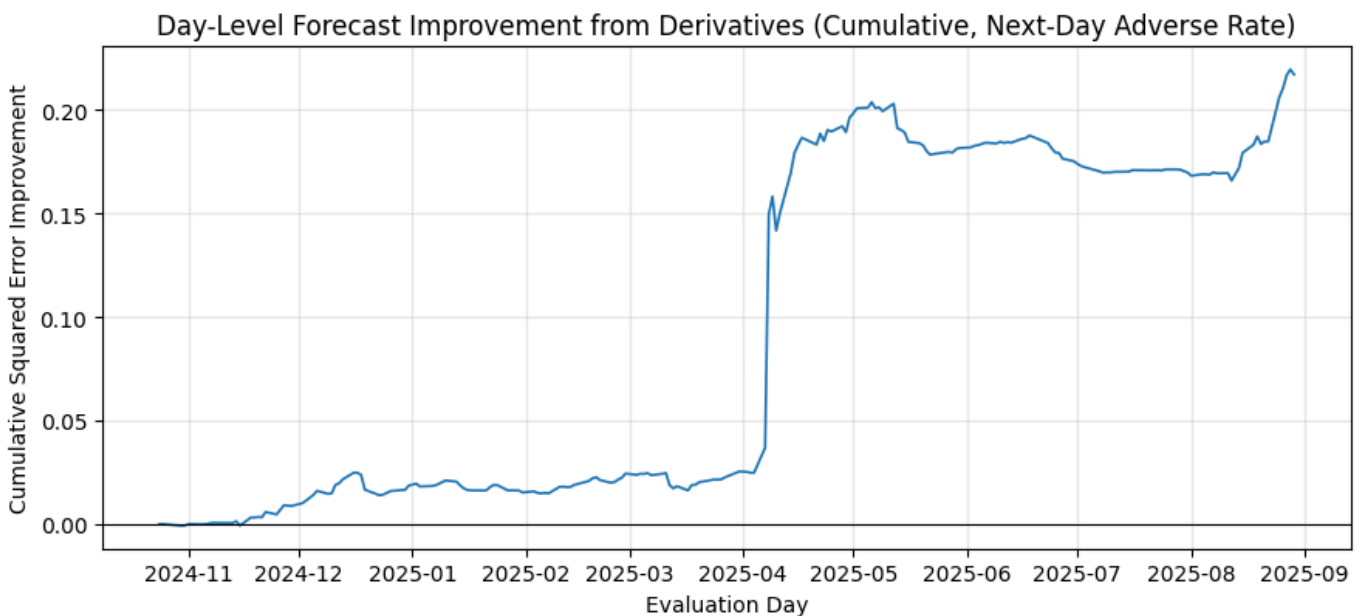
ax.plot(pd.to_datetime(e_lob.index), np.cumsum(d_se), color='tab:blue', lw=1.2)
ax.axhline(0, color='k', lw=0.8)
ax.set_xlabel('Evaluation Day')
ax.set_ylabel('Cumulative Squared Error Improvement')
ax.set_title('Day-Level Forecast Improvement from Derivatives (Cumulative, Next-Day Adverse Rate)')
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig(os.path.join(FIG_DIR, 'day_level_value.png'), dpi=150, bbox_inches='tight')
plt.show()

```

OOS R2, next-day adverse rate: LOB lags +0.296 LOB lags + derivatives +0.444
DM on squared day-level forecast errors (NW lag 5): +1.57 p=0.115
Block-bootstrap (L=10) mean squared-error improvement: 0.001038 [-0.000072, 0.002675]
Clark-West MSPE-adjusted statistic (NW lag 5): +2.04 one-sided p=0.021

Leave-group-out attribution (OOS R2 of the full set minus a group)

drop none (full set)	: OOS R2 = +0.444
drop surface derivatives (6)	: OOS R2 = +0.293
drop option flow (5)	: OOS R2 = +0.458
drop all derivatives (11)	: OOS R2 = +0.296
drop LOB day lags	: OOS R2 = +0.168



In [45]:

```

# ridge control for the day-level forecast (overfitting check)
from sklearn.linear_model import RidgeCV
from sklearn.pipeline import make_pipeline

alphas = np.logspace(-4, 4, 25)

def ridge_oos_r2(cols, start=200):
    preds, actual = [], []
    for i in range(start, len(X_day_all)):
        # standardise inside each window, otherwise the ridge penalty
        # depends on the very different units of the predictors
        reg = make_pipeline(StandardScaler(), RidgeCV(alphas=alphas)).fit(
            X_day_all[cols].iloc[:i], Y_adv.iloc[:i])
        preds.append(reg.predict(X_day_all[cols].iloc[i:i + 1])[0])
        actual.append(Y_adv.iloc[i])
    preds, actual = np.array(preds), np.array(actual)
    return 1 - ((actual - preds) ** 2).sum() / denom

```

```
print('Ridge expanding OOS R2, next-day adverse rate')
for sname, scols in (('LOB day lags', lag_cols),
                    ('Surface derivatives (6)', DERIV_FEATURES),
                    ('LOB lags + all derivatives', lag_cols + DERIV11)):
    print(f' {sname:28s}: {ridge_oos_r2(scols):+.3f}')
```

```
Ridge expanding OOS R2, next-day adverse rate
LOB day lags           : +0.330
Surface derivatives (6) : +0.126
LOB lags + all derivatives : +0.457
```

Section 18: EOD Mitigation and the Attribution Control

The core design limitation is that OptionMetrics is end-of-day, so every derivatives feature so far is day-constant. This section addresses the limitation directly, then runs the control that keeps the result honest.

Mitigation 1, making the EOD anchor intraday: The feature `vrp_run` is the previous-day ATM IV minus the running annualised realised volatility of the session so far. It varies within the day, is fully causal (the running volatility needs at least six bins, so the first roughly 50 seconds of each session carry no value and those rows are excluded rather than backfilled), and represents the unfolding gap between implied and realised risk, an intraday variance-risk-premium proxy.

Mitigation 2, using everything OptionMetrics has: The previously unused parameters are the 25-delta butterfly (smile curvature), the long-tenor term slope, near-ATM theta, and an open-interest-weighted gamma proxy `gex` (gamma times open interest, calls minus puts), a simplified form of the positioning measure used in the ODTE literature. With these, every informative parameter set in `vsurfd` and `opprcd` has been used.

Retired diagnostic: The day-level fit printed below joins these four features by their own end-of-day date rather than through the previous-trading-day map `om_lookup` used everywhere else in the project, so the day `T` surface is paired with the day `T` outcome. The resulting R2 values are same-day fits, not causal next-day forecasts, and they are not used in the report. The day-level result the project does report is the correctly lagged one in Section 17.

The attribution control: `vrp_run` contains a pure LOB quantity, the running session volatility. If `LOB + rv_run` matches `LOB + vrp_run`, the apparent derivatives gain is just a better LOB volatility feature. The control compares `LOB + rv_run` (no options data), `LOB + vrp_run`, and `LOB + rv_run + iv` at 10 seconds, 30 minutes and 2 hours.

```
In [46]: # new surface and Greeks features, and their day-level forecasting power
vs18 = pd.concat([
    pd.read_parquet(os.path.join(OM_DIR, 'vsurfd2024.parquet')),
    pd.read_parquet(os.path.join(OM_DIR, 'vsurfd2025.parquet')),
], ignore_index=True)
vs18 = vs18[vs18['secid'] == SPX_SECID].copy()
vs18['date'] = pd.to_datetime(vs18['date']).dt.date.astype(str)
vs18 = vs18[vs18['dispersion'] <= disp_95] # same training-window cutoff as Section 3
long_tenor = max(vs18['days'].unique())
```

```

def surf_point(days_, delta_, cp=None):
    m = vs18[(vs18['days'] == days_) & (vs18['delta'] == delta_)]
    if cp:
        m = m[m['cp_flag'] == cp]
    return m.set_index('date')['impl_volatility']

new_surf = pd.DataFrame({
    'bf25_30': (surf_point(30, -25) + surf_point(30, 25)) / 2 - surf_point(30, 50, 'C'),
    'ts_long': surf_point(30, 50, 'C') - surf_point(long_tenor, 50, 'C'),
})

cols18 = ['secid', 'date', 'cp_flag', 'open_interest', 'impl_volatility',
          'delta', 'gamma', 'theta', 'best_bid']
op18 = pd.concat([
    pd.read_parquet(os.path.join(OM_DIR, 'opprcd2024.parquet'), columns=cols18),
    pd.read_parquet(os.path.join(OM_DIR, 'opprcd2025.parquet'), columns=cols18),
], ignore_index=True)
op18 = op18[(op18['secid'] == SPX_SECID) & op18['impl_volatility'].notna()
            & (op18['best_bid'] > 0)].copy()
op18['date'] = pd.to_datetime(op18['date']).dt.date.astype(str)
new_surf['theta_atm'] = op18[op18['delta'].abs().between(0.40, 0.60)].groupby('date')
['theta'].mean()
liq18 = op18[op18['open_interest'] > 0]
gex_c = liq18[liq18['cp_flag'] == 'C'].groupby('date').apply(
    lambda d: (d['gamma'] * d['open_interest']).sum())
gex_p = liq18[liq18['cp_flag'] == 'P'].groupby('date').apply(
    lambda d: (d['gamma'] * d['open_interest']).sum())
new_surf['gex'] = (gex_c - gex_p) / 1e3
new_surf = new_surf.dropna()
NEW_SURF = ['bf25_30', 'ts_long', 'theta_atm', 'gex']
print(f'New features: {new_surf.shape} (long tenor used: {long_tenor} days)')

# day-level forecasting with the new features
X_day18 = X_day_all.join(new_surf.rename_axis('date_str'), how='inner').dropna()
Y_day18 = Y_day_all.loc[X_day18.index]
for tgt in ('adv_rate', 'rv_bps'):
    print(f'\nTarget: next-day {tgt}')
    for sname, scols in (('Surface derivatives (6)', DERIV_FEATURES),
                        ('New features (4, incl. GEX)', NEW_SURF),
                        ('All surface + Greeks (10)', DERIV_FEATURES + NEW_SURF)):
        print(f' {sname:28s}: OOS R2 = {oos_r2(X_day18[scols], Y_day18[tgt]):+.3f}')

```

New features: (417, 4) (long tenor used: 730.0 days)

Target: next-day adv_rate

Surface derivatives (6)	: OOS R2 = +0.129
New features (4, incl. GEX)	: OOS R2 = +0.030
All surface + Greeks (10)	: OOS R2 = +0.113

Target: next-day rv_bps

Surface derivatives (6)	: OOS R2 = +0.594
New features (4, incl. GEX)	: OOS R2 = +0.650
All surface + Greeks (10)	: OOS R2 = +0.630

In [47]: *# the VRP attribution control at 10s, 30m and 2h*
ctrl = lob.copy()


```

ctrl['date_str'] = ctrl['date'].astype(str)
ctrl['quote_dir'] = np.sign(ctrl['signed_flow'])
ret18 = ctrl.groupby('date_str')['mid'].pct_change().fillna(0.0)
ctrl['_r'], ctrl['_r2'] = ret18, ret18 ** 2
cs_r = ctrl.groupby('date_str')['_r'].cumsum()
cs_r2 = ctrl.groupby('date_str')['_r2'].cumsum()
n_obs = ctrl.groupby('date_str').cumcount() + 1
var_run = (cs_r2 - cs_r ** 2 / n_obs) / np.maximum(n_obs - 1, 1)
rv_series = np.sqrt(np.maximum(var_run, 0)) * np.sqrt(23400 / 10 * 252)
rv_series[n_obs < 6] = np.nan
# no backfill: the first five bins of each day have no running vol and are
# dropped by the dropna calls downstream, keeping the feature strictly causal
ctrl['rv_run'] = rv_series
ctrl['vrp_run'] = ctrl['iv_atm_30'] - ctrl['rv_run']
ctrl = ctrl.drop(columns=['_r', '_r2'])

SETS18 = [('LOB', LOB_FEATURES),
          ('LOB+rv_run', LOB_FEATURES + ['rv_run']),
          ('LOB+vrp', LOB_FEATURES + ['vrp_run']),
          ('LOB+rv_run+iv', LOB_FEATURES + ['rv_run', 'iv_atm_30'])]

for h, hname in [(1, '10s'), (180, '30m'), (720, '2h')]:
    print(f'\nHorizon {hname}', flush=True)
    fr = ctrl.copy()
    fr['mid_h'] = fr.groupby('date_str')['mid'].shift(-h)
    fr['y'] = np.where(
        fr['quote_dir'] != 0,
        ((fr['mid_h'] - fr['mid']) * fr['quote_dir'] >= 0.5 *
fr['half_spread']).astype(int), 0)
    fr = fr.dropna(subset=['mid_h', 'vrp_run', 'iv_atm_30'])
    tr_f, va_f = slice_ds(fr, TRAIN_START, TRAIN_END), slice_ds(fr, VAL_START, VAL_END)
    te_f = slice_ds(fr, TEST_START, TEST_END)
    tv_f = pd.concat([tr_f, va_f], ignore_index=True)
    spw = float((tr_f['y'] == 0).sum()) / max((tr_f['y'] == 1).sum(), 1)
    days18 = te_f['date_str'].values
    DA18 = {}
    for sname, scols in SETS18:
        p, _ = fit_xgb_consistent(tr_f, va_f, tv_f, te_f, scols, spw)
        DA18[sname] = daily_auc_generic(te_f['y'].values, p, days18)
        print(f' {sname:14s}: pooled={roc_auc_score(te_f["y"].values, p):.4f} '
              f'within-day={DA18[sname].mean():.4f}', flush=True)
    for a, b, label in (('LOB', 'LOB+rv_run', 'rv_run alone (pure LOB)'),
                        ('LOB', 'LOB+vrp', 'vrp (IV minus rv_run)'),
                        ('LOB+rv_run', 'LOB+vrp', 'IV increment beyond rv_run'),
                        ('LOB+rv_run', 'LOB+rv_run+iv', 'adding iv as a column')):
        m_, lo, hi, frac = gain_ci(DA18[b] - DA18[a])
        print(f' {label:28s}: {m_:+.4f} [{lo:+.4f}, {hi:+.4f}] days+={100 * frac:.0f}%',
flush=True)

```

Horizon 10s

LOB	: pooled=0.6330	within-day=0.6104	
LOB+rv_run	: pooled=0.6365	within-day=0.6121	
LOB+vrp	: pooled=0.6351	within-day=0.6120	
LOB+rv_run+iv	: pooled=0.6365	within-day=0.6120	
rv_run alone (pure LOB)	: +0.0016	[+0.0005, +0.0025]	days+=71%
vrp (IV minus rv_run)	: +0.0015	[+0.0007, +0.0024]	days+=67%
IV increment beyond rv_run	: -0.0001	[-0.0010, +0.0009]	days+=49%
adding iv as a column	: -0.0000	[-0.0002, +0.0001]	days+=52%

Horizon 30m

LOB	:	pooled=0.6587	within-day=0.6335	
LOB+rv_run	:	pooled=0.6597	within-day=0.6361	
LOB+vrp	:	pooled=0.6648	within-day=0.6450	
LOB+rv_run+iv	:	pooled=0.6573	within-day=0.6348	
rv_run alone (pure LOB)	:	+0.0026	[-0.0003, +0.0055]	days+=58%
vrp (IV minus rv_run)	:	+0.0115	[+0.0039, +0.0196]	days+=57%
IV increment beyond rv_run	:	+0.0089	[+0.0026, +0.0157]	days+=49%
adding iv as a column	:	-0.0013	[-0.0040, +0.0008]	days+=13%

Horizon 2h

LOB	:	pooled=0.7017	within-day=0.6677	
LOB+rv_run	:	pooled=0.7073	within-day=0.6829	
LOB+vrp	:	pooled=0.6719	within-day=0.6358	
LOB+rv_run+iv	:	pooled=0.6956	within-day=0.6768	
rv_run alone (pure LOB)	:	+0.0152	[+0.0029, +0.0275]	days+=58%
vrp (IV minus rv_run)	:	-0.0319	[-0.0740, +0.0098]	days+=55%
IV increment beyond rv_run	:	-0.0471	[-0.0893, -0.0074]	days+=40%
adding iv as a column	:	-0.0061	[-0.0177, +0.0049]	days+=51%

18.1 Attribution results

Reading the printed labels: IV increment beyond rv_run compares LOB+vrp against LOB+rv_run, and because vrp_run REPLACES rv_run rather than joining it, that line is a functional-form comparison between two same-sized feature sets, not a conditional test of implied volatility. The clean incremental test is adding iv as a column, which holds rv_run and adds iv_atm_30 beside it.

Adding IV beyond causal running volatility is null at all three horizons (-0.0000 at 10s, -0.0013 at 30m, -0.0061 at 2h, every CI including zero). The apparent VRP gain at 10 seconds is almost entirely accounted for by rv_run, the pure LOB component, while the clean IV increment is not statistically distinguishable from zero, so the 10-second null survives its strongest challenge.

At 30 minutes the difference form beats plain rv_run (+0.0089 with CI [+0.0026, +0.0157]), but supplying iv_atm_30 as a separate column beside rv_run adds no detectable increment (-0.0013, CI including zero). Read together, the two lines locate the 30-minute effect in the functional form rather than in an isolated implied-volatility contribution, which is a practical lesson about how a feature is written down. rv_run on its own is also no longer significant at this horizon.

At 2 hours the difference form is significantly worse than plain rv_run (-0.0471 with CI [-0.0893, -0.0074]), so the engineered form actively hurts at long horizons while the clean IV increment stays null.

Other findings: rv_run has positive point estimates at all three attribution horizons (+0.0016, +0.0026 and +0.0152), with confidence intervals excluding zero at 10 seconds and 2 hours, and it joins the recommended production set in Section 20. The day-level fit of the four extra parameters is retired for the timing reason given in Section 18, so the defensible day-level case rests on the correctly lagged Section 17 forecasts, which improve both next-day adverse-selection conditions and next-day realised volatility.

Section 19: Integrity Audit and Cboe VIX

A full alignment audit of the data pipeline, run after the modelling phase on the project owner's request, plus the integration of Cboe daily VIX data as an independent cross-vendor check and a freshness test.

Audit checks: The timezone chain (raw UTC to stored America/New_York, every bin inside regular trading hours), calendar alignment (stored dates equal NYSE trading days minus ex-dividend dates, the previous-trading-day lookup spot-checked across holidays), cross-vendor date conventions (OptionMetrics ATM IV against VIX close must correlate maximally at lag zero), and corporate actions (option adjustment factors, overnight gap scan, a split would appear as an enormous apparent gap).

This audit is what found the half-day bug: On NYSE early-close days the synthetic EQU5.MINI BBO keeps carrying abnormal one-sided quotes after 13:00 ET, and the original pipeline binned them to 16:00, contaminating six days including one inside the primary test window. The Section 2 loop now trims every session to its actual market close and the affected days were regenerated. Every number in this notebook is computed on the cleaned data.

VIX integration: The VIX open is printed at 09:30 ET, a same-morning derivatives reading, fresher than the previous-day OptionMetrics surface. Two questions. Does the morning print improve day-level forecasts, and does a fresher anchor revive the intraday VRP channel at 30 minutes (the one horizon where the EOD anchor showed an increment).

```
In [48]: # integrity audit
from datetime import time as dttime

lob_files19 = sorted(glob.glob(os.path.join(LOB_PROC, 'lob_*.parquet')))

# 1 timezone chain on one raw file and the stored output
raw_file = sorted(glob.glob(os.path.join(DBN_DIR, '*.dbn.zst')))[250]
df_raw = db.DBNStore.from_file(raw_file).to_df()
tz_raw = str(df_raw['ts_event'].dt.tz)
print(f'1. raw ts_event timezone: {tz_raw} {"PASS" if tz_raw == "UTC" else "FAIL"}')
sample19 = pd.read_parquet(lob_files19[250])
tz_stored = str(sample19.index.tz)
print(f'   stored index timezone: {tz_stored} '
      f'{"PASS" if tz_stored == "America/New_York" else "FAIL"}')

bad_days = 0
for f in lob_files19:
    idx = pd.read_parquet(f, columns=['mid']).index
    if idx.time.min() < dttime(9, 30) or idx.time.max() > dttime(15, 59, 50):
        bad_days += 1
print(f'   all {len(lob_files19)} days inside regular trading hours: '
      f'{"PASS" if bad_days == 0 else f"FAIL ({bad_days} days)"}')

# 2 calendar alignment
sched19 = nyse.schedule(start_date='2024-01-02', end_date='2025-12-31')
expected = {str(d.date()) for d in sched19.index} - EX_DIV_DATES
stored = {os.path.basename(f)[4:14] for f in lob_files19}
print(f'2. stored dates equal NYSE days minus ex-div: {"PASS" if stored == expected else "FAIL"}')
```

```

        f'({len(stored)} stored, {len(expected)} expected)')
spot = {'2024-01-16': '2024-01-12', '2024-07-08': '2024-07-05',
        '2025-04-21': '2025-04-17', '2025-05-27': '2025-05-23'}
ok_lookup = all(om_lookup.get(k) == v for k, v in spot.items())
print(f'    previous-trading-day lookup holiday spot-checks: {"PASS" if ok_lookup else
"FAIL"}')

# 3 cross-vendor date convention
vix = pd.read_csv(CBOE_CSV)
vix['Date'] = pd.to_datetime(vix['Date']).dt.date.astype(str)
vix = vix.set_index('Date').sort_index()
deriv_chk = pd.read_parquet(os.path.join(OM_PROC, 'deriv_features.parquet'))
deriv_chk.index = deriv_chk.index.astype(str)
both = deriv_chk[['iv_atm_30']].join(vix[['vix', 'vixo']], how='inner')
print(f'3. OM and VIX overlapping dates: {len(both)}')
for lag in (-1, 0, 1):
    c = both['iv_atm_30'].corr(both['vix'].shift(lag))
    print(f'    corr(iv_atm_30_t, vix_t{lag:+d}) = {c:.4f}')
print('    PASS expected as a peak at lag 0 (same-day EOD conventions)')

# 4 splits and adjustment factors
op_cfadj = pd.read_parquet(os.path.join(OM_DIR, 'opprcd2024.parquet'), columns=['secid',
'cfadj'])
op_cfadj = op_cfadj[op_cfadj['secid'] == SPX_SECID]
vals = set(op_cfadj['cfadj'].dropna().unique())
print(f'4. oprpcd cfadj unique values: {sorted(vals)} '
      f'{"PASS (no adjustments)" if vals <= {1.0} else "CHECK"}')
closes, opens_, dates19 = [], [], []
for f in lob_files19:
    s = pd.read_parquet(f, columns=['mid'])
    closes.append(s['mid'].iloc[-1])
    opens_.append(s['mid'].iloc[0])
    dates19.append(os.path.basename(f)[4:14])
gaps = pd.Series(np.abs(np.array(opens_[1:]) / np.array(closes[:-1]) - 1),
                  index=dates19[1:]).sort_values(ascending=False)
print('    largest SPY overnight gaps (a split would be near 50 percent):')
print(gaps.head(5).round(4).to_string())
print(f'    max gap {gaps.max():.2%}: '
      f'{"PASS (no splits, SPY last split 1997)" if gaps.max() < 0.10 else "CHECK"}')

```

1. raw ts_event timezone: UTC PASS
stored index timezone: America/New_York PASS
all 494 days inside regular trading hours: PASS
2. stored dates equal NYSE days minus ex-div: PASS (494 stored, 494 expected)
previous-trading-day lookup holiday spot-checks: PASS
3. OM and VIX overlapping dates: 416
corr(iv_atm_30_t, vix_t-1) = 0.8945
corr(iv_atm_30_t, vix_t+0) = 0.9815
corr(iv_atm_30_t, vix_t+1) = 0.9032
PASS expected as a peak at lag 0 (same-day EOD conventions)
4. oprpcd cfadj unique values: [np.float64(1.0)] PASS (no adjustments)
largest SPY overnight gaps (a split would be near 50 percent):

2024-08-05	0.0418
2025-04-03	0.0342
2025-04-08	0.0327
2025-04-07	0.0326
2025-05-12	0.0306

max gap 4.18%: PASS (no splits, SPY last split 1997)

```

In [49]: # VIX integration, day level and the 30-minute freshness test
vixf = pd.DataFrame({
    'vix_prev': vix['vix'].shift(1) / 100,
    'vix_open': vix['vixo'] / 100,
    'vix_overnight': (vix['vixo'] - vix['vix'].shift(1)) / 100,
}).dropna()

vctrl = ctrl.join(vixf, on='date_str')
vctrl['vrp_om'] = vctrl['iv_atm_30'] - vctrl['rv_run']
vctrl['vrp_vix'] = vctrl['vix_open'] - vctrl['rv_run']

# day-level OOS, same-day risk from information known at or before the open
day19 = vctrl.groupby('date_str').agg(
    iv_prev=('iv_atm_30', 'first'), vix_prev=('vix_prev', 'first'),
    vix_open=('vix_open', 'first'), vix_on=('vix_overnight', 'first'))
fr19 = vctrl.copy()
fr19['mid_h'] = fr19.groupby('date_str')['mid'].shift(-1)
fr19['y'] = np.where(
    fr19['quote_dir'] != 0,
    ((fr19['mid_h'] - fr19['mid']) * fr19['quote_dir'] >= 0.5 *
fr19['half_spread']).astype(int), 0)
fr19 = fr19.dropna(subset=['mid_h'])
day19['adv_rate'] = fr19.groupby('date_str')['y'].mean()
day19['rv_bps'] = fr19.groupby('date_str')['mid'].apply(lambda s: s.pct_change().std() *
1e4)
day19 = day19.dropna()
print(f'Day-level OOS, same-day risk (n={len(day19)} days)')
SETS_VIX = {
    'OM prev-day IV only': ['iv_prev'],
    'VIX prev close only': ['vix_prev'],
    'VIX open (same morning)': ['vix_open'],
    'VIX open + overnight change': ['vix_open', 'vix_on'],
    'OM + VIX open + overnight': ['iv_prev', 'vix_open', 'vix_on'],
}
for tgt in ('adv_rate', 'rv_bps'):
    print(f' target same-day {tgt}:')
    for sname, scols in SETS_VIX.items():
        print(f' {sname:30s}: OOS R2 = {oos_r2(day19[scols], day19[tgt]):+.3f}')

# 30-minute freshness attribution (the one horizon with an EOD IV increment)
h, hname = 180, '30m'
fr = vctrl.copy()
fr['mid_h'] = fr.groupby('date_str')['mid'].shift(-h)
fr['y'] = np.where(
    fr['quote_dir'] != 0,
    ((fr['mid_h'] - fr['mid']) * fr['quote_dir'] >= 0.5 * fr['half_spread']).astype(int), 0)
fr = fr.dropna(subset=['mid_h', 'vrp_om', 'vrp_vix', 'vix_overnight'])
tr_f, va_f = slice_ds(fr, TRAIN_START, TRAIN_END), slice_ds(fr, VAL_START, VAL_END)
te_f = slice_ds(fr, TEST_START, TEST_END)
tv_f = pd.concat([tr_f, va_f], ignore_index=True)
spw = float((tr_f['y'] == 0).sum()) / max((tr_f['y'] == 1).sum(), 1)
days19i = te_f['date_str'].values
SETS19 = [('LOB+rv_run', LOB_FEATURES + ['rv_run']),
          ('LOB+vrp_om', LOB_FEATURES + ['vrp_om']),
          ('LOB+vrp_vix', LOB_FEATURES + ['vrp_vix']),
          ('LOB+vrp_vix+on', LOB_FEATURES + ['vrp_vix', 'vix_overnight'])]
DA19 = {}

```

```

print(f'\nHorizon {hname}, fresher anchor attribution')
for sname, scols in SETS19:
    p, _ = fit_xgb_consistent(tr_f, va_f, tv_f, te_f, scols, spw)
    DA19[sname] = daily_auc_generic(te_f['y'].values, p, days19i)
    print(f' {sname:16s}: pooled={roc_auc_score(te_f["y"].values, p):.4f} '
          f'within-day={DA19[sname].mean():.4f}', flush=True)
for a, b, label in (('LOB+rv_run', 'LOB+vrp_om', 'OM EOD anchor increment'),
                   ('LOB+rv_run', 'LOB+vrp_vix', 'VIX morning anchor increment'),
                   ('LOB+vrp_om', 'LOB+vrp_vix', 'freshness gain (VIX vs OM)')):
    m_, lo, hi, frac = gain_ci(DA19[b] - DA19[a])
    print(f' {label:30s}: {m_:+.4f} [{lo:+.4f}, {hi:+.4f}] days+={100 * frac:.0f}%')

```

Day-level OOS, same-day risk (n=409 days)

```

target same-day adv_rate:
    OM prev-day IV only          : OOS R2 = +0.070
    VIX prev close only          : OOS R2 = +0.041
    VIX open (same morning)      : OOS R2 = +0.039
    VIX open + overnight change  : OOS R2 = +0.030
    OM + VIX open + overnight    : OOS R2 = +0.093
target same-day rv_bps:
    OM prev-day IV only          : OOS R2 = +0.626
    VIX prev close only          : OOS R2 = +0.639
    VIX open (same morning)      : OOS R2 = +0.704
    VIX open + overnight change  : OOS R2 = +0.691
    OM + VIX open + overnight    : OOS R2 = +0.677

```

Horizon 30m, fresher anchor attribution

```

LOB+rv_run      : pooled=0.6597  within-day=0.6361
LOB+vrp_om      : pooled=0.6648  within-day=0.6450
LOB+vrp_vix     : pooled=0.6620  within-day=0.6378
LOB+vrp_vix+on  : pooled=0.6584  within-day=0.6343
OM EOD anchor increment      : +0.0089  [+0.0026, +0.0157]  days+=49%
VIX morning anchor increment : +0.0017  [-0.0013, +0.0050]  days+=41%
freshness gain (VIX vs OM)  : -0.0072  [-0.0145, +0.0001]  days+=51%

```

19.1 Audit and VIX conclusions

All alignment checks pass on the cleaned data. The OptionMetrics surface and the Cboe VIX agree at lag zero with correlation near 0.98, confirming both vendors use the same EOD date convention and the previous-day broadcast is aligned correctly.

The morning VIX open is the strongest single day-level forecaster of same-day realised volatility, beating the previous-evening OptionMetrics surface, and the OM plus VIX combination is best for the adverse rate. Intraday, the fresher VIX-open anchor does not beat the smooth OM surface anchor at 30 minutes (freshness gain negative), so the 30-minute freshness test is null. The OM anchor increment at 30 minutes is significant only in the engineered difference form and disappears once implied volatility is supplied as a separate column, so no robust incremental IV effect appears at the three attribution horizons tested, which leaves the case for acquiring an intraday IV stream weak.

19.2 Preprocessing verification

Three checks interrogate the preprocessing choices of Section 3 and the descriptive sample rather than produce results.

The dispersion quality filter is estimated on the training window: Section 3 sets the OptionMetrics dispersion cutoff at the 95th percentile of the training period alone and applies that fixed threshold to every period, so no validation or test observation influences the filter. The check below quantifies how much that choice matters by rebuilding the four surface features under the alternative full-sample cutoff and comparing them.

Vega units: Reported vega is checked against the Black-Scholes identity $\text{vega} = \gamma * S^2 * \sigma * T$ on near-ATM contracts, which fixes the vendor convention and therefore the units quoted in the descriptive table of the report.

```
In [50]: # preprocessing verification: dispersion cutoff sensitivity and the vega unit convention
vs_chk = pd.concat([
    pd.read_parquet(os.path.join(OM_DIR, 'vsurfd2024.parquet')),
    pd.read_parquet(os.path.join(OM_DIR, 'vsurfd2025.parquet')),
], ignore_index=True)
vs_chk = vs_chk[vs_chk['secid'] == SPX_SECID].copy()
vs_chk['date'] = pd.to_datetime(vs_chk['date']).dt.date

cut_train = vs_chk[vs_chk['date'] <= pd.Timestamp(TRAIN_END).date()]
['dispersion'].quantile(0.95)
cut_full = vs_chk['dispersion'].quantile(0.95)

def surface_under(cut):
    """Rebuild the four vsurfd features under a given dispersion cutoff."""
    v = vs_chk[vs_chk['dispersion'] <= cut]
    atm30 = (v[(v['days'] == 30) & (v['delta'] == 50) & (v['cp_flag'] == 'C')])
        .set_index('date')[['impl_volatility']]
        .rename(columns={'impl_volatility': 'iv_atm_30'})
    put25 = v[(v['days'] == 30) & (v['delta'] == -25)].set_index('date')['impl_volatility']
    call25 = v[(v['days'] == 30) & (v['delta'] == 25)].set_index('date')['impl_volatility']
    atm91 = (v[(v['days'] == 91) & (v['delta'] == 50) & (v['cp_flag'] == 'C')])
        .set_index('date')['impl_volatility']
    d = atm30.copy()
    d['skew'] = put25 - call25
    d['term_structure'] = d['iv_atm_30'] - atm91
    d['delta_iv'] = d['iv_atm_30'].diff()
    return d[['iv_atm_30', 'delta_iv', 'skew', 'term_structure']].dropna()

f_train, f_full = surface_under(cut_train), surface_under(cut_full)
shared = f_train.index.intersection(f_full.index)
used_pts = vs_chk[((vs_chk['days'] == 30) & (vs_chk['delta'].isin([50, 25, -25])))
    | ((vs_chk['days'] == 91) & (vs_chk['delta'] == 50))]
lo, hi = min(cut_train, cut_full), max(cut_train, cut_full)
between = used_pts[(used_pts['dispersion'] > lo) & (used_pts['dispersion'] <= hi)]

print('1. Dispersion quality filter')
print(f'    cutoff in use (training window) : {cut_train:.4f}')
print(f'    alternative full-sample cutoff   : {cut_full:.4f}')
print(f'    surface points the features read: {len(used_pts):,}')
print(f'    points between the two cutoffs   : {len(between):,}')
for _, r in between.iterrows():
    print(f"        {r['date']} days={r['days']:.0f} delta={r['delta']:.0f} "
          f"dispersion={r['dispersion']:.6f}")
```

```

print(f'    days retained, training cutoff {len(f_train)} vs full-sample {len(f_full)}')
print(f'    max absolute feature difference on shared days: '
      f'{(f_train.loc[shared] - f_full.loc[shared]).abs().to_numpy().max():g}')

# vega units, via the Black-Scholes identity  $vega = \gamma * S^2 * \sigma * T$ 
vcols = ['secid', 'date', 'exdate', 'delta', 'gamma', 'vega', 'impl_volatility']
opv = pd.read_parquet(os.path.join(OM_DIR, 'opprcd2024.parquet'), columns=vcols)
opv = opv[(opv['secid'] == SPX_SECID) & opv['impl_volatility'].notna()
          & opv['gamma'].notna() & opv['vega'].notna() & (opv['gamma'] > 0)].copy()
opv['date'] = pd.to_datetime(opv['date'])
opv['exdate'] = pd.to_datetime(opv['exdate'])
opv['T'] = (opv['exdate'] - opv['date']).dt.days / 365.0
opv = opv[(opv['T'] > 0.02) & opv['delta'].abs().between(0.40, 0.60)]

spx_chk = pd.read_parquet(os.path.join(OM_DIR, 'secprd2024.parquet'))
spx_chk = spx_chk[spx_chk['secid'] == SPX_SECID].copy()
spx_chk['date'] = pd.to_datetime(spx_chk['date'])
opv = opv.merge(spx_chk[['date', 'close']], on='date', how='left').dropna(subset=['close'])
vratio = opv['vega'] / (opv['gamma'] * opv['close'] ** 2 * opv['impl_volatility'] *
opv['T'])

print('\n2. Vega units')
print(f'    near-ATM contracts checked: {len(opv):,}')
print(f'    vega / ( $\gamma * S^2 * \sigma * T$ ): median {np.median(vratio):.4f}, '
      f'5th {np.quantile(vratio, 0.05):.4f}, 95th {np.quantile(vratio, 0.95):.4f}')
print('    A ratio near 1 means vega is quoted per UNIT change in implied volatility, so a one')
print('    percentage point move is one hundredth of the reported value.')

```

1. Dispersion quality filter

```

cutoff in use (training window) : 0.0524
alternative full-sample cutoff   : 0.0494
surface points the features read: 1,668
points between the two cutoffs   : 1
    2025-04-08 days=30 delta=-25 dispersion=0.050689
days retained, training cutoff 416 vs full-sample 415
max absolute feature difference on shared days: 0

```

2. Vega units

```

near-ATM contracts checked: 480,773
vega / ( $\gamma * S^2 * \sigma * T$ ): median 0.9999, 5th 0.9862, 95th 1.0004
A ratio near 1 means vega is quoted per UNIT change in implied volatility, so a one
percentage point move is one hundredth of the reported value.

```

Residual wide-spread rows: the descriptive table of the report notes how many predictive-sample intervals carry a quoted spread above two dollars. The count comes from the same stored per-day feature files as the descriptive statistics, so it is produced by the short script `verify_wide_spreads.py` rather than by a cell of this notebook. Its source and output are reproduced here for completeness of the code record. The script rebuilds the predictive sample the same way Section 5 does, by dropping the final bin of each session, which has no forward markout.

""Counts the residual wide-spread rows quoted in Appendix B.

Reads only the stored per-day feature files, so it needs no kernel state and runs in a few seconds. It rebuilds the predictive sample the same way the notebook does, by dropping the final bin of each session, which has no forward markout, and then counts rows whose quoted spread exceeds two dollars.


```

"""
import glob
import os

import pandas as pd

ROOT = os.path.dirname(os.path.abspath(__file__))
TEST_START, TEST_END = "2025-05-01", "2025-08-29"

files = sorted(glob.glob(os.path.join(ROOT, "Data", "Processed", "Databento",
"lob_*.parquet")))
sel = [f for f in files if TEST_START <= os.path.basename(f)[4:14] <= TEST_END]
lob = pd.concat([pd.read_parquet(f) for f in sel])

lob["mid_next"] = lob.groupby("date")["mid"].shift(-1)
pred = lob.dropna(subset=["mid_next"])

wide = pred[pred["spread"] > 2.0]
print(f"test-window sessions           : {len(sel)}")
print(f"predictive sample rows           : {len(pred):,}")
print(f"rows with spread above $2          : {len(wide)}")
print(f"their times of day                 : {[str(t)[11:19] for t in wide.index]}")
print(f"all before noon                    : {all(t.hour < 12 for t in wide.index)}")
test-window sessions           : 83
predictive sample rows         : 193,057
rows with spread above $2      : 7
their times of day             : ['09:59:50', '10:00:00', '10:00:10', '10:01:10',
'10:17:20', '10:22:30', '09:59:50']
all before noon                 : True

```

The seven rows all fall between 09:59 and 10:23. An eighth interval above two dollars exists in the raw grid at 15:59:50 on 2025-08-11, but it is the final bin of that session and is dropped for want of a forward markout, so it is not part of the predictive sample.

Section 20: Order-Flow Imbalance and the LOB+ Production Feature Set

A second pass over the raw tick data (`process_ofi.py` , resume-safe, output in `Data/Processed/Databento_OFI/`) computed the canonical Cont, Kukanov and Stoikov level-1 quote-based order-flow imbalance per 10-second bin, plus gross traded volume and quote-update intensity. Two questions:

1. How much does the strongest constructible LOB feature set improve on the locked six? `LOB+` is the production base (the locked six minus the unpopulated `queue_imb`) plus `ofi` , `gross_vol` , `n_quotes` and `rv_run` , nine features.
2. Does the derivatives null survive against that strongest LOB set? If derivatives add nothing beyond `LOB+` , the headline conclusion is conservative with respect to LOB feature completeness.

Script source: the full source of `process_ofi.py` , the resume-safe second pass over the raw ticks that builds the order-flow imbalance, gross volume and quote intensity inputs used below, is reproduced

here for completeness of the code record.

```
"""
```

Second pass over the raw Databento ticks: quote-based order-flow imbalance (Cont, Kukanov and Stoikov, 2014) plus gross trade volume and quote intensity on the 10-second grid.

Motivation: the corrected pipeline showed signed trade flow dominates the 10-second signal. CKS show the QUOTE-based order-flow imbalance at the best bid and ask is the canonical microstructure driver of short-horizon price changes, and it is the one major LOB feature the original pipeline did not build. This script is resume-safe (skips existing per-day outputs) and can be relaunched any number of times until all 494 days are done.

Output: Data/Processed/Databento_OFI/ofi_YYYY-MM-DD.parquet with columns

```
ofi          level-1 order-flow imbalance summed per 10s bin
gross_vol     total traded size per bin
n_trades      trade count per bin
n_quotes      book-update count per bin (quote intensity)
"""
```

```
import glob
import os
import sys
import warnings
from datetime import time as dttime, date as ddate

import numpy as np
import pandas as pd
import pandas_market_calendars as mcal
import databento as db

sys.stdout.reconfigure(encoding='utf-8')
warnings.filterwarnings('ignore')

ROOT      = os.path.expanduser(r'~\Desktop\Imperial College London\MSc Risk Management
& Financial Engineering\Applied Project\Applied Project Analysis')
DBN_DIR   = os.path.join(ROOT, 'Data', 'Raw', 'Databento')
OUT_DIR   = os.path.join(ROOT, 'Data', 'Processed', 'Databento_OFI')
os.makedirs(OUT_DIR, exist_ok=True)

EX_DIV = {'2024-03-15', '2024-06-21', '2024-09-20', '2024-12-20',
          '2025-03-21', '2025-06-20', '2025-09-19', '2025-12-19'}
nyse = mcal.get_calendar('NYSE')
sched = nyse.schedule(start_date='2024-01-02', end_date='2025-12-31')
trading_days = set(sched.index.date)
# trim sessions to the ACTUAL market close (13:00 on half-days, the synthetic
# EQUUS.MINI BBO carries abnormal one-sided quotes after early closes)
close_map = {d.date(): c.tz_convert('America/New_York').time()
              for d, c in sched['market_close'].items()}

files = sorted(glob.glob(os.path.join(DBN_DIR, '*.dbn.zst')))
print(f'{len(files)} raw files', flush=True)

done = skipped = 0
```

```

for f in files:
    date_str = os.path.basename(f).split('.')[0].split('-')[-1]
    d = ddate(int(date_str[:4]), int(date_str[4:6]), int(date_str[6:]))
    out = os.path.join(OUT_DIR, f'ofi_{d}.parquet')
    if os.path.exists(out):
        done += 1
        continue
    if d not in trading_days or str(d) in EX_DIV:
        skipped += 1
        continue

    df = db.DBNStore.from_file(f).to_df()
    df['ts_et'] = df['ts_event'].dt.tz_convert('America/New_York')
    t_col = df['ts_et'].dt.time
    df = df[(t_col >= datetime(9, 30)) & (t_col < close_map.get(d, datetime(16, 0)))]

    q = df[df['action'] != 'T'].set_index('ts_et').sort_index()
    t = df[df['action'] == 'T'].set_index('ts_et').sort_index()

    pb, pa = q['bid_px_00'].values, q['ask_px_00'].values
    qb = q['bid_sz_00'].values.astype(np.int64)
    qa = q['ask_sz_00'].values.astype(np.int64)
    pb1, pa1 = np.roll(pb, 1), np.roll(pa, 1)
    qb1, qa1 = np.roll(qb, 1), np.roll(qa, 1)
    # Cont-Kukanov-Stoikov Level-1 OFI event contribution
    e = ((pb >= pb1) * qb - (pb <= pb1) * qb1
          - (pa <= pa1) * qa + (pa >= pa1) * qa1).astype(np.float64)
    e[0] = 0.0
    ofi = pd.Series(e, index=q.index).resample('10s', label='left',
closed='left').sum()
    n_quotes = pd.Series(1.0, index=q.index).resample('10s', label='left',
closed='left').sum()

    if len(t):
        gross = t['size'].resample('10s', label='left', closed='left').sum()
        ntr = pd.Series(1.0, index=t.index).resample('10s', label='left',
closed='left').sum()
    else:
        gross = pd.Series(dtype=float)
        ntr = pd.Series(dtype=float)

    res = pd.DataFrame({'ofi': ofi, 'n_quotes': n_quotes})
    res['gross_vol'] = gross.reindex(res.index).fillna(0.0)
    res['n_trades'] = ntr.reindex(res.index).fillna(0.0)
    res = res[(res.index.time >= datetime(9, 30)) & (res.index.time < datetime(16, 0))]
    res.to_parquet(out)
    done += 1
    if done % 25 == 0:
        print(f' {done} days done...', flush=True)

print(f'OFI pass complete: {done} done, {skipped} skipped', flush=True)

```

```

In [51]: # OFI join, LOB+ set, and the conservative-null check
NEWLOB = ['ofi', 'gross_vol', 'n_quotes', 'rv_run']

```

```

LOB_PLUS = LOB_PROD_BASE + NEWLOB
ofi_files = sorted(glob.glob(os.path.join(OFI_DIR, 'ofi_*.parquet')))
print(f'OFI files: {len(ofi_files)} (process_ofi.py regenerates them if missing)')
ofi_df = pd.concat([pd.read_parquet(f) for f in ofi_files]).sort_index()

lobp = ctrl.join(ofi_df, how='left')
miss = lobp['ofi'].isna().sum()
print(f'OFI missing after join: {miss:,} ({100 * miss / len(lobp):.3f} percent, forward
filled within day)')
for c in ['ofi', 'gross_vol', 'n_quotes', 'n_trades']:
    lobp[c] = lobp.groupby('date_str')[c].ffill().fillna(0.0)

lobp['mid_next'] = lobp.groupby('date_str')['mid'].shift(-1)
lobp['mid_change'] = lobp['mid_next'] - lobp['mid']
lobp['y'] = np.where(lobp['quote_dir'] != 0,
                    ((lobp['mid_change'] * lobp['quote_dir'])
                     >= 0.5 * lobp['half_spread']).astype(int), 0)
lobp = lobp.dropna(subset=['mid_next'])

SETS20 = [('LOB6', LOB_FEATURES),
          ('LOB+ (9)', LOB_PLUS),
          ('LOB+ + Deriv6', LOB_PLUS + DERIV_FEATURES)]
need20 = LOB_PLUS + DERIV_FEATURES
base20 = lobp.dropna(subset=need20)
tr20 = slice_ds(base20, TRAIN_START, TRAIN_END)
va20 = slice_ds(base20, VAL_START, VAL_END)
te20 = slice_ds(base20, TEST_START, TEST_END)
tv20 = pd.concat([tr20, va20], ignore_index=True)
rob20 = slice_ds(lobp.dropna(subset=LOB_PLUS), ROB_START, ROB_END)
spw20 = float((tr20['y'] == 0).sum()) / max((tr20['y'] == 1).sum(), 1)
days20 = te20['date_str'].values
print(f'rows: train {len(tr20):,} test {len(te20):,} robustness {len(rob20):,}')

DA20, MODELS20, SCALERS20 = {}, {}, {}
for sname, scols in SETS20:
    best_auc, best_lr, best_it = 0, 0.05, 100
    sc_tr = StandardScaler().fit(tr20[scols])
    for lr in (0.05, 0.1):
        m = XGBClassifier(max_depth=3, learning_rate=lr, subsample=0.8,
                          n_estimators=400, scale_pos_weight=spw20, eval_metric='auc',
                          early_stopping_rounds=30, random_state=42, verbosity=0)
        m.fit(sc_tr.transform(tr20[scols]), tr20['y'].values,
              eval_set=[(sc_tr.transform(va20[scols]), va20['y'].values)], verbose=False)
        a = roc_auc_score(va20['y'].values, m.predict_proba(sc_tr.transform(va20[scols]))[:,
1])

        if a > best_auc:
            best_auc, best_lr, best_it = a, lr, max(m.best_iteration + 1, 1)
    sc_tv = StandardScaler().fit(tv20[scols])
    mf = XGBClassifier(max_depth=3, learning_rate=best_lr, subsample=0.8,
                      n_estimators=best_it, scale_pos_weight=spw20,
                      random_state=42, verbosity=0)
    mf.fit(sc_tv.transform(tv20[scols]), tv20['y'].values)
    p = mf.predict_proba(sc_tv.transform(te20[scols]))[:, 1]
    DA20[sname] = daily_auc_generic(te20['y'].values, p, days20)
    MODELS20[sname], SCALERS20[sname] = mf, sc_tv
    print(f'{sname:14s}: pooled={roc_auc_score(te20["y"].values, p):.4f} '
          f'within-day={DA20[sname].mean():.4f} (lr={best_lr}, trees={best_it})',
flush=True)

```

```

for a, b, label in (('LOB6', 'LOB+ (9)', 'OFI + volume + intensity + rv_run'),
                   ('LOB+ (9)', 'LOB+ + Deriv6', 'derivatives beyond strongest LOB')):
    m_, lo, hi, frac = gain_ci(DA20[b] - DA20[a])
    print(f'within-day gain, {label}: {m_:.4f} [{lo:.4f}, {hi:.4f}] days+={100 *
frac:.0f}%')

imp20 = pd.Series(MODELS20['LOB+ (9)'].feature_importances_, index=LOB_PLUS)
print('\nLOB+ gain-importance shares')
print(imp20.sort_values(ascending=False).round(4).to_string())

cols20 = LOB_PLUS
p_rob20 = MODELS20['LOB+ (9)'].predict_proba(
    SCALERS20['LOB+ (9)'].transform(rob20[cols20]))[:, 1]
days_r20 = rob20['date_str'].values
y_rob20 = rob20['y'].values
wd20 = np.mean([roc_auc_score(y_rob20[days_r20 == d], p_rob20[days_r20 == d])
                 for d in np.unique(days_r20)
                 if y_rob20[days_r20 == d].min() != y_rob20[days_r20 == d].max()])
print(f'\nLOB+ out of sample Sep to Dec: pooled={roc_auc_score(y_rob20, p_rob20):.4f} '
      f'within-day={wd20:.4f}')

```

OFI files: 494 (process_ofi.py regenerates them if missing)
 OFI missing after join: 0 (0.000 percent, forward filled within day)
 rows: train 525,324 test 192,642 robustness 191,562
 LOB6 : pooled=0.6328 within-day=0.6103 (lr=0.1, trees=77)
 LOB+ (9) : pooled=0.6380 within-day=0.6136 (lr=0.05, trees=247)
 LOB+ + Deriv6 : pooled=0.6367 within-day=0.6131 (lr=0.05, trees=243)
 within-day gain, OFI + volume + intensity + rv_run: +0.0033 [+0.0020, +0.0047] days+=75%
 within-day gain, derivatives beyond strongest LOB: -0.0006 [-0.0010, -0.0001] days+=43%

LOB+ gain-importance shares

signed_flow	0.6180
gross_vol	0.3258
n_quotes	0.0200
imbalance	0.0124
spread	0.0079
rv_run	0.0072
ofi	0.0034
depth	0.0031
realised_vol	0.0021

LOB+ out of sample Sep to Dec: pooled=0.7372 within-day=0.7251

In [52]:

```

# composition of the activity convention on the test window
sf_test = slice_ds(lobp, TEST_START, TEST_END)
ct = pd.crosstab(sf_test['gross_vol'] > 0, sf_test['signed_flow'].abs() > 0,
                 normalize='all').round(4)
ct.index = ['No Traded Volume', 'Traded Volume > 0']
ct.columns = ['Signed Flow = 0', 'Directional Signed Flow']
print('Share of test bins by trade activity and signed-flow direction')
print(ct.to_string())
print('\nBins with traded volume but zero signed flow carry ambiguous or fully')
print('netted direction (side N or offsetting buys and sells). They are excluded')
print('from the active-flow definition because the exposure direction is undefined.')

```

Share of test bins by trade activity and signed-flow direction

	Signed Flow = 0	Directional Signed Flow
No Traded Volume	0.0180	0.0000

Traded Volume > 0

0.1237

0.8583

Bins with traded volume but zero signed flow carry ambiguous or fully netted direction (side N or offsetting buys and sells). They are excluded from the active-flow definition because the exposure direction is undefined.

20.1 OFI conclusions

The strongest LOB set beats the locked six within-day by about +0.003 with the CI excluding zero, on roughly three quarters of days, and generalises out of sample. Gross traded volume carries the largest gain-importance share among the newly added features, far above the quote-based OFI itself (share below 1%), so at 10 seconds signed trade flow appears to carry most of the order-flow story and the CKS quote-level refinement adds little on top. Measured against `LOB+`, all three derivatives variants are significantly negative, so the headline null is conservative with respect to LOB feature completeness.

`LOB+` is the recommended production feature set, with `queue_imb` dropped from it. Order-count fields are not populated in EQUUS.MINI, so queue-position features are excluded rather than carried as a zero column.

20.2 The minimal results hierarchy

The single table that answers the research question, replacing the eight-model grid with the production-relevant feature hierarchy on one engine (XGBoost, identical protocol throughout). The locked baseline, the strongest constructible order-flow and top-of-book set, and that set augmented by each derivatives variant. The thesis in three lines:

1. `LOB+` beats `LOB6`.
2. No derivatives variant beats `LOB+` at 10 seconds, and all three are significantly negative against it.
3. After the causality fix of Section 18, no pure derivatives increment is statistically reliable at any of the three attribution horizons tested with the causal running-volatility control.

```
In [53]: # minimal hierarchy on one engine, within-day AUC and gain vs LOB+
hier_frame = lobb.copy()
hier_frame['om_date'] = hier_frame['date_str'].map(om_lookup)
hier_frame = hier_frame.join(deriv_all_ext[DERIV_EXT].rename_axis('om_date'),
                             on='om_date', how='left')
hier_frame = hier_frame.drop(columns=['om_date'])

HIER = [
    ('LOB6 (locked baseline)', LOB_FEATURES),
    ('LOB+ (production set)', LOB_PLUS),
    ('LOB+ and surface (6)', LOB_PLUS + DERIV_FEATURES),
    ('LOB+ and surface + flow (11)', LOB_PLUS + DERIV11),
    ('LOB+ and VRP difference', LOB_PLUS + ['vrp_run']),
]
need_h = sorted(set(sum((c for _, c in HIER), [])))
base_h = hier_frame.dropna(subset=need_h)
tr_h = slice_ds(base_h, TRAIN_START, TRAIN_END)
va_h = slice_ds(base_h, VAL_START, VAL_END)
te_h = slice_ds(base_h, TEST_START, TEST_END)
tv_h = pd.concat([tr_h, va_h], ignore_index=True)
sp_h = float((tr_h['y'] == 0).sum()) / max((tr_h['y'] == 1).sum(), 1)
```

```

days_h = te_h['date_str'].values
print(f'rows: train {len(tr_h[:,}) test {len(te_h[:,})}')

DA_h = {}
for hname, hcols in HIER:
    best_auc, best_lr, best_it = 0, 0.05, 100
    sc_tr = StandardScaler().fit(tr_h[hcols])
    for lr in (0.05, 0.1):
        m = XGBClassifier(max_depth=3, learning_rate=lr, subsample=0.8,
                           n_estimators=400, scale_pos_weight=sp_h, eval_metric='auc',
                           early_stopping_rounds=30, random_state=42, verbosity=0)
        m.fit(sc_tr.transform(tr_h[hcols]), tr_h['y'].values,
              eval_set=[(sc_tr.transform(va_h[hcols]), va_h['y'].values)], verbose=False)
        a = roc_auc_score(va_h['y'].values, m.predict_proba(sc_tr.transform(va_h[hcols]))[:,
1]))
        if a > best_auc:
            best_auc, best_lr, best_it = a, lr, max(m.best_iteration + 1, 1)
    sc_tv = StandardScaler().fit(tv_h[hcols])
    mf = XGBClassifier(max_depth=3, learning_rate=best_lr, subsample=0.8,
                       n_estimators=best_it, scale_pos_weight=sp_h,
                       random_state=42, verbosity=0)
    mf.fit(sc_tv.transform(tv_h[hcols]), tv_h['y'].values)
    p = mf.predict_proba(sc_tv.transform(te_h[hcols]))[:, 1]
    DA_h[hname] = daily_auc_generic(te_h['y'].values, p, days_h)
    print(f'{hname:30s}: pooled={roc_auc_score(te_h["y"].values, p):.4f} '
          f'within-day={DA_h[hname].mean():.4f}', flush=True)

print('\nTable 9, Minimal Hierarchy (within-day gain vs LOB+)')
ref = DA_h['LOB+ (production set)']
for hname, _ in HIER:
    if hname == 'LOB+ (production set)':
        continue
    m_, lo, hi, fr_ = gain_ci(DA_h[hname] - ref)
    print(f' {hname:30s}: {m_:+.4f} [{lo:+.4f}, {hi:+.4f}] days+={100 * fr_: .0f}%')

```

rows: train 525,324 test 192,642

LOB6 (locked baseline)	: pooled=0.6328	within-day=0.6103
LOB+ (production set)	: pooled=0.6380	within-day=0.6136
LOB+ and surface (6)	: pooled=0.6367	within-day=0.6131
LOB+ and surface + flow (11)	: pooled=0.6369	within-day=0.6127
LOB+ and VRP difference	: pooled=0.6375	within-day=0.6130

Table 9, Minimal Hierarchy (within-day gain vs LOB+)

LOB6 (locked baseline)	: -0.0033	[-0.0047, -0.0020]	days+=25%
LOB+ and surface (6)	: -0.0006	[-0.0010, -0.0001]	days+=43%
LOB+ and surface + flow (11)	: -0.0009	[-0.0014, -0.0004]	days+=40%
LOB+ and VRP difference	: -0.0006	[-0.0009, -0.0003]	days+=37%

Section 21: Sequence Model (Temporal Convolutional Network)

The last untested model class. A small causal dilated TCN (three convolutional layers, 32 channels, about 7,000 parameters) reads a rolling 30-bin (5-minute) window of features per prediction, the sequence representation that the tabular models cannot see. Windows never span a day boundary. Features are standardised with a train-only scaler, because the network never refits on the validation window, which it

uses solely for early stopping. Training follows the standard deep learning protocol. Fit on the training window, early stopping on validation AUC, one evaluation on test. Following the Section 15 lesson, each configuration is a three-seed ensemble (seeds 42, 7, 123) with per-seed results printed for transparency.

There are two configurations, TCN on the LOB+ features (9 inputs) and TCN on LOB+ plus the six surface derivatives (15 inputs).

```
In [54]: # TCN on rolling 5-minute windows, torch CPU
torch.set_num_threads(os.cpu_count())
W21 = 30

tcn_frame = lobb.dropna(subset=need20).copy()
tcn_frame['bin_in_day'] = tcn_frame.groupby('date_str').cumcount()

class TCN(nn.Module):
    def __init__(self, f_in, ch=32):
        super().__init__()
        self.net = nn.Sequential(
            nn.Conv1d(f_in, ch, 3, padding=2, dilation=1), nn.ReLU(),
            nn.Conv1d(ch, ch, 3, padding=4, dilation=2), nn.ReLU(),
            nn.Conv1d(ch, ch, 3, padding=8, dilation=4), nn.ReLU(),
        )
        self.head = nn.Linear(ch, 1)

    def forward(self, x):
        h = self.net(x)[: , : , :x.shape[2]]
        return self.head(h[: , : , -1]).squeeze(-1)

def run_tcn_seed(label, feats, seed):
    torch.manual_seed(seed)
    print(f'\n{label} seed {seed}', flush=True)
    tr_mask = ((tcn_frame['date_str'] >= TRAIN_START)
               & (tcn_frame['date_str'] <= TRAIN_END))
    sc = StandardScaler().fit(tcn_frame.loc[tr_mask, feats])
    X_all = sc.transform(tcn_frame[feats]).astype(np.float32)
    y_all = tcn_frame['y'].values.astype(np.float32)
    dts = tcn_frame['date_str'].values
    okwin = tcn_frame['bin_in_day'].values >= W21 - 1

    def split_idx(s, e):
        return np.where(okwin & (dts >= s) & (dts <= e))[0]

    idx_tr, idx_va, idx_te = (split_idx(TRAIN_START, TRAIN_END),
                             split_idx(VAL_START, VAL_END),
                             split_idx(TEST_START, TEST_END))
    print(f'windows: train {len(idx_tr):,} val {len(idx_va):,} test {len(idx_te):,}',
          flush=True)

    X_t = torch.from_numpy(X_all)
    offs = torch.arange(-(W21 - 1), 1)

    def batch_windows(ix):
        rows = torch.from_numpy(ix).unsqueeze(1) + offs.unsqueeze(0)
        return X_t[rows].permute(0, 2, 1)
```

```

model = TCN(len(feats))
pos_w = torch.tensor((y_all[idx_tr] == 0).sum() / max((y_all[idx_tr] == 1).sum(), 1))
lossf = nn.BCEWithLogitsLoss(pos_weight=pos_w)
opt = torch.optim.Adam(model.parameters(), lr=1e-3)

```

```

def predict(ix, bs=8192):
    model.eval()
    out = []
    with torch.no_grad():
        for k in range(0, len(ix), bs):
            out.append(torch.sigmoid(model(batch_windows(ix[k:k + bs]))).numpy())
    return np.concatenate(out)

```

```

best_va, best_state, patience = 0.0, None, 0
rng21 = np.random.RandomState(42)
for epoch in range(15):
    model.train()
    perm = rng21.permutation(idx_tr)
    for k in range(0, len(perm), 2048):
        ix = perm[k:k + 2048]
        opt.zero_grad()
        loss = lossf(model(batch_windows(ix)), torch.from_numpy(y_all[ix]))
        loss.backward()
        opt.step()
    va = roc_auc_score(y_all[idx_va], predict(idx_va))
    print(f'epoch {epoch + 1}: val AUC={va:.4f}', flush=True)
    if va > best_va + 1e-4:
        best_va, patience = va, 0
        best_state = {k: v.clone() for k, v in model.state_dict().items()}
    else:
        patience += 1
        if patience >= 3:
            break
model.load_state_dict(best_state)
p_te = predict(idx_te)
pooled = roc_auc_score(y_all[idx_te], p_te)
wd = daily_auc_generic(y_all[idx_te], p_te, dts[idx_te]).mean()
print(f'{label} seed {seed}: test pooled={pooled:.4f} within-day={wd:.4f}', flush=True)
return p_te, y_all[idx_te], dts[idx_te]

```

```

def run_tcn(label, feats):
    ps = []
    for seed in SEEDS:
        p, y_te21, d_te21 = run_tcn_seed(label, feats, seed)
        ps.append(p)
    p_ens = np.mean(ps, axis=0)
    pooled = roc_auc_score(y_te21, p_ens)
    wd = daily_auc_generic(y_te21, p_ens, d_te21).mean()
    print(f'{label} ENSEMBLE ({len(SEEDS)} seeds): pooled={pooled:.4f} within-day=
{wd:.4f}', flush=True)
    return pooled, wd

```

```

tcn_lobp = run_tcn('TCN LOB+', LOB_PLUS)
tcn_full = run_tcn('TCN LOB+ Deriv', LOB_PLUS + DERIV_FEATURES)
print(f'\nDerivatives gain in the sequence class (seed ensembles): '

```



```

        f'pooled {tcn_full[0] - tcn_lobp[0]:+.4f} within-day {tcn_full[1] -
tcn_lobp[1]:+.4f}')
print(f'Reference, XGBoost on the same LOB+ features: '
      f'pooled {roc_auc_score(te20["y"].values, MODELS20["LOB+
(9)"].predict_proba(SCALERS20["LOB+ (9)"].transform(te20[cols20]))[:, 1]):.4f}')

```

TCN LOB+ seed 42

windows: train 518,770 val 229,420 test 190,235

epoch 1: val AUC=0.6052

epoch 2: val AUC=0.6134

epoch 3: val AUC=0.6162

epoch 4: val AUC=0.6179

epoch 5: val AUC=0.6188

epoch 6: val AUC=0.6197

epoch 7: val AUC=0.6204

epoch 8: val AUC=0.6192

epoch 9: val AUC=0.6231

epoch 10: val AUC=0.6276

epoch 11: val AUC=0.6303

epoch 12: val AUC=0.6338

epoch 13: val AUC=0.6357

epoch 14: val AUC=0.6411

epoch 15: val AUC=0.6437

TCN LOB+ seed 42: test pooled=0.6152 within-day=0.5782

TCN LOB+ seed 7

windows: train 518,770 val 229,420 test 190,235

epoch 1: val AUC=0.6043

epoch 2: val AUC=0.6136

epoch 3: val AUC=0.6166

epoch 4: val AUC=0.6177

epoch 5: val AUC=0.6185

epoch 6: val AUC=0.6190

epoch 7: val AUC=0.6194

epoch 8: val AUC=0.6170

epoch 9: val AUC=0.6189

epoch 10: val AUC=0.6194

TCN LOB+ seed 7: test pooled=0.5999 within-day=0.5572

TCN LOB+ seed 123

windows: train 518,770 val 229,420 test 190,235

epoch 1: val AUC=0.6060

epoch 2: val AUC=0.6127

epoch 3: val AUC=0.6162

epoch 4: val AUC=0.6167

epoch 5: val AUC=0.6175

epoch 6: val AUC=0.6173

epoch 7: val AUC=0.6189

epoch 8: val AUC=0.6157

epoch 9: val AUC=0.6190

epoch 10: val AUC=0.6186

epoch 11: val AUC=0.6199

epoch 12: val AUC=0.6232

epoch 13: val AUC=0.6223

epoch 14: val AUC=0.6272

epoch 15: val AUC=0.6285

TCN LOB+ seed 123: test pooled=0.6039 within-day=0.5645

TCN LOB+ ENSEMBLE (3 seeds): pooled=0.6128 within-day=0.5732

```
TCN LOB+ Deriv seed 42
windows: train 518,770 val 229,420 test 190,235
epoch 1: val AUC=0.6032
epoch 2: val AUC=0.6109
epoch 3: val AUC=0.6140
epoch 4: val AUC=0.6144
epoch 5: val AUC=0.6157
epoch 6: val AUC=0.6173
epoch 7: val AUC=0.6165
epoch 8: val AUC=0.6175
epoch 9: val AUC=0.6179
epoch 10: val AUC=0.6174
epoch 11: val AUC=0.6129
epoch 12: val AUC=0.6168
TCN LOB+ Deriv seed 42: test pooled=0.6062 within-day=0.5577
```

```
TCN LOB+ Deriv seed 7
windows: train 518,770 val 229,420 test 190,235
epoch 1: val AUC=0.5995
epoch 2: val AUC=0.6099
epoch 3: val AUC=0.6139
epoch 4: val AUC=0.6164
epoch 5: val AUC=0.6154
epoch 6: val AUC=0.6168
epoch 7: val AUC=0.6151
epoch 8: val AUC=0.6168
epoch 9: val AUC=0.6159
TCN LOB+ Deriv seed 7: test pooled=0.6038 within-day=0.5567
```

```
TCN LOB+ Deriv seed 123
windows: train 518,770 val 229,420 test 190,235
epoch 1: val AUC=0.6000
epoch 2: val AUC=0.6086
epoch 3: val AUC=0.6137
epoch 4: val AUC=0.6124
epoch 5: val AUC=0.6107
epoch 6: val AUC=0.6159
epoch 7: val AUC=0.6166
epoch 8: val AUC=0.6177
epoch 9: val AUC=0.6157
epoch 10: val AUC=0.6149
epoch 11: val AUC=0.6139
TCN LOB+ Deriv seed 123: test pooled=0.6043 within-day=0.5565
TCN LOB+ Deriv ENSEMBLE (3 seeds): pooled=0.6077 within-day=0.5598
```

Derivatives gain in the sequence class (seed ensembles): pooled -0.0051 within-day -0.0134
Reference, XGBoost on the same LOB+ features: pooled 0.6380

21.1 Sequence-model conclusions

The TCN underperforms tabular boosting on the same features (within-day 0.5732 against 0.6136 for the LOB+ ensembles), and the derivatives increment in the sequence class is negative (-0.0134 within-day between seed ensembles). The 5-minute window representation adds nothing at the 10-second horizon, consistent with `signed_flow` summarising much of the relevant recent history at this horizon. A

compact TCN robustness check did not overturn the tabular result. This is a robustness check, not an exhaustive deep architecture search. Richer architectures remain future work.

Section 22: Final Answer and Desk Recommendation

At the primary 10-second target, derivatives-implied information provides no robust incremental value, and it shows no consistent benefit across the tested intraday horizons. The information-set sweep contains one isolated significant positive at one minute that does not replicate across adjacent horizons or specifications, and separate exploratory checks on the label threshold and on feature form produce further positive estimates, none of them stable enough to overturn the intraday conclusion. The remaining positive evidence sits at the day level, where the nested-model test supports incremental predictive content but the decision layer cannot confirm its economic conversion.

Frequency	Derivatives increment	Evidence
10 Seconds to 5 Minutes	No consistent increment	Five model classes, surface and flow features, the VRP attribution control
30 Minutes	No robust benefit	Sweep not significant, clean IV increment null (-0.0013), difference-form estimate unstable
2 Hours	No robust benefit	Attribution control, adding IV beyond running volatility not significant (-0.0061, CI includes zero)
1 Day	Positive, surface-driven, nested test significant	OOS R-squared 0.30 to 0.44 beyond own LOB lags (Clark-West $p = 0.021$ on 209 days, unadjusted DM $p = 0.115$), and derivatives also improve the volatility forecast
Decision Layer	Not identified	Rule gain is almost perfectly ordered by flag rate, so pairwise differences measure coverage, not information set

Scope of the discrimination claim: Conditional on a bin having directional flow (the economically exposed case), within-day AUC is roughly 0.50 to 0.52 across model classes and the retrained derivatives gain stays negative without excluding zero (Section 15.4). A large share of the headline 0.61 within-day AUC is exposure prediction, identifying intervals without net directional signed flow, which are mechanically safe under the target convention, which the calibrated rules monetise through safe-window tightening. The label-threshold robustness (Section 15.5) shows the null also holds at the break-even threshold, with a small exploratory positive derivatives gain (+0.0021) appearing only at the severe threshold, consistent with the volatility-regime channel.

Answer to the research question: At the primary 10-second target, no robust incremental derivatives value is found. Every stable estimate is null, tiny or fragile across five model classes (logistic regression, XGBoost, MLP ensembles, HistGB, TCN), multiple derivatives feature constructions, and every validation lens, and several rounds of data correction reduced the apparent gains further. At the day level, the evidence is positive. End-of-day surface features improve next-day adverse-selection forecasts beyond the desk's own day-level history (0.296 to 0.444, surface-driven), and they also improve the next-day volatility forecast, 0.605 for the derivatives block alone against 0.571 for the desk's own day lags. Because the comparison is nested, the appropriate test is the Clark-West adjusted statistic, which reaches +2.04

with a one-sided p of 0.021 and rejects equal predictive accuracy on 209 evaluation days, while the unadjusted comparison does not at $p = 0.115$. The decision layer cannot rescue the claim either. Once calibration and threshold selection are out of sample, the rules sit at different coverage and their measured gain is almost perfectly ordered by flag rate, so the pairwise comparison identifies selectivity rather than information and the rule-level gain reported under the earlier in-sample calibration protocol was a further evaluation distortion.

Desk recommendation: Quote at 10 seconds off the LOB-only boosted engine on the **LOB+** feature set (the production base plus OFI, gross volume, quote intensity and the causal running volatility, nine features). Do not buy the derivatives feed for intraday signal, and treat the decision layer as silent on the question rather than as support. The channel worth monitoring is day-ahead forecasting of adverse-selection conditions, where the surface improves the forecast with statistical support from the nested-model test, but with its economic conversion still unvalidated. The highest-value data acquisition question is intraday options flow (OPRA-level), not another daily surface.

Literature reconciliation: The sceptical quote-frequency findings (Muravyev, Pearson and Broussard, 2013. Options quotes adjust to stocks, not the reverse) and the supportive daily-frequency findings (Pan and Poteshman, 2006, Cremers and Weinbaum, 2010, Xing, Zhang and Zhao, 2010) are both consistent with the results of one pipeline, with the clearest positive evidence for derivatives information at the day level.

The methodological contribution: The naive pipeline produced an apparent yes. A validation layer designed to falsify it (placebo permutation, seed ensembling, within-day decomposition, a day-ID control) dismantled the apparent gains and exposed a final-evaluation scaler mismatch that no metric-level reasoning could have found, and making the decision layer's calibration and threshold selection strictly out of sample removed the last apparent economic gain. The six resulting lessons (consistent scaling between training and evaluation, pooled versus within-day metrics with day-constant features, seed ensembling for neural networks, placebo testing as standard practice, out-of-sample calibration for decision rules, coverage matching before comparing threshold rules) generalise to any study that broadcasts low-frequency features to high-frequency grids.

Limitations:

Limitation	Consequence for interpretation
EQU5.MINI MBP-1 is top-of-book (L1), not full depth or MBO	No queue position and no order-level dynamics. Depth and imbalance are L1 quantities
No fill-probability, queue-priority or inventory model	All dollar figures are mean spread-capture-proxy improvements among retained intervals, not realised PnL
OptionMetrics is end-of-day, broadcast to the following session	The design tests prior-day regime information by construction. Intraday options flow is untested (OPRA-level data would be required)
SPX options map to SPY ETF risk state, not the same instrument	The derivatives features proxy the index-level volatility and skew environment, not SPY-specific option flow
Common derivatives sample ends 2025-08-29	The Sep to Dec 2025 window supports LOB-only engine stability claims, not information-set comparisons

Limitation	Consequence for interpretation
Target convention sets $y = 0$ without net directional signed flow	Headline AUC mixes exposure prediction with conditional adverse selection (Section 15.4 quantifies both)
The TCN is one compact sequence architecture, not a search	The sequence-class null is a robustness check, not an exhaustive deep learning result

Reproducibility: The core modelling pipeline uses fixed seeds (42, and 7/123 for ensembles), pinned environment of Python 3.10.8, numpy 2.2.6, pandas 2.2.2, scikit-learn 1.5.0, xgboost 2.1.4 (pinned below 3, see Section 7), shap 0.49.1, statsmodels 0.14.2, databento 0.79.0, pyarrow 24.0.0, torch 2.12.0+cpu (imported before numpy) and pandas_market_calendars 5.4.0. Inputs are the raw files under `Data/Raw/`, outputs are the processed per-day features, saved predictions, figures and tables, and the saved test predictions are reproducible under the pinned environment and fixed seeds.